



# Microsoft Power BI

Mohammad Ghaheri



## Microsoft Power BI Desktop

With the Power BI Desktop you can visually explore your data through a free-form drag-and-drop canvas, a broad range of modern data visualizations, and an easy-to-use report authoring experience.

[Download >](#)

[Advanced download options >](#)



## On-premises reporting with Power BI Report Server

Deploy and distribute interactive Power BI reports—and traditional paginated reports—within the boundaries of your organization's firewall.



## Microsoft Power BI Mobile

Access your data anywhere, anytime. These native apps provide live, interactive, mobile access to your important business information.



## Microsoft Power BI Report Builder

Create pixel-perfect paginated reports for printing or electronic distribution with a familiar experience relied on by thousands of report authors.

[Download >](#)



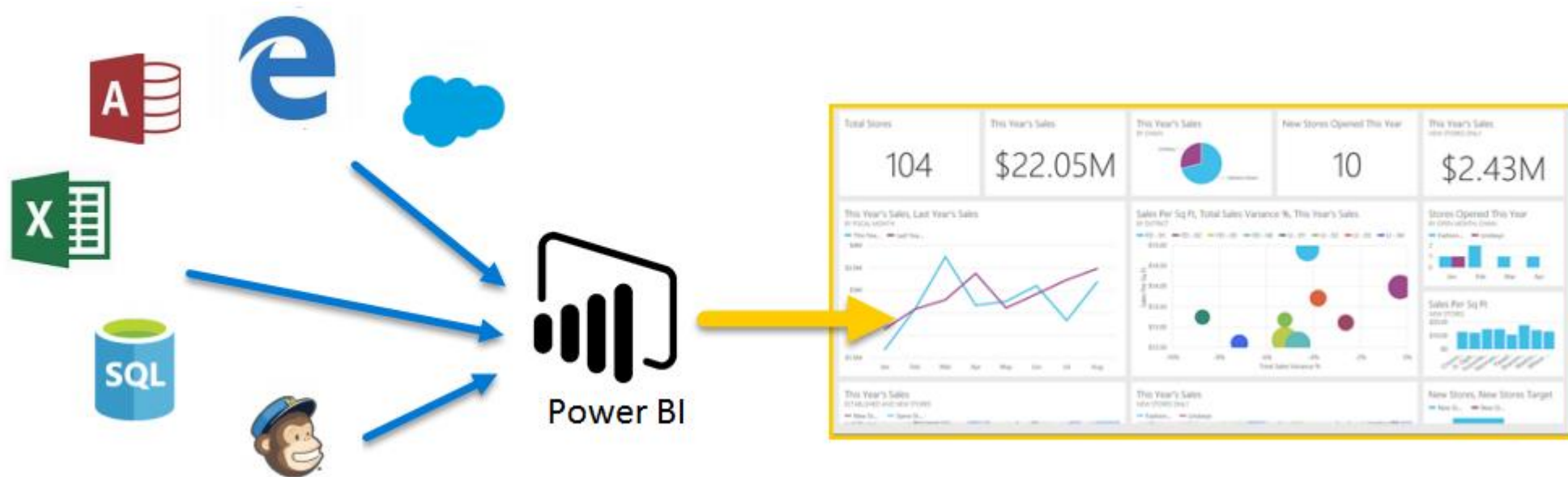
## Microsoft on-premises data gateway

Keep your dashboards and reports up to date by connecting to your on-premises data sources—without the need to move the data.

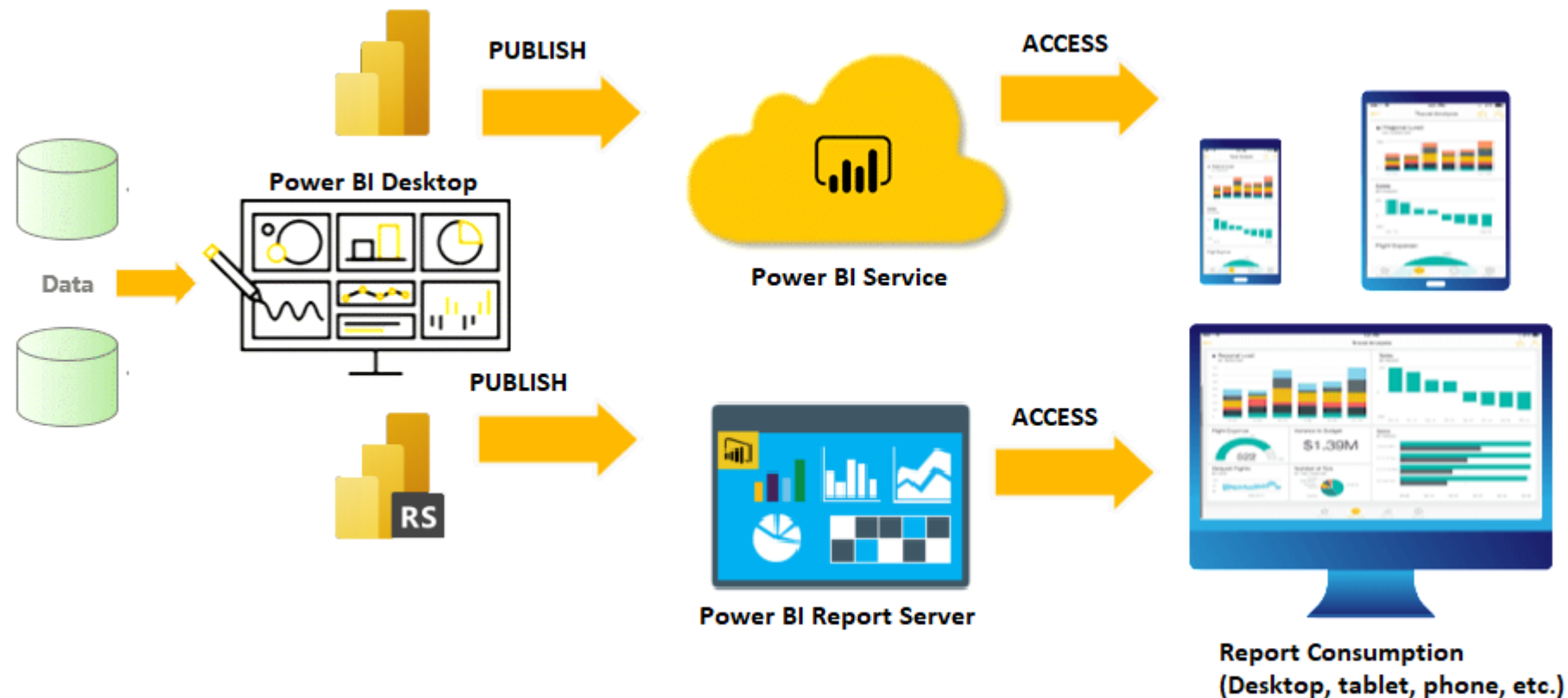
[Download standard mode >](#)

[Download personal mode >](#)

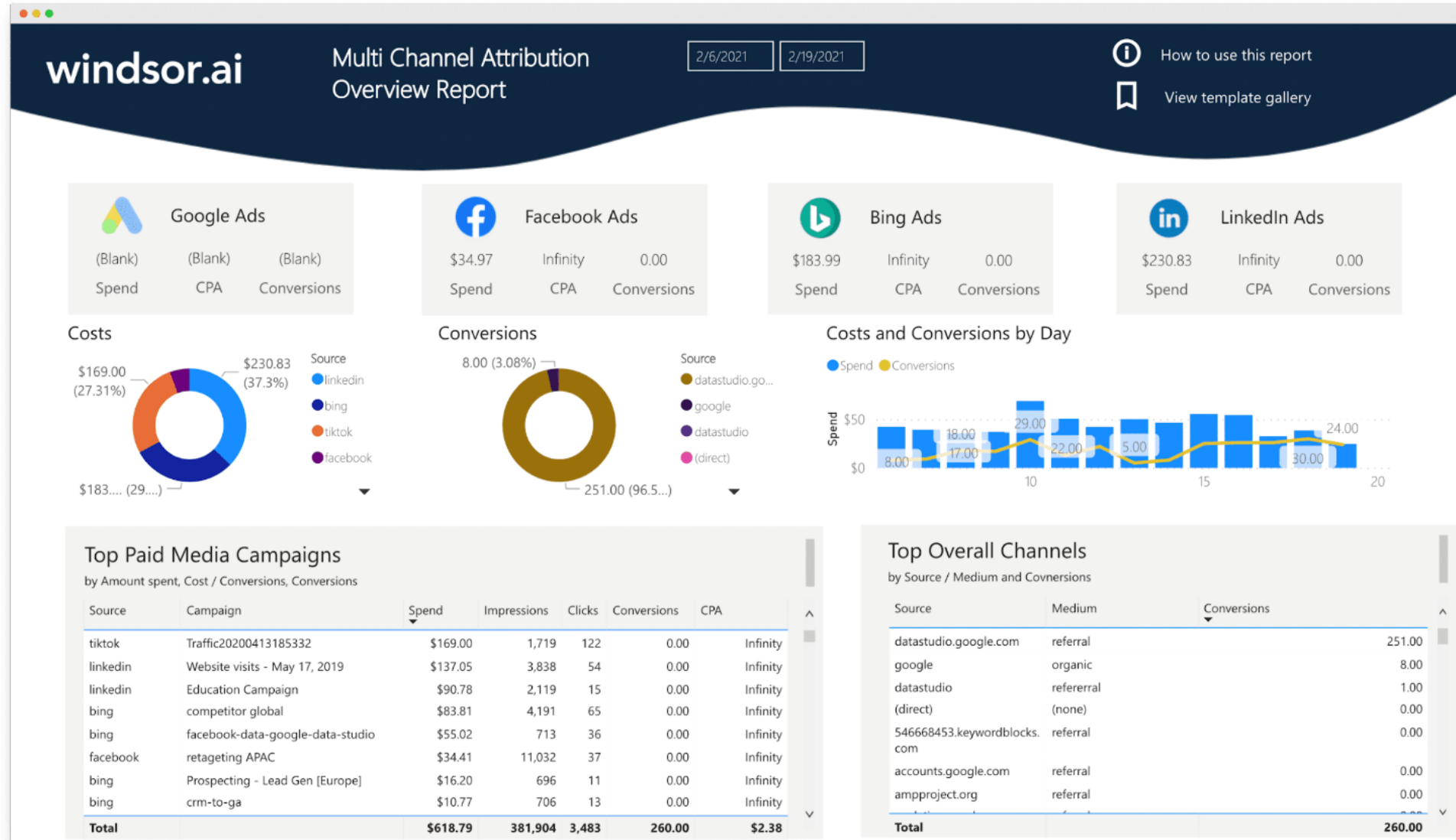
# Converting Data into Insights



# Power BI Desktop vs Power BI Desktop Report Server



# Dashboards



# Course Index

- What is Power BI
- Power BI installation
- My first dashboard in Power BI
- Get data in Power BI
- Clean, transform, and load data in Power BI
- Design a data model in Power BI
- Work with Power BI visuals

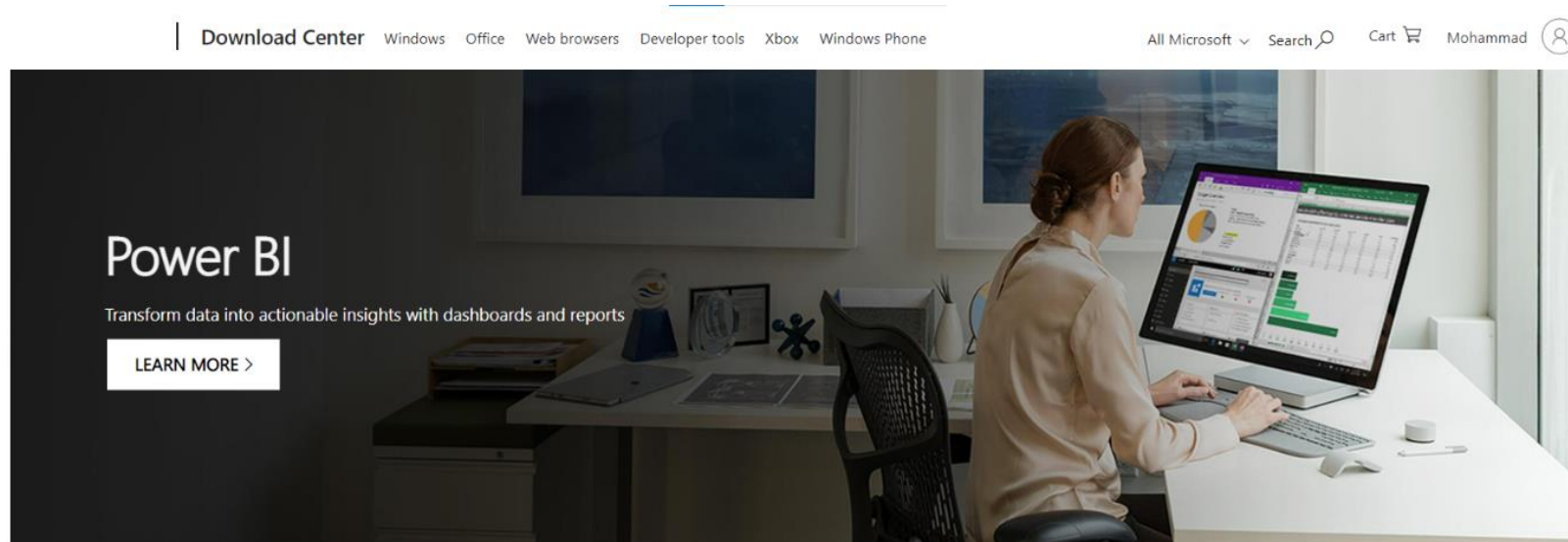


# Course Index

- Shamsi dates & dates dimension in Power Bi
- Introduction to creating measures using DAX in Power BI
- Create a data-driven story with Power BI reports
- Create dashboards in Power BI
- Perform analytics in Power BI
- Power BI advanced topics



# Install Power BI



## Microsoft Power BI Desktop

*Important!* Selecting a language below will dynamically change the complete page content to that language.

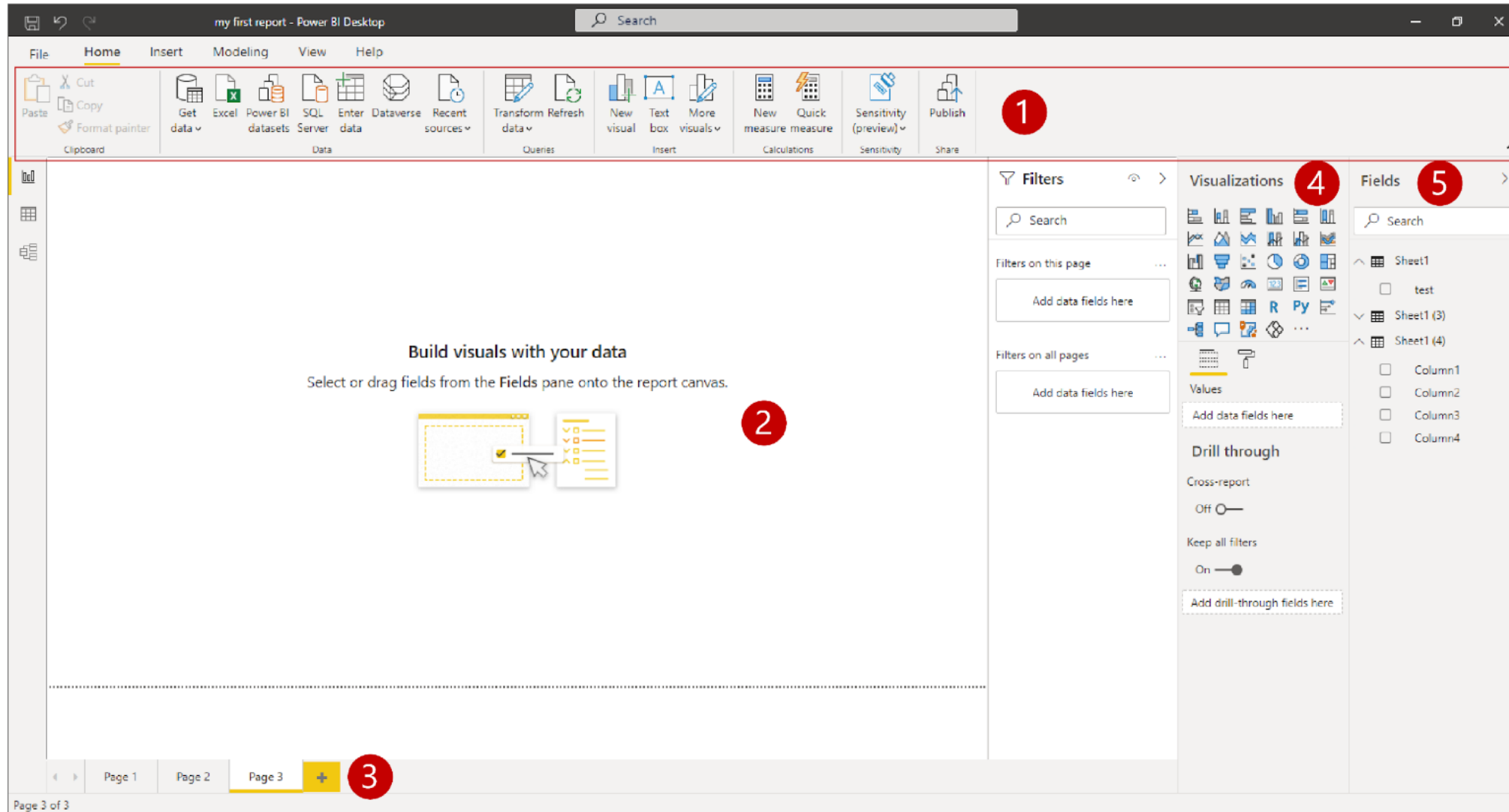
Select Language:

English

Download

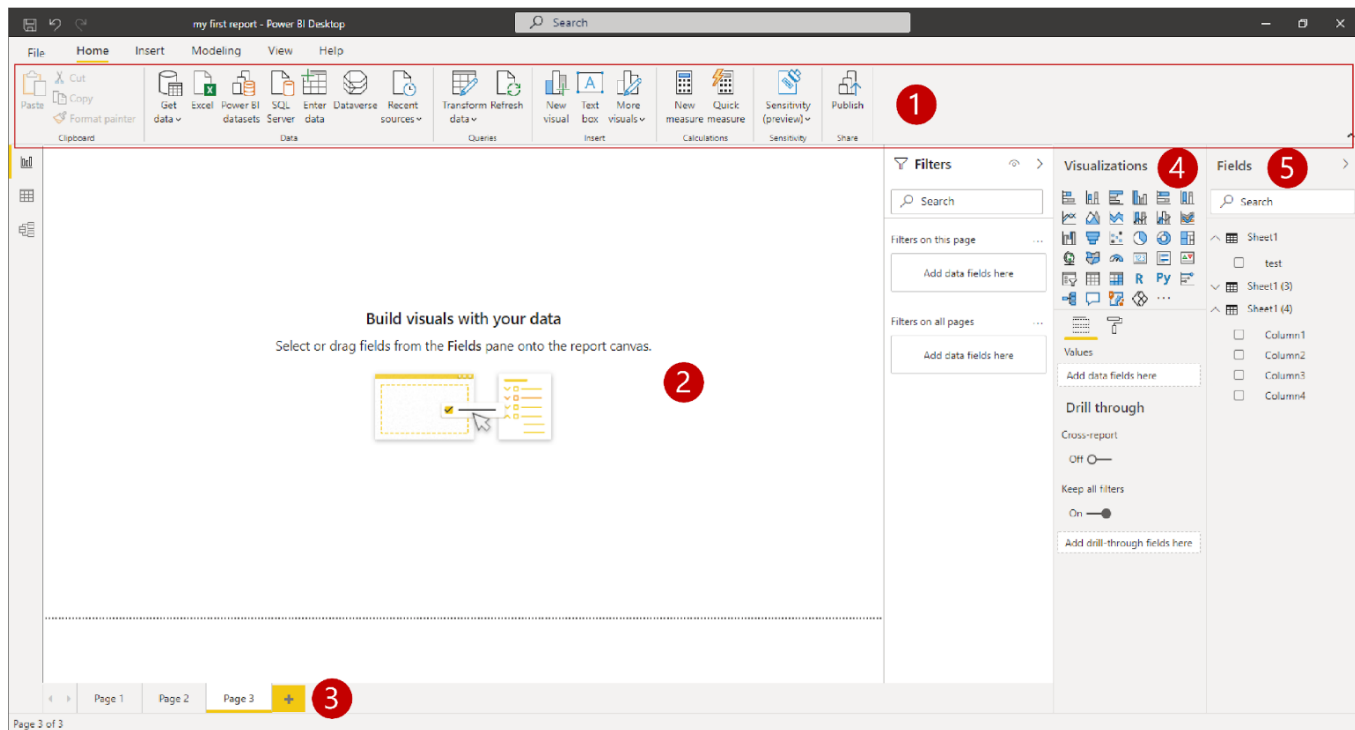
Microsoft Power BI Desktop is built for the analyst. It combines state-of-the-art interactive visualizations, with industry-leading data query and modeling built-in. Create and publish your reports to Power BI. Power BI Desktop helps you empower others with timely critical insights anytime, anywhere.

# Power BI Desktop

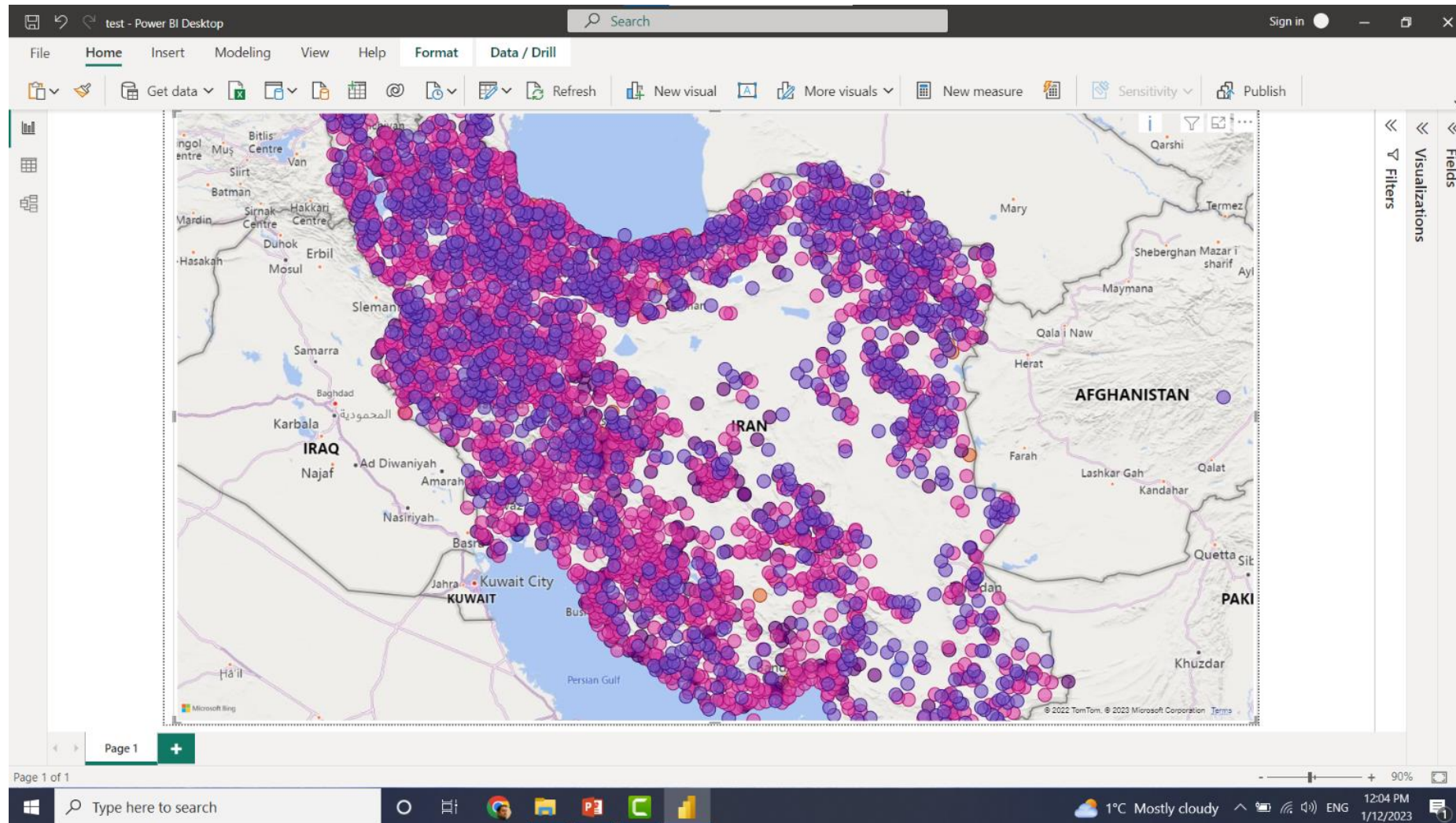


# Power BI Desktop

1. Ribbon
2. Report view, or canvas
3. Pages tab
4. Visualizations pane
5. Fields pane



# My First Dashboard



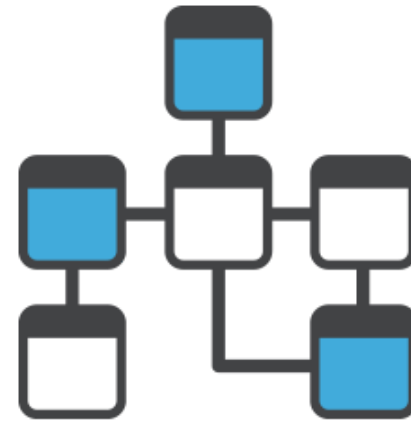
# Our RoadMap



Getting Data



Clean &  
Transform  
Data



Design a  
Data Model



Making Visuals



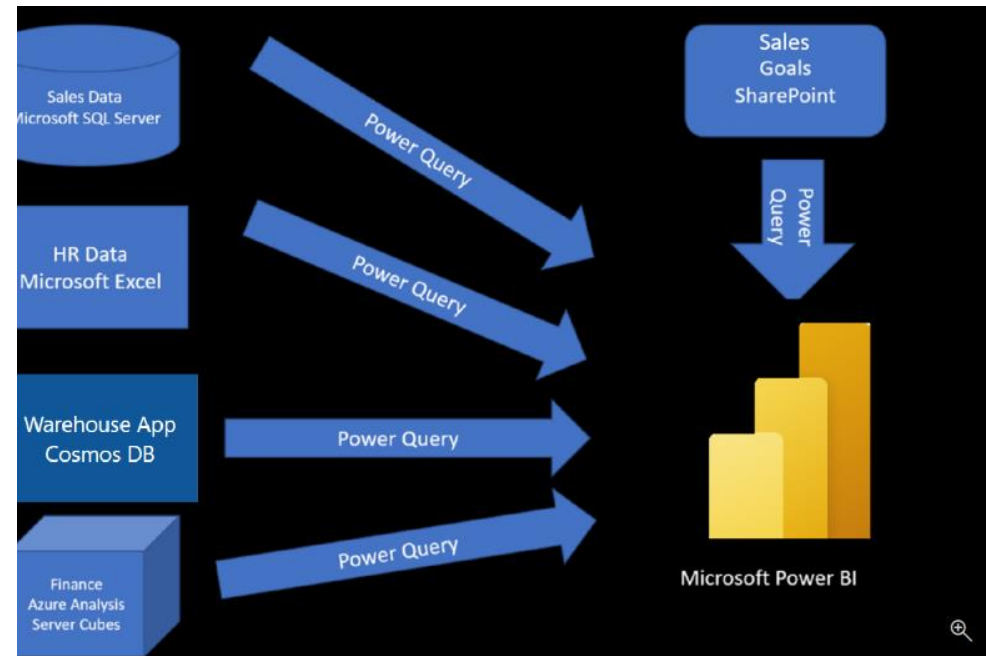
# 1<sup>st</sup> Step



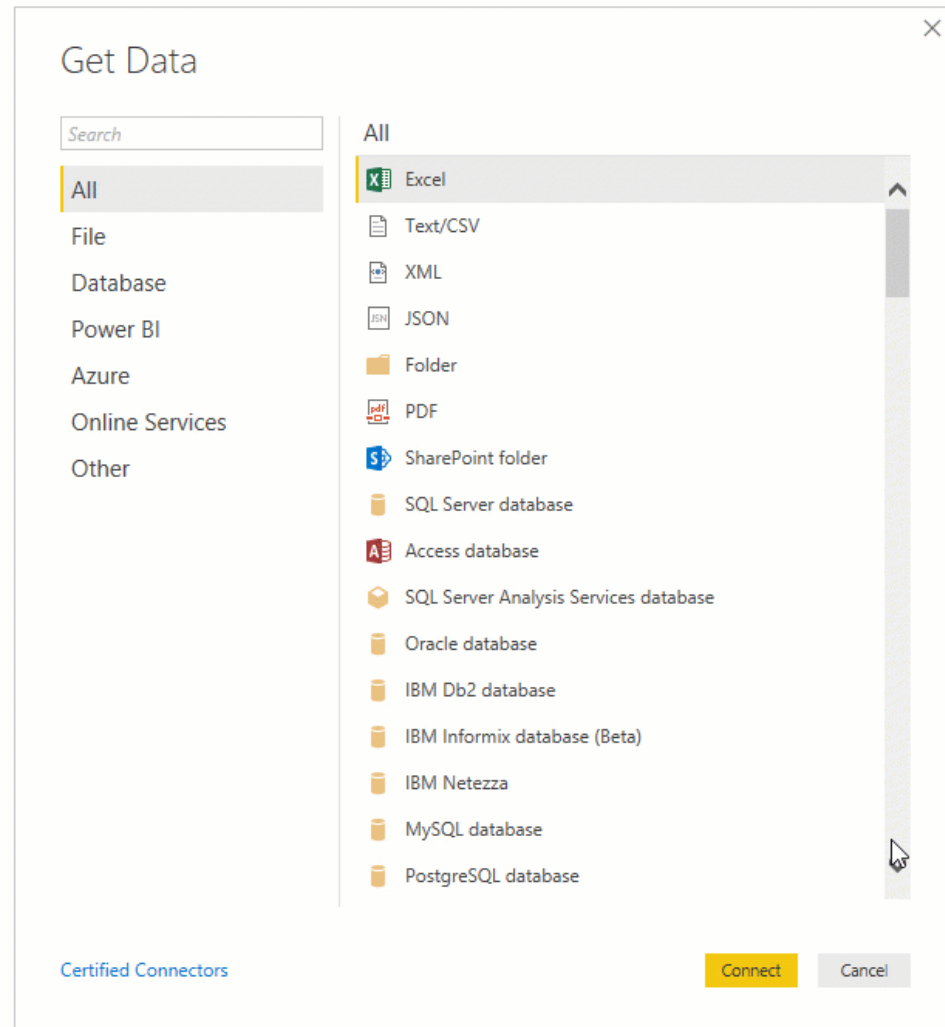
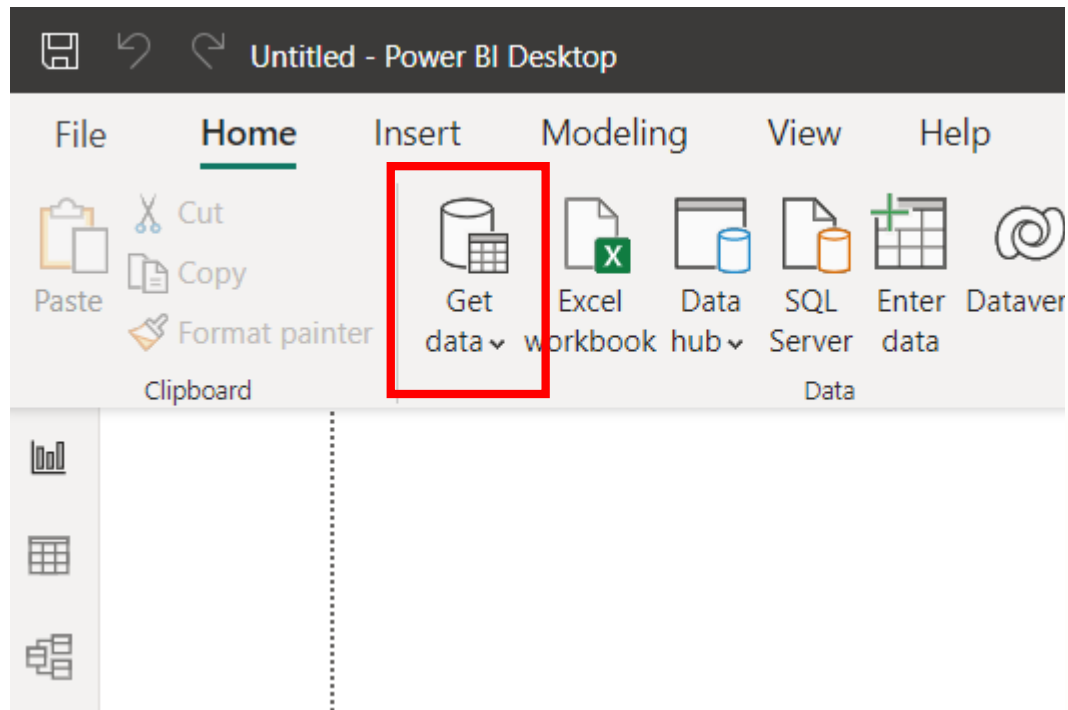
Getting Data

# Get data in Power BI

- Identify and connect to a data source
- Get data from a file, such as Microsoft Excel
- Get data from a relational database, such as Microsoft SQL Server
- Get data online
- Select a storage mode



# Get data in Power BI



# Get data in Power BI

## CSV File

cars - Notepad

File Edit Format View Help

Date of purchase,Make,RowID,Date of manufacture,Su  
01/06/2006,BMW,0001,1999,Jones,"45,500"  
01/06/2005,BMW,0002,2003,Chau,"75,200"  
02/01/2006,Lexus,0003,2006,Smith,"25,365"  
30/09/2007,Mazda,0004,2007,Mukherjee,"35,000"  
09/09/2009,Toyota,0005,2009,Barker,400

cars.txt - Notepad

File Edit Format View Help

|          |        |      |      |           |        |
|----------|--------|------|------|-----------|--------|
| 20060601 | BMW    | 0001 | 1999 | Jones     | 45,500 |
| 20050601 | BMW    | 0002 | 2003 | Chau      | 75,200 |
| 20060102 | Lexus  | 0003 | 2006 | Smith     | 25,365 |
| 20070930 | Mazda  | 0004 | 2007 | Mukherjee | 35,000 |
| 20090909 | Toyota | 0005 | 2009 | Barker    | 400    |

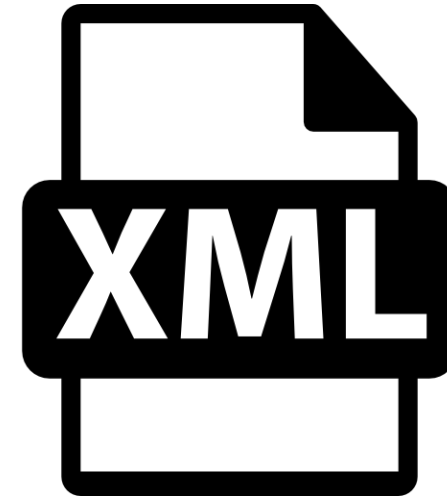
4



# Get data in Power BI

## XML File

```
<?xml version="1.0" encoding="UTF-8"?>
- <EmployeeData>
  - <employee id="34594">
    <firstName>Heather</firstName>
    <lastName>Banks</lastName>
    <hireDate>1/19/1998</hireDate>
    <deptCode>BB001</deptCode>
    <salary>72000</salary>
  </employee>
  - <employee id="34593">
    <firstName>Tina</firstName>
    <lastName>Young</lastName>
    <hireDate>4/1/2010</hireDate>
    <deptCode>BB001</deptCode>
    <salary>65000</salary>
  </employee>
</EmployeeData>
```



Get data in Power BI

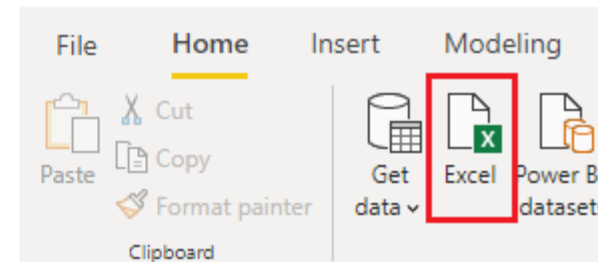
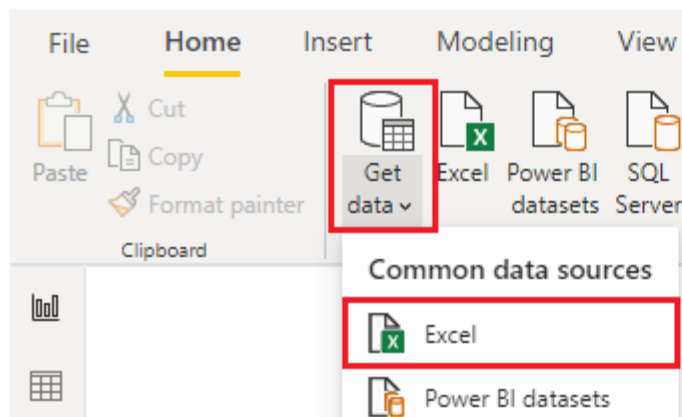
## JSON File

```
Cars.json x
1  [
2      {
3          "Car Name": "Honda City",
4          "Car Model": "City",
5          "Car Maker": "Honda",
6          "Car Price": "20,000 USD"
7      },
8      {
9          "Car Name": "Bugatti Chiron",
10         "Car Model": "Chiron",
11         "Car Maker": "Bugatti",
12         "Car Price": "3 Million USD"
13     }
14 ]
```



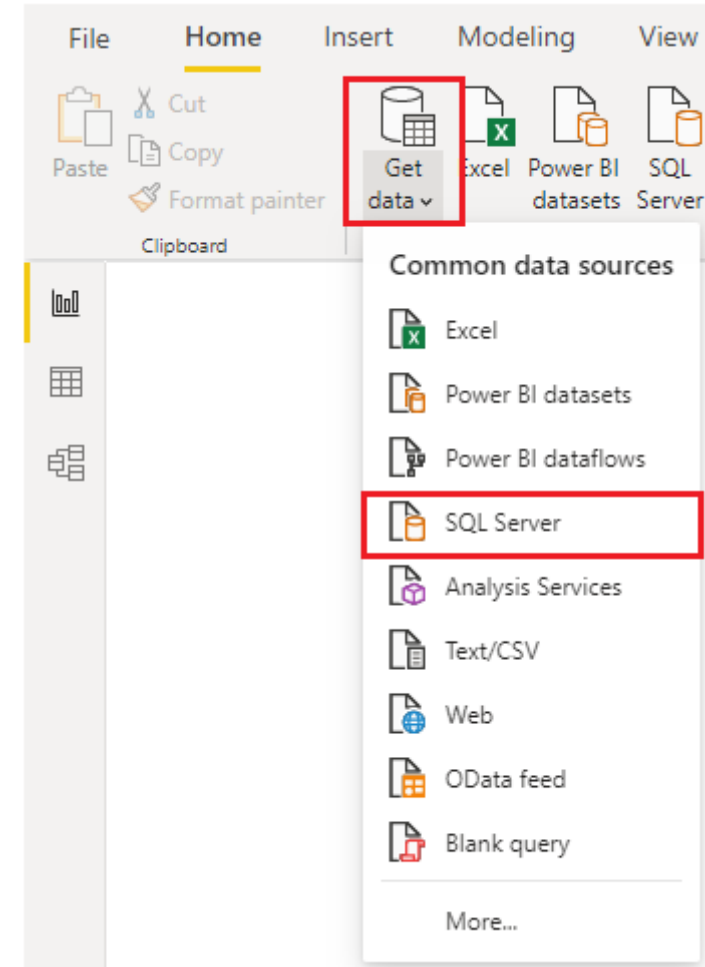
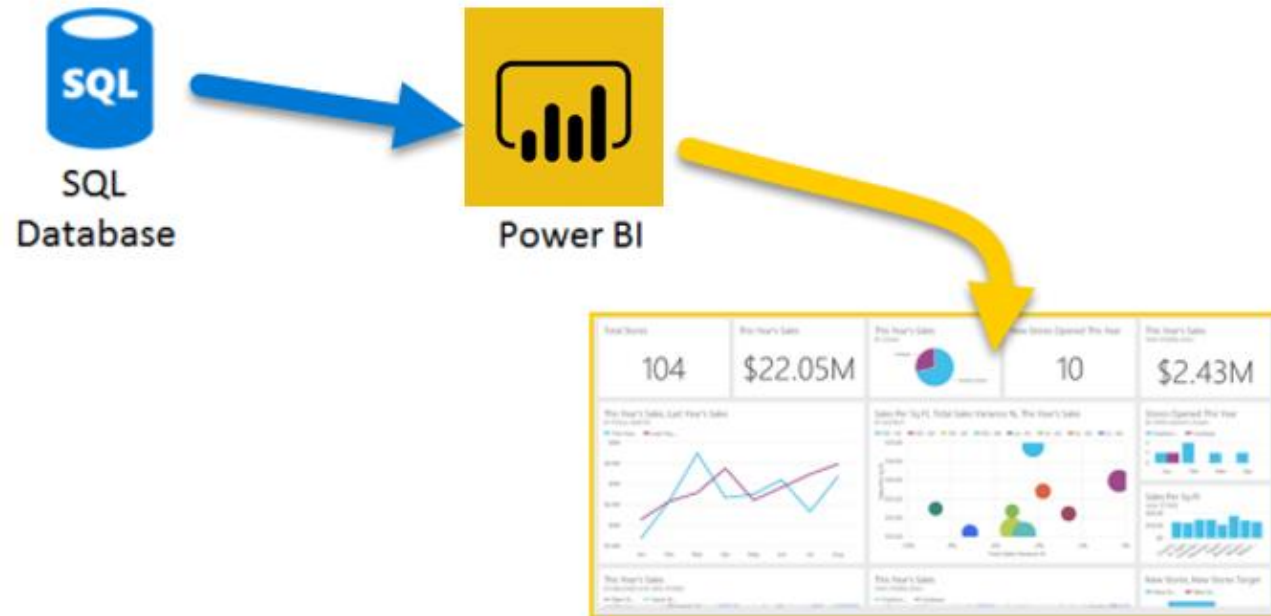
# Get data in Power BI

## Get data from files



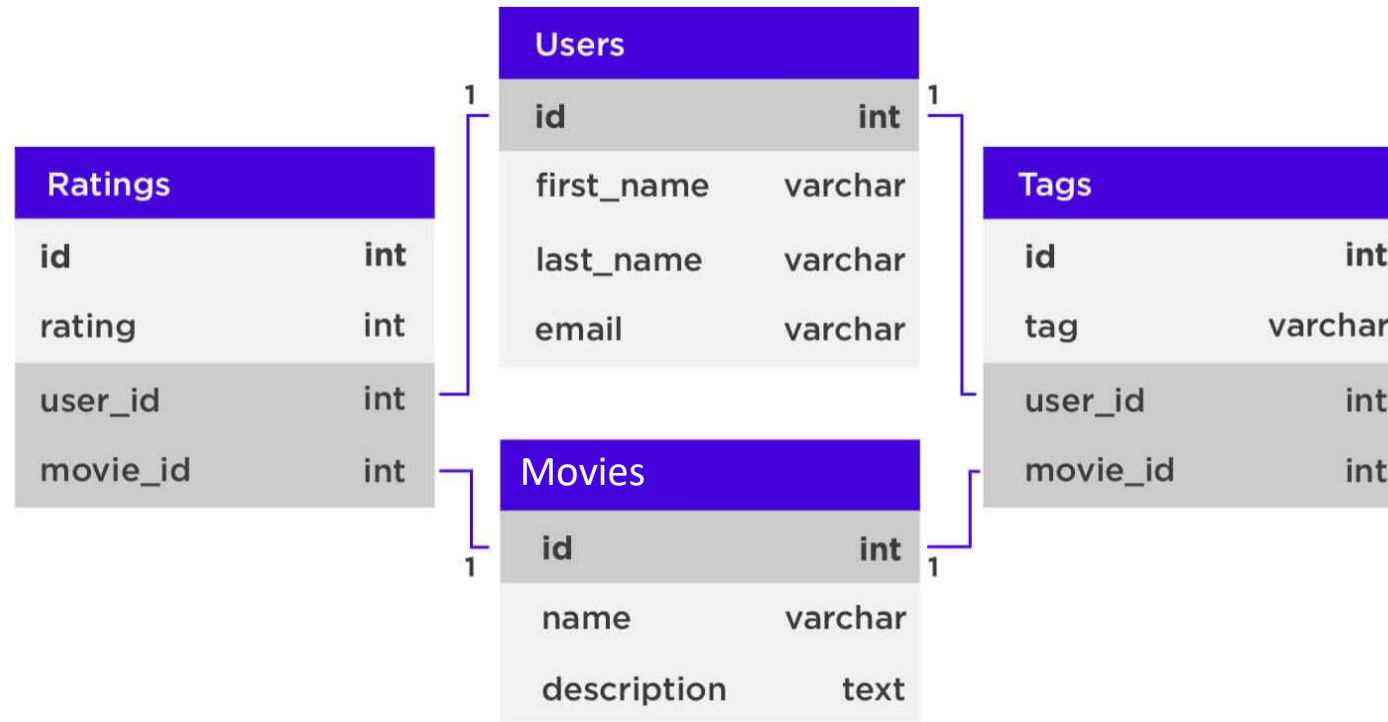
Get data in Power BI

# Get data from relational data sources



# Get data in Power BI

## Relational Database

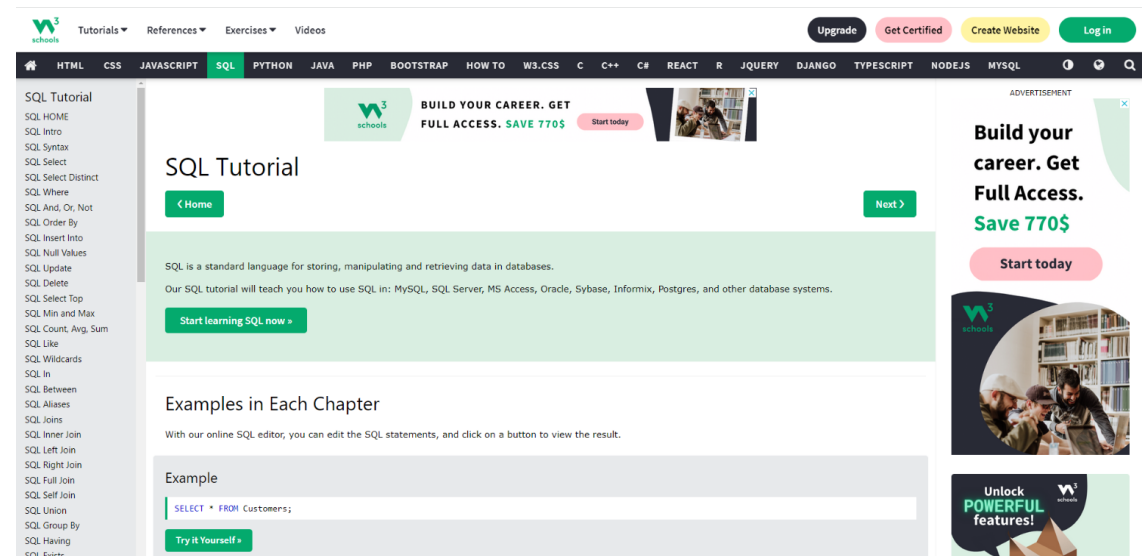


# Get data in Power BI

## SQL Query

```
SELECT * FROM table_name;
Select * from Apps
select count(*) from Apps
select * from Apps where AppName = 'MoneyControl'
select AppName, AppCategory from Apps
select distinct AppName from Apps
select AppName from Apps where AppPrice > 60
```

- <https://www.w3schools.com/sql/>



The screenshot shows the W3 Schools website's SQL Tutorial page. The top navigation bar includes links for various technologies like HTML, CSS, JavaScript, SQL, Python, Java, PHP, Bootstrap, and more. The main content area is titled 'SQL Tutorial' and includes a 'Home' button and a 'Next' button. Below the title, there is a brief introduction to SQL as a standard language for storing, manipulating, and retrieving data in databases. A 'Start learning SQL now' button is prominently displayed. The page also features a sidebar with a list of SQL topics, an 'Examples in Each Chapter' section with an online SQL editor, and an advertisement on the right side promoting career building and full access to the site's features.

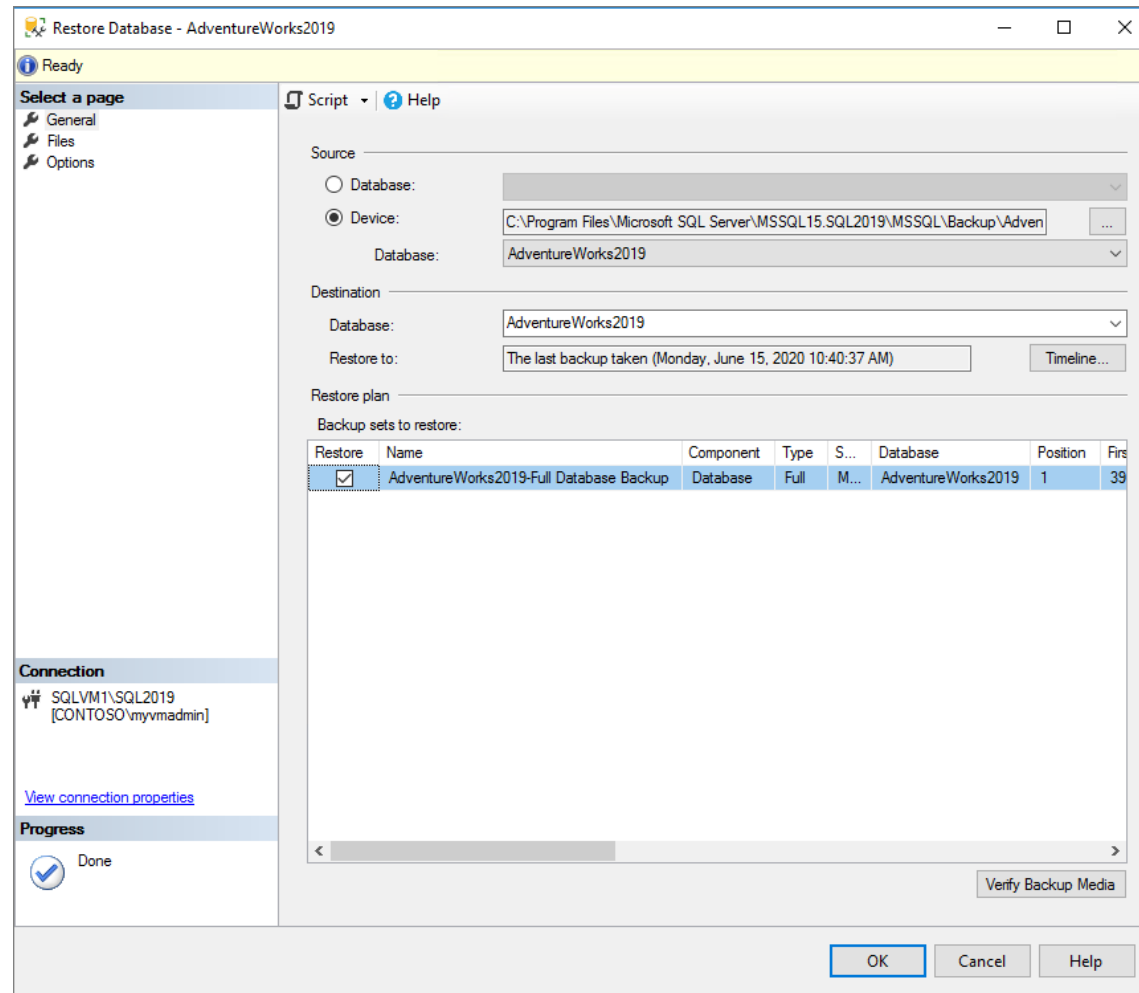
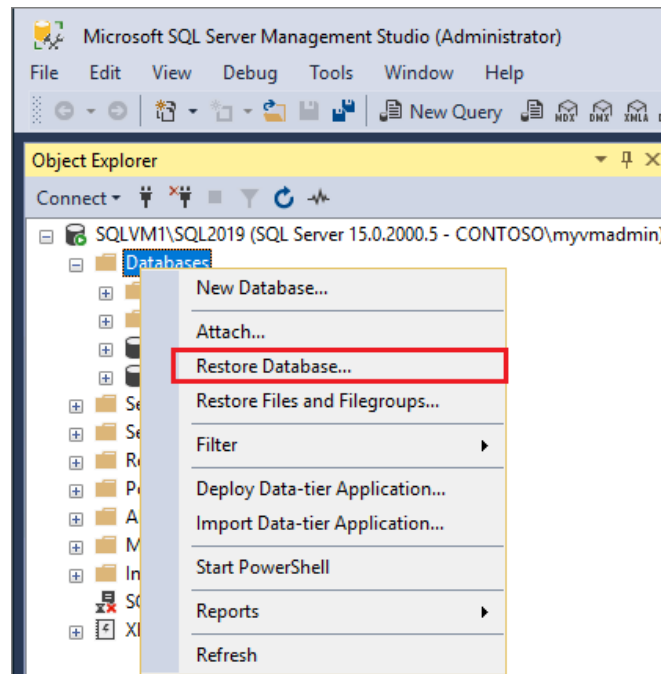
# Install SQL Server & SSMS



Microsoft®  
**SQL Server®**

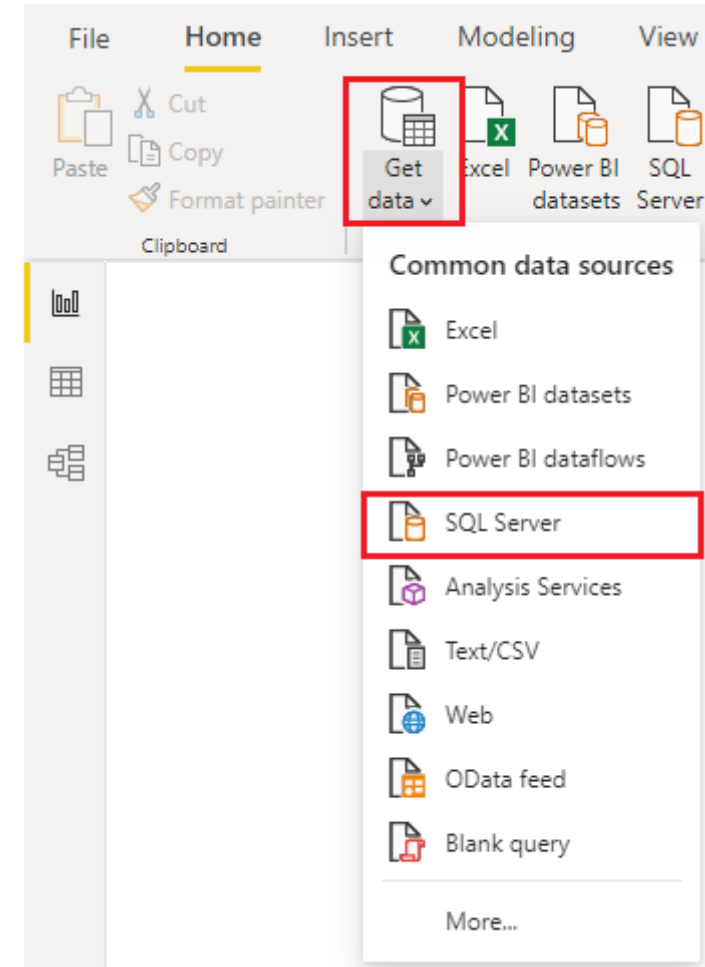
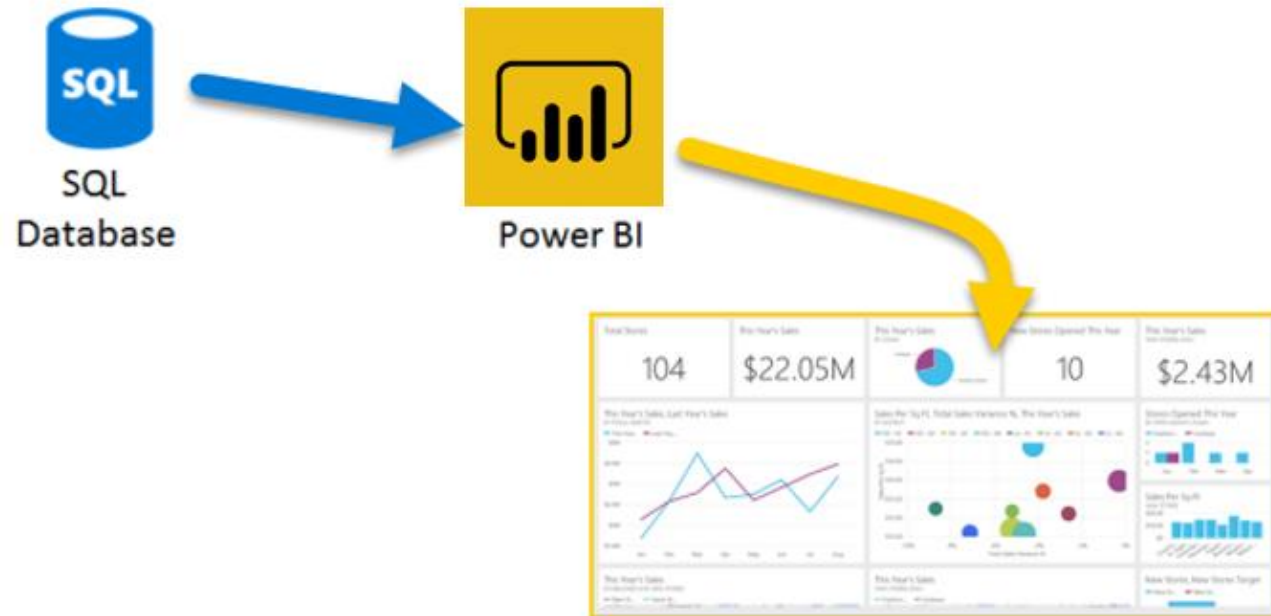


# Install Adventure Works DB



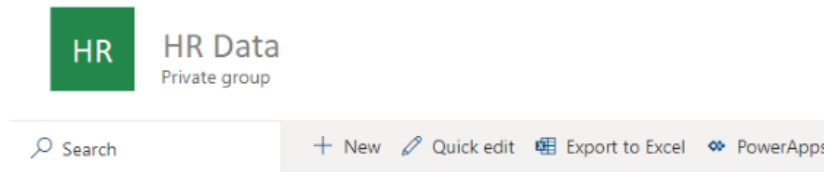
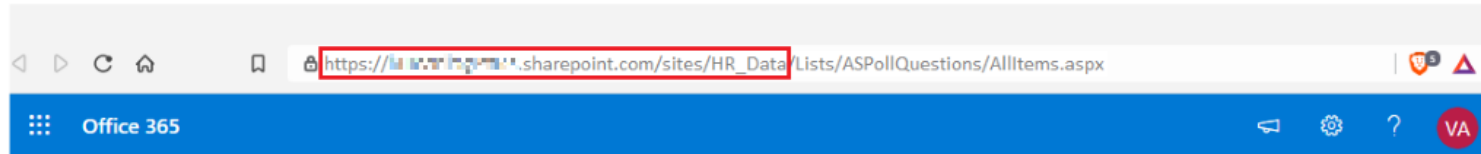
Get data in Power BI

# Get data from relational data sources

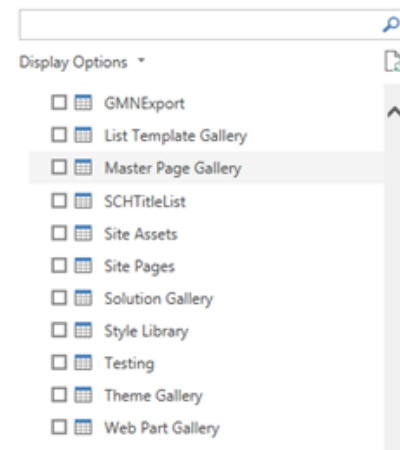


Get data in Power BI

# Get data from online services



Navigator



Master Page Gallery

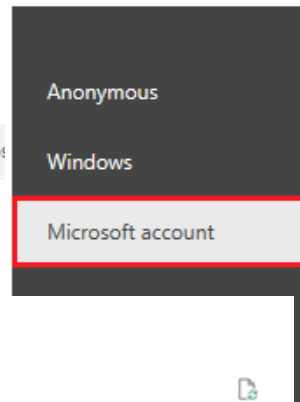
| FileSystemObjectType | Id | ServerRedirectedEmbedUri | ServerRedirectedEmbed |
|----------------------|----|--------------------------|-----------------------|
| 1                    | 7  |                          | null                  |
| 1                    | 8  |                          | null                  |
| 1                    | 55 |                          | null                  |
| 1                    | 63 |                          | null                  |
| 1                    | 64 |                          | null                  |
| 1                    | 68 |                          | null                  |
| 1                    | 72 |                          | null                  |
| 1                    | 76 |                          | null                  |
| 1                    | 80 |                          | null                  |
| 1                    | 84 |                          | null                  |

The data in the preview has been truncated due to size limits.

Load

Transform Data

Cancel



SharePoint

https://[redacted].sharepoint.com/sites/HR\_Data

You are currently signed in.

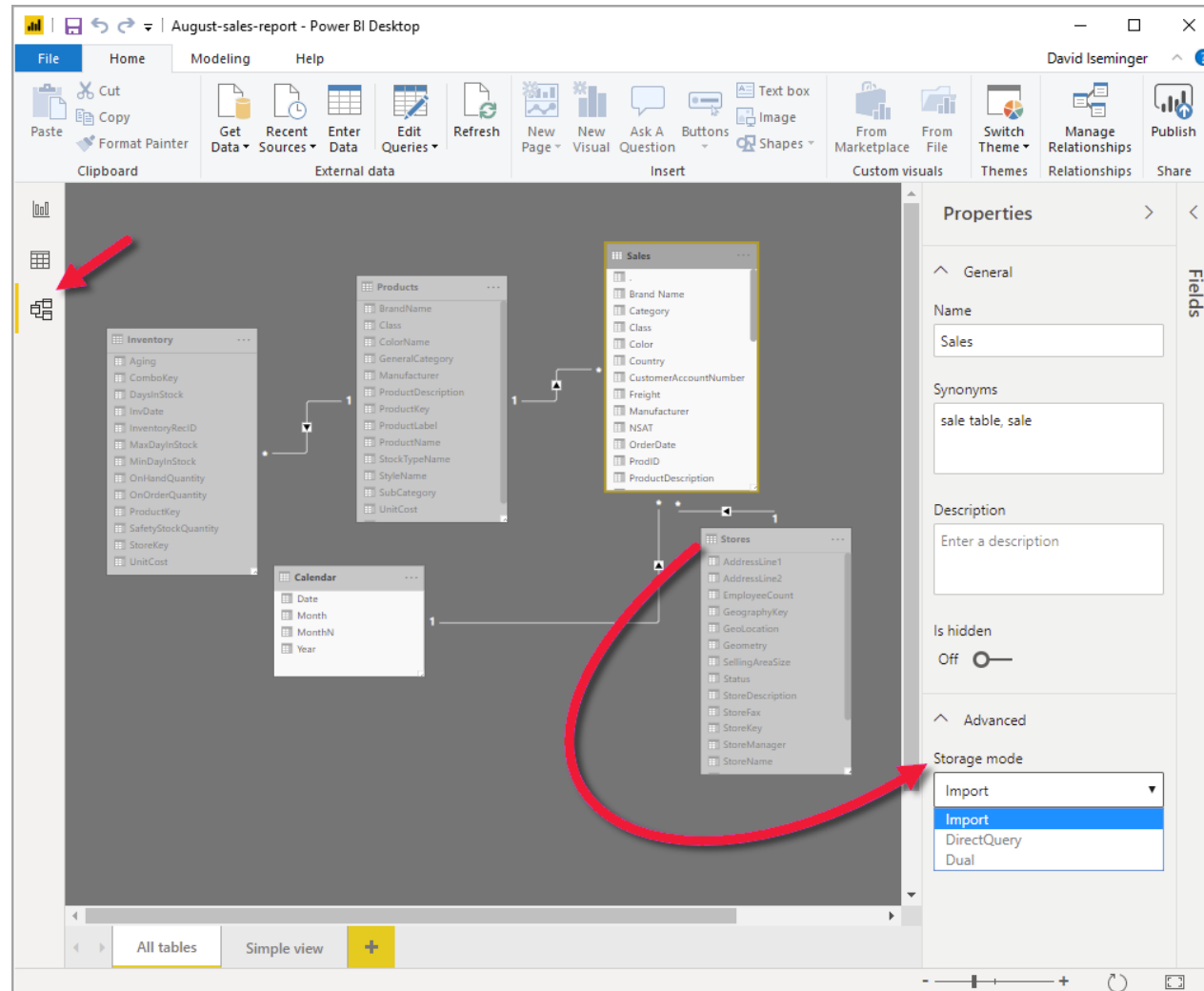
Sign in as different user

Connect

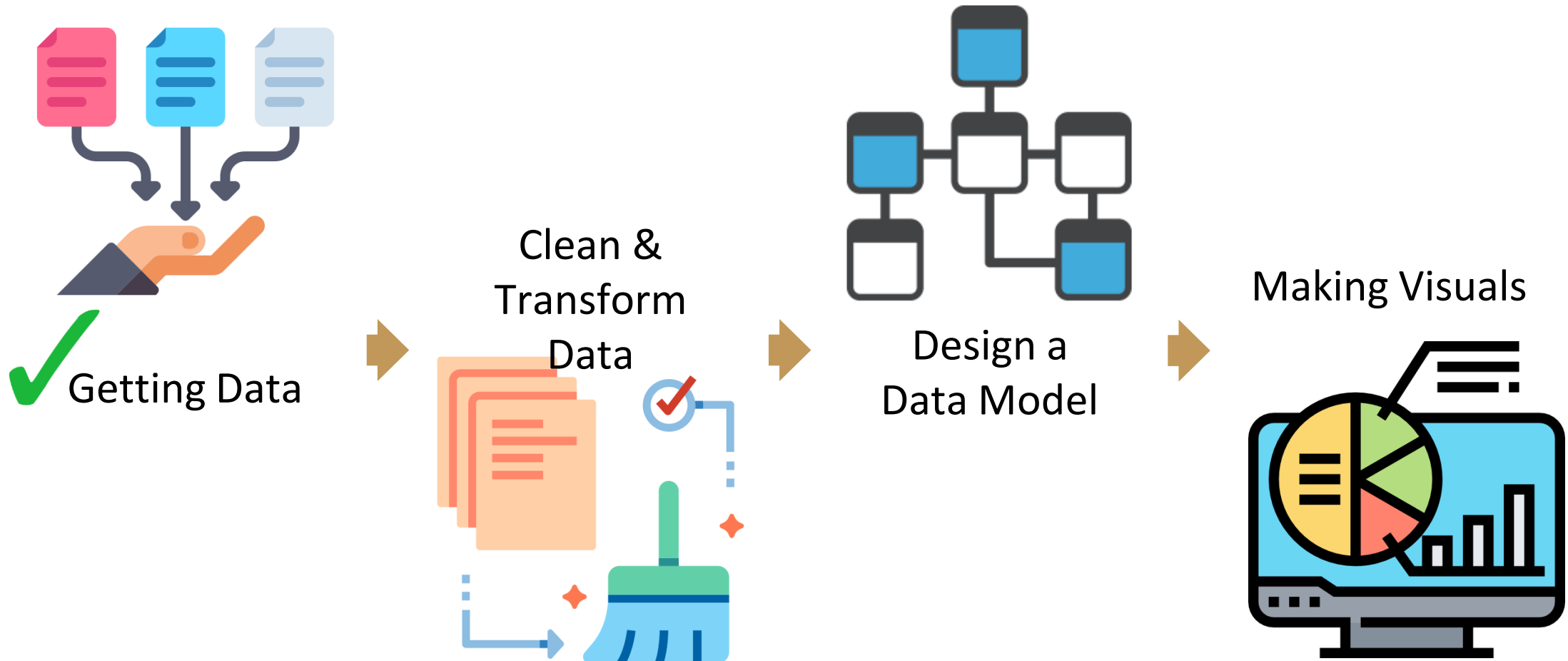
Cancel

# Get data in Power BI

## Select a storage mode



# Our RoadMap

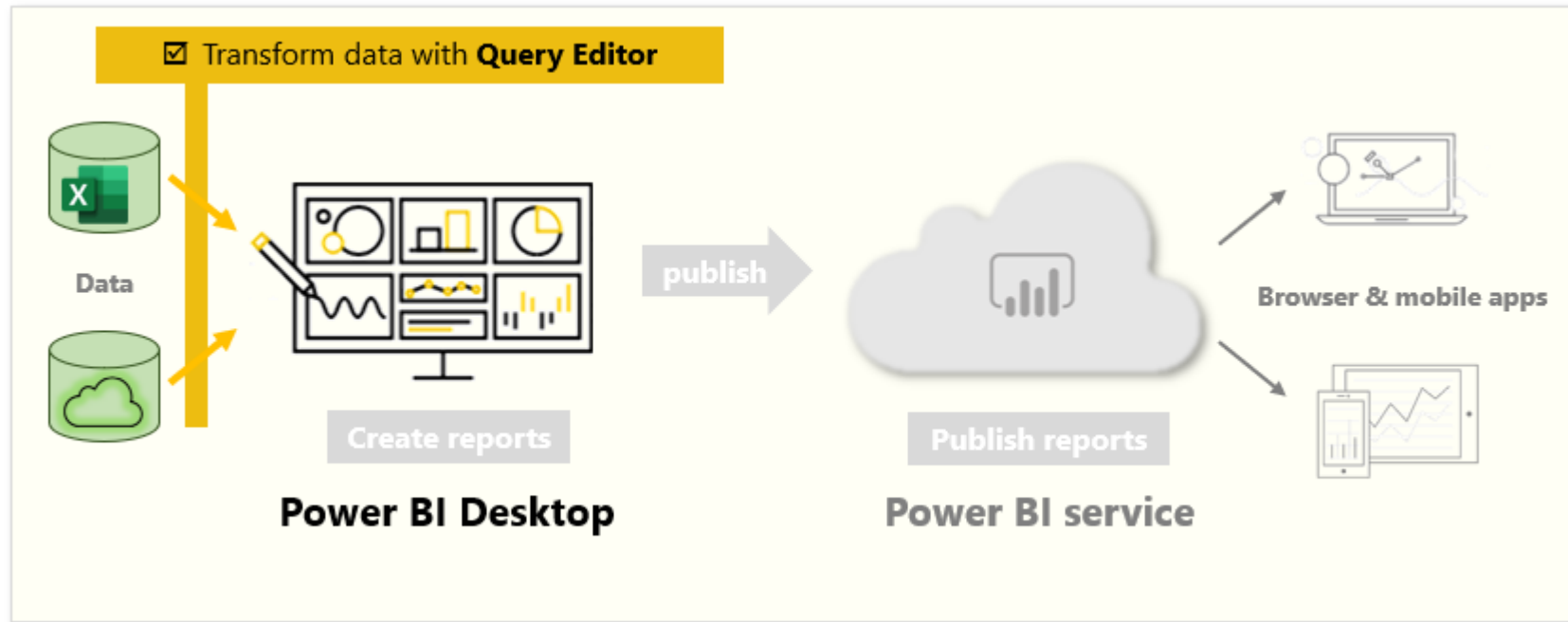


## 2<sup>nd</sup> Step

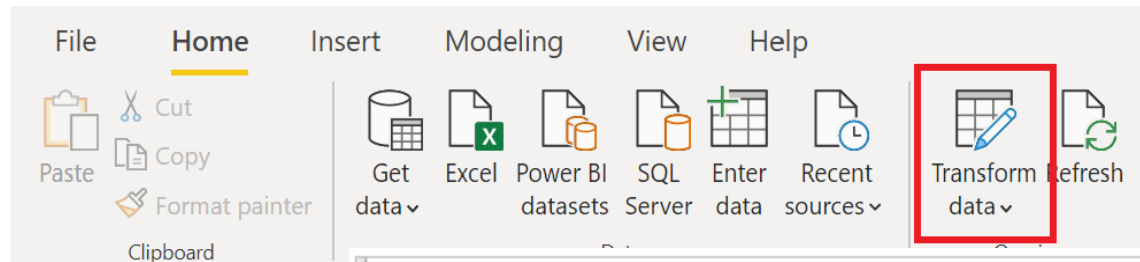


Clean &  
Transform Data

# Data Transforming



# Power Query



The screenshot displays the Power Query Editor interface. The 'Transform' tab is selected in the ribbon. The 'Queries' list on the left shows a query named 'Ranking of best and'. The main area displays a table with the following data:

|   | State          | Overall rank |
|---|----------------|--------------|
| 1 | South Dakota   | 1            |
| 2 | Utah           | 2            |
| 3 | Idaho          | 3            |
| 4 | New Hampshire  | 4            |
| 5 | Florida        | 5            |
| 6 | Montana        | 6            |
| 7 | North Carolina | 6            |
| 8 | Wyoming        | 8            |
| 9 | Nebraska       | 9            |

The 'Query Settings' pane on the right shows the 'Name' as 'Ranking of best and worst states for retire' and the 'Applied Steps' list.

**PROPERTIES**

Name  
Ranking of best and worst states for retire

[All Properties](#)

**APPLIED STEPS**

- Source
- Navigation
- Changed Type

# Append Tables

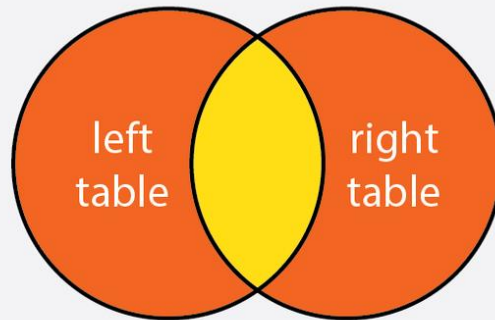
|  1 <sup>2</sup> <sub>3</sub> id | A <sup>B</sup> <sub>C</sub> company | A <sup>B</sup> <sub>C</sub> name | 1 <sup>2</sup> <sub>3</sub> phone |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------|-----------------------------------|
| 1                                                                                                                | 96351 cus145                        | Mike Poi                         | 1626258811                        |
| 2                                                                                                                | 64826 cus134                        | John Likec                       | 1363718361                        |
| 3                                                                                                                | 92647 cus018                        | Theresa Yulia                    | 2467355553                        |

|  1 <sup>2</sup> <sub>3</sub> id | A <sup>B</sup> <sub>C</sub> company | A <sup>B</sup> <sub>C</sub> name | 1 <sup>2</sup> <sub>3</sub> phone |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------|-----------------------------------|
| 1                                                                                                                | 86563 sup126                        | Mike Firt                        | 7163615121                        |
| 2                                                                                                                | 96352 sup889                        | Josie Lind                       | 1831635671                        |
| 3                                                                                                                | 48256 sup761                        | Joanna Threven                   | 2936827338                        |

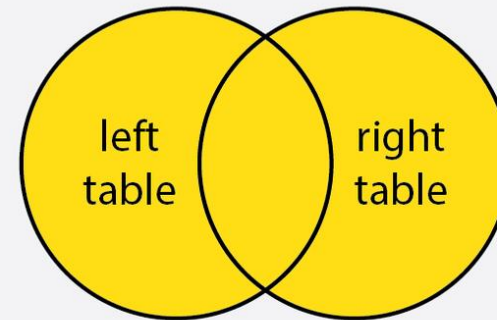
|  1 <sup>2</sup> <sub>3</sub> id | A <sup>B</sup> <sub>C</sub> company | A <sup>B</sup> <sub>C</sub> name | 1 <sup>2</sup> <sub>3</sub> phone |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------|-----------------------------------|
| 1                                                                                                                  | 12347 emp124                        | John Kate                        | 1625535511                        |
| 2                                                                                                                  | 76478 emp273                        | Luke John                        | 1028777351                        |
| 3                                                                                                                  | 82482 emp291                        | Michael Uji                      | 1245245672                        |

# Merge Tables

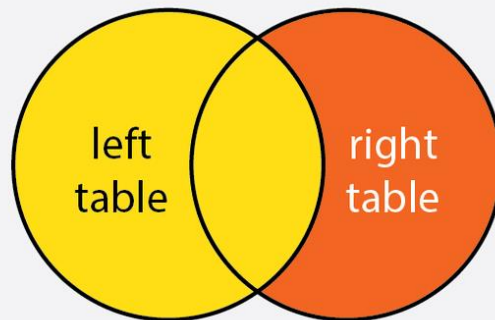
INNER JOIN



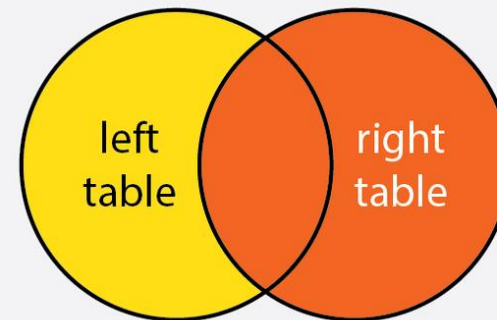
FULL JOIN



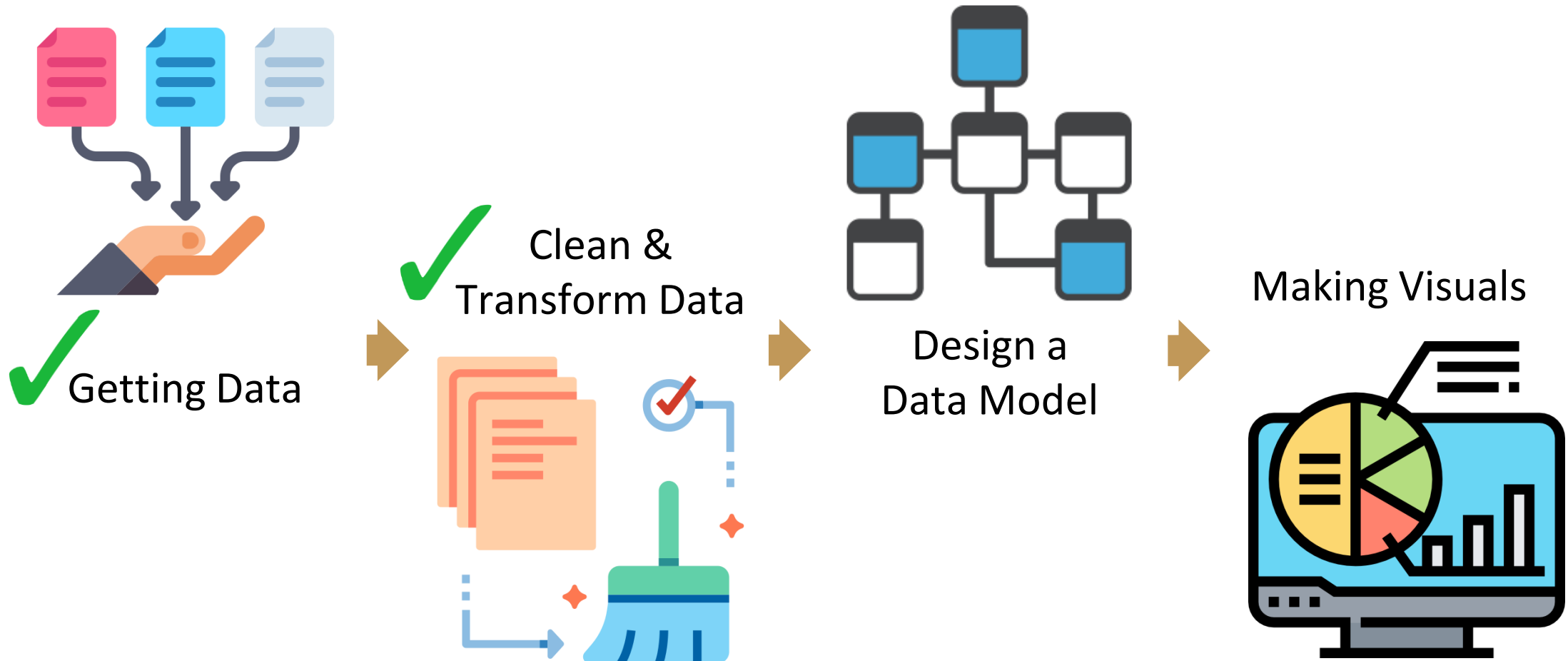
LEFT JOIN



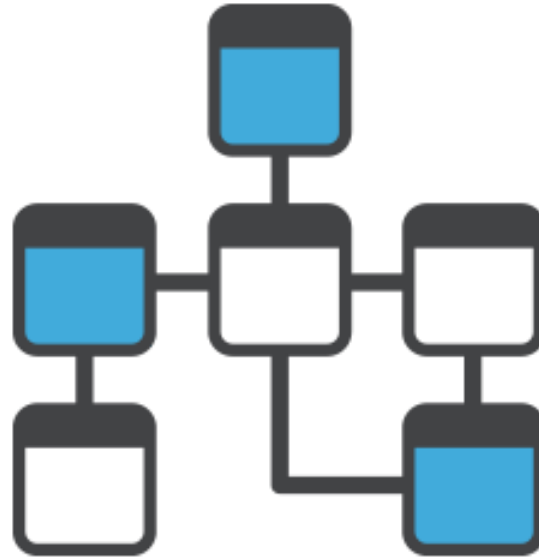
RIGHT JOIN



# Our RoadMap

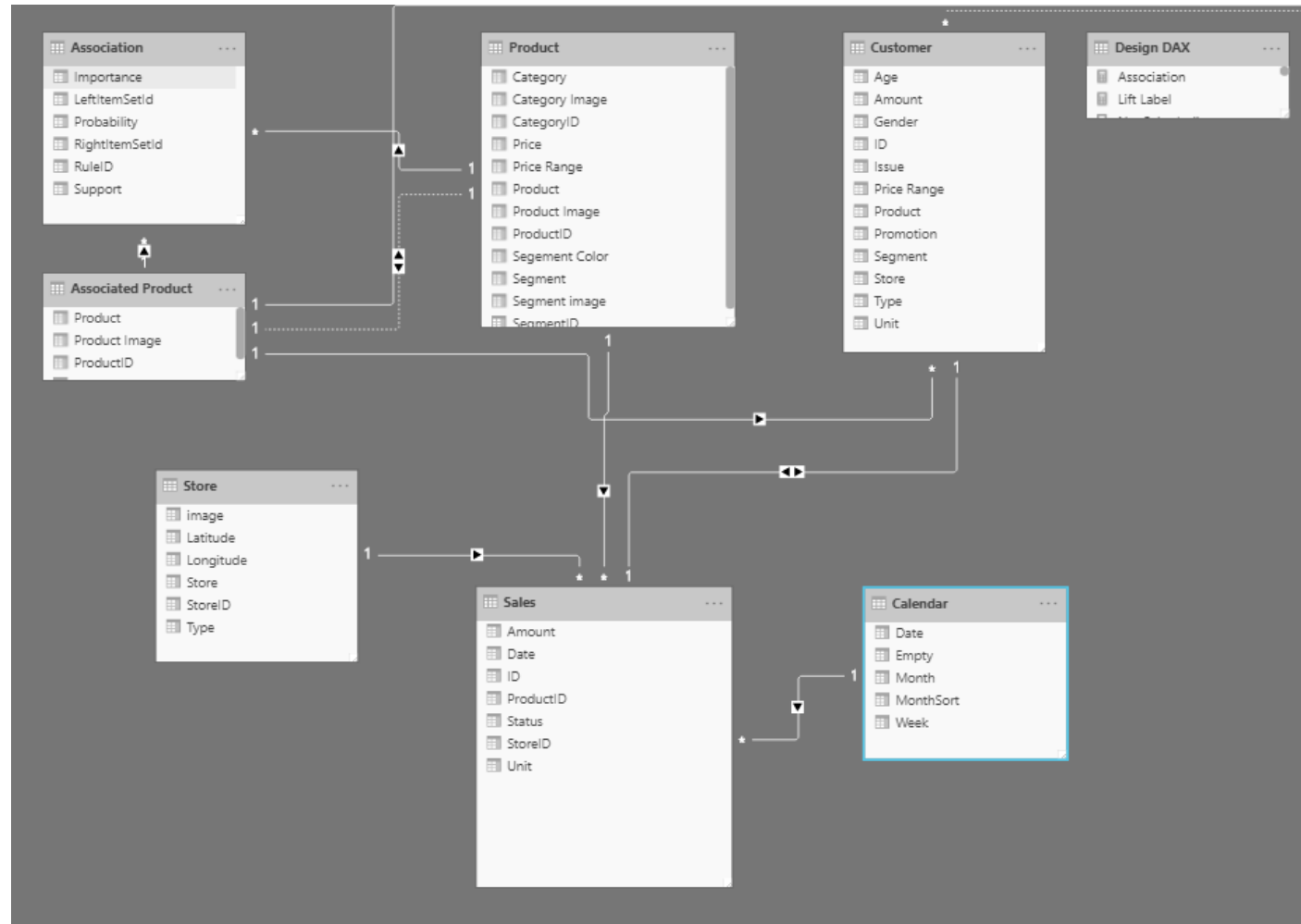


## 3<sup>rd</sup> Step



Design a  
Data Model

# Data Modeling



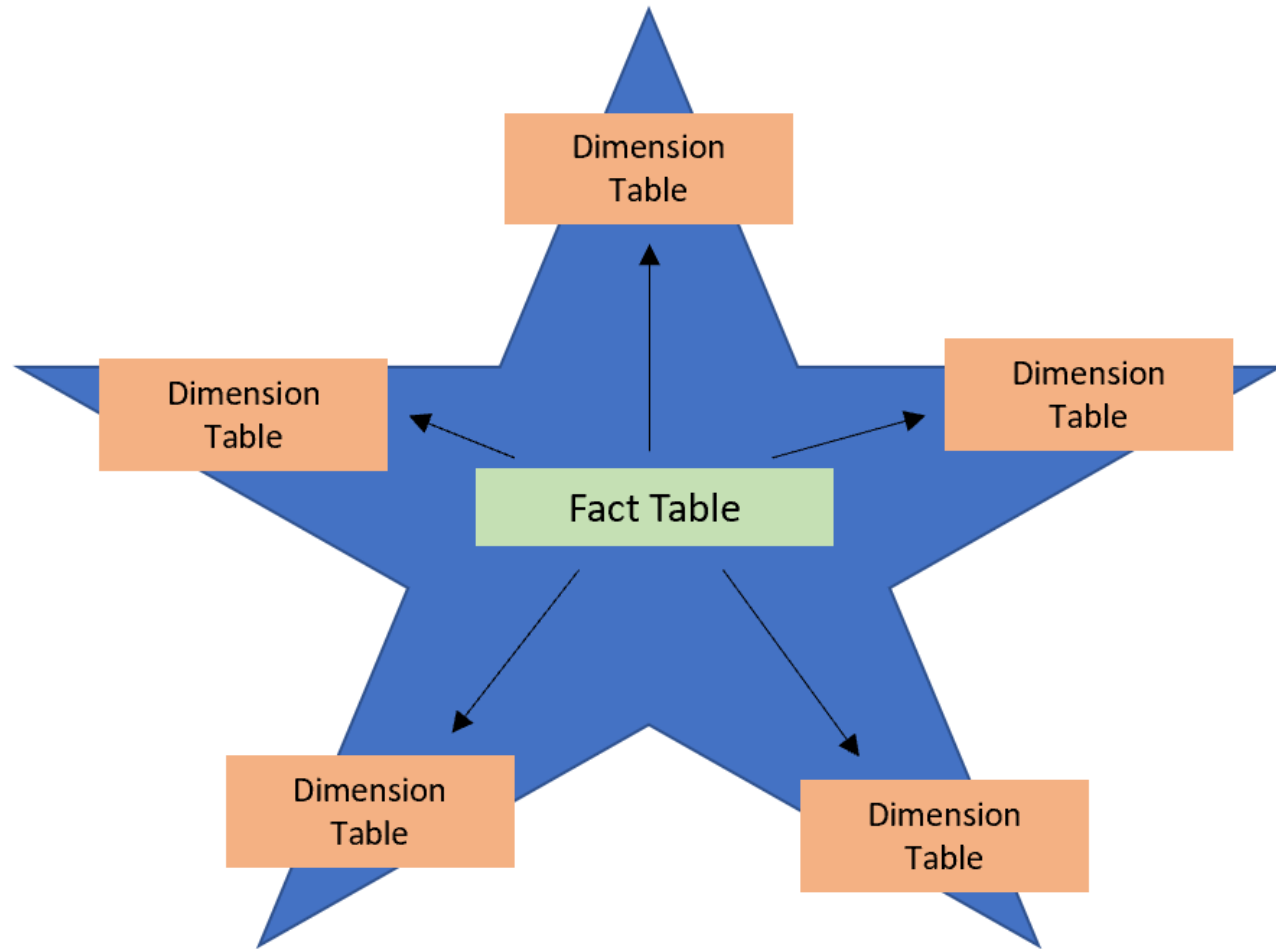
# Good Model Advantages

- Data exploration is faster.
- Aggregations are simpler to build.
- Reports are more accurate.
- Writing reports takes less time.
- Reports are easier to maintain in the future.



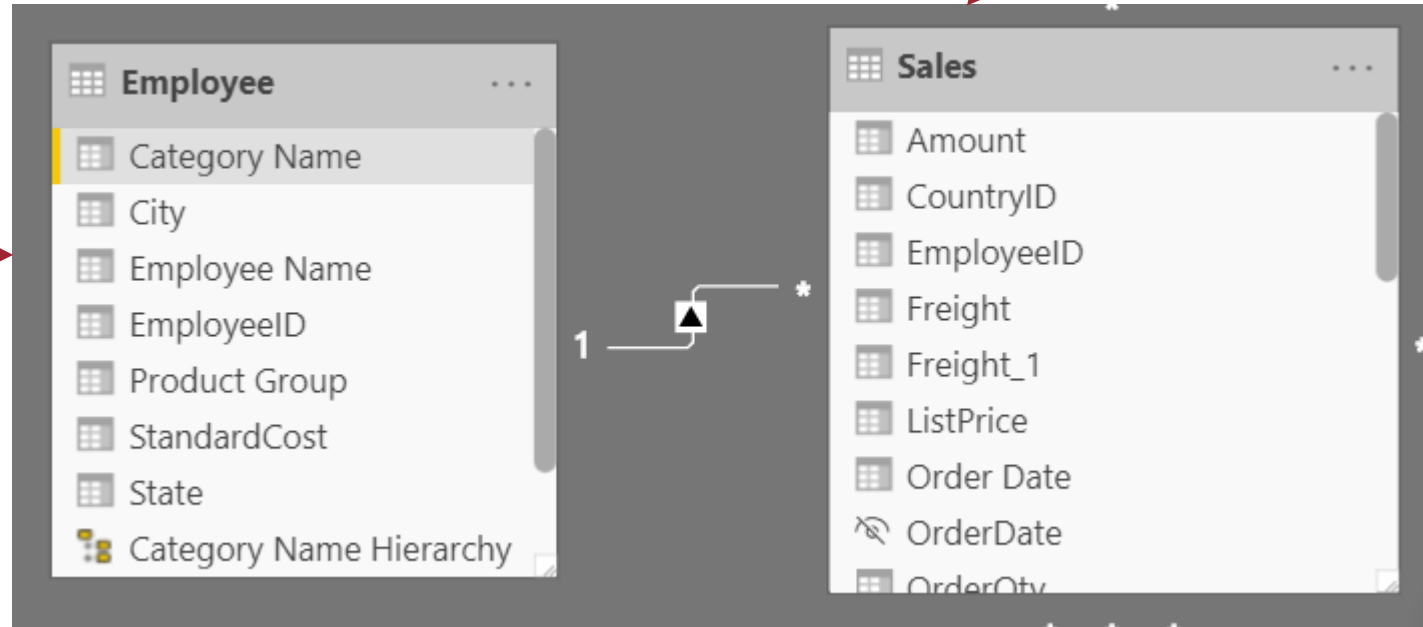
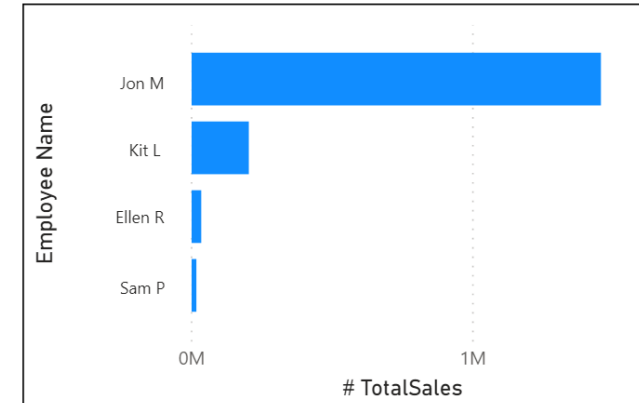
# Star schemas

- **Fact tables**  
contain observational or event data values
- **Dimension tables**  
contain the details about the data in fact tables



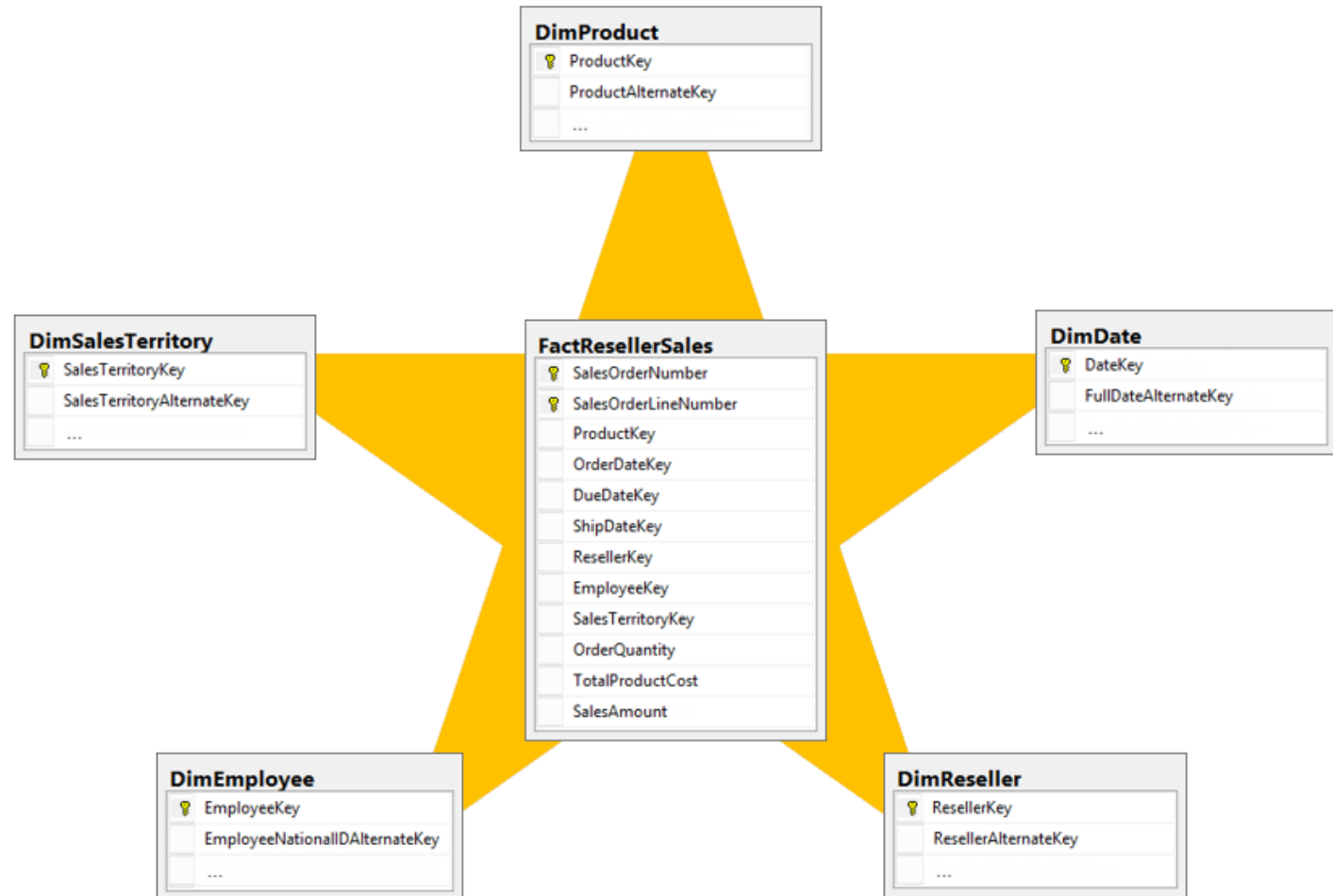
# Star schemas

- **Fact tables**  
contain observational  
or event data values
- **Dimension tables**  
contain the details  
about the data in fact  
tables



# Star schemas

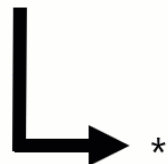
- **Fact tables**  
contain observational or event data values
- **Dimension tables**  
contain the details about the data in fact tables



### Category

|       |             |
|-------|-------------|
| Cat-A | Jenny Jones |
| Cat-B | John Smith  |

1



### Product

|   |        |       |        |
|---|--------|-------|--------|
| 1 | Prod-A | Cat-A | Red    |
| 2 | Prod-B | Cat-B | Yellow |
| 3 | Prod-C | Cat-A | Blue   |

1



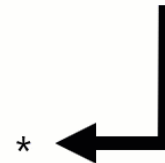
### Sales

|   |        |    |
|---|--------|----|
| 2 | CY2017 | 10 |
| 1 | CY2017 | 3  |
| 2 | CY2018 | 5  |
| 2 | CY2018 | 2  |
| 3 | CY2018 | 11 |

### Year

|        |
|--------|
| CY2017 |
| CY2018 |
| CY2019 |
| CY2020 |

1



# Hierarchies

|         |   |   |
|---------|---|---|
| Date    | ▽ | × |
| Year    |   | × |
| Quarter |   | × |
| Month   |   | × |
| Day     |   | × |

Product

☐ Category Name

Check

New hierarchy

New measure

New hierarchy

Category Name Hierar...

☐ Category Name

☐ SubCategory Name

Axis

Category Name Hierarcl ▾ ×

Category Name ×

SubCategory Name ×

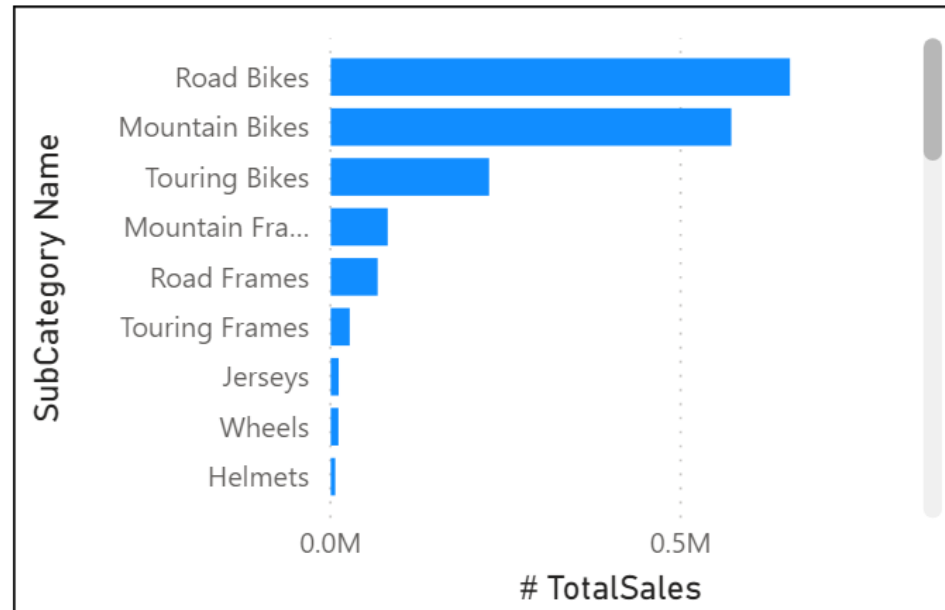
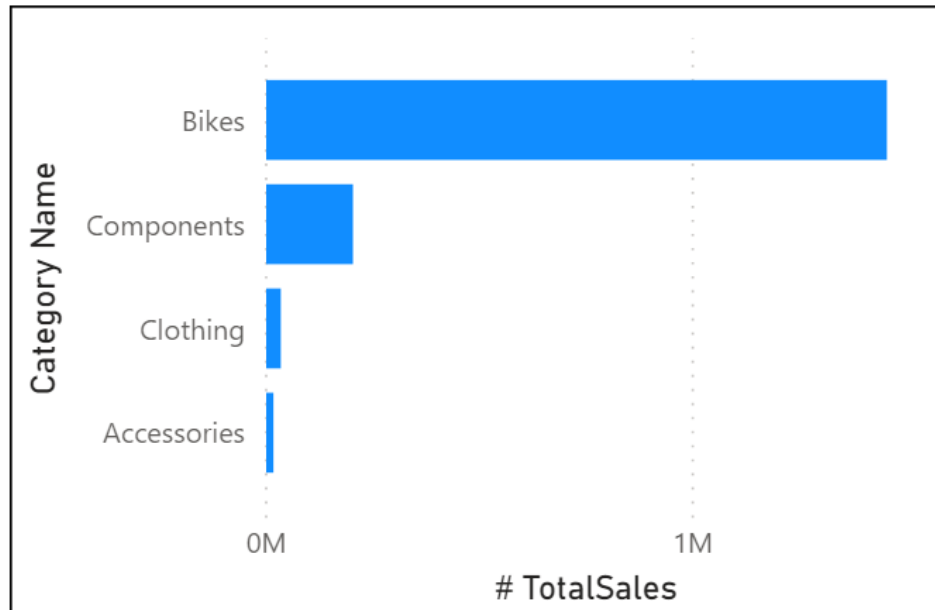
Legend

Add data fields here

Values

# TotalSales ▾ ×

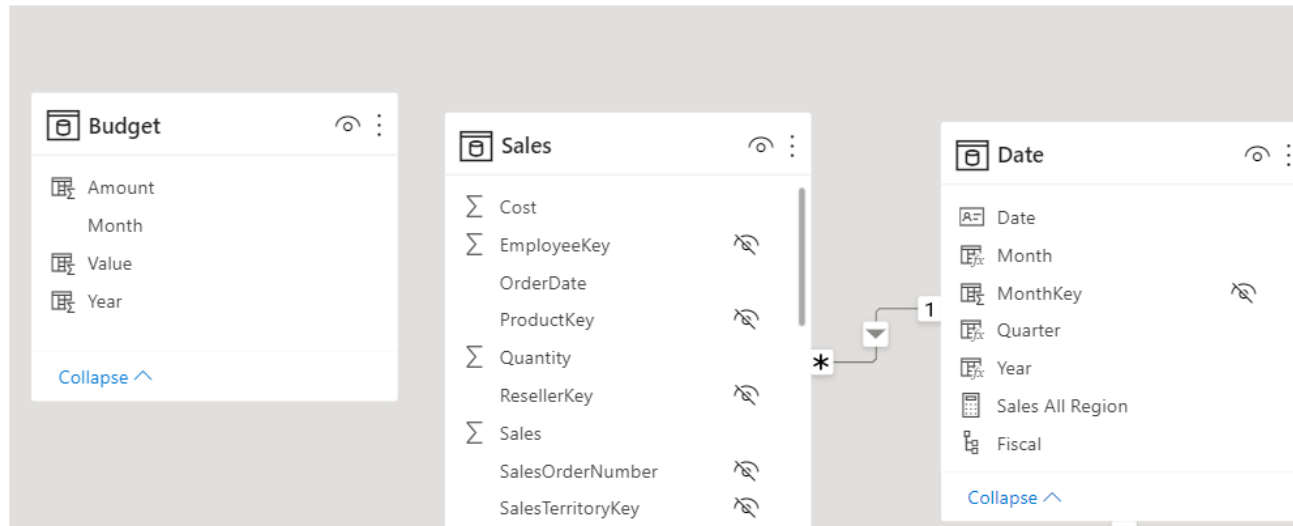
# Hierarchies



# Parent-child hierarchy

Path = PATH(Employee[Employee ID], Employee[Manager ID])

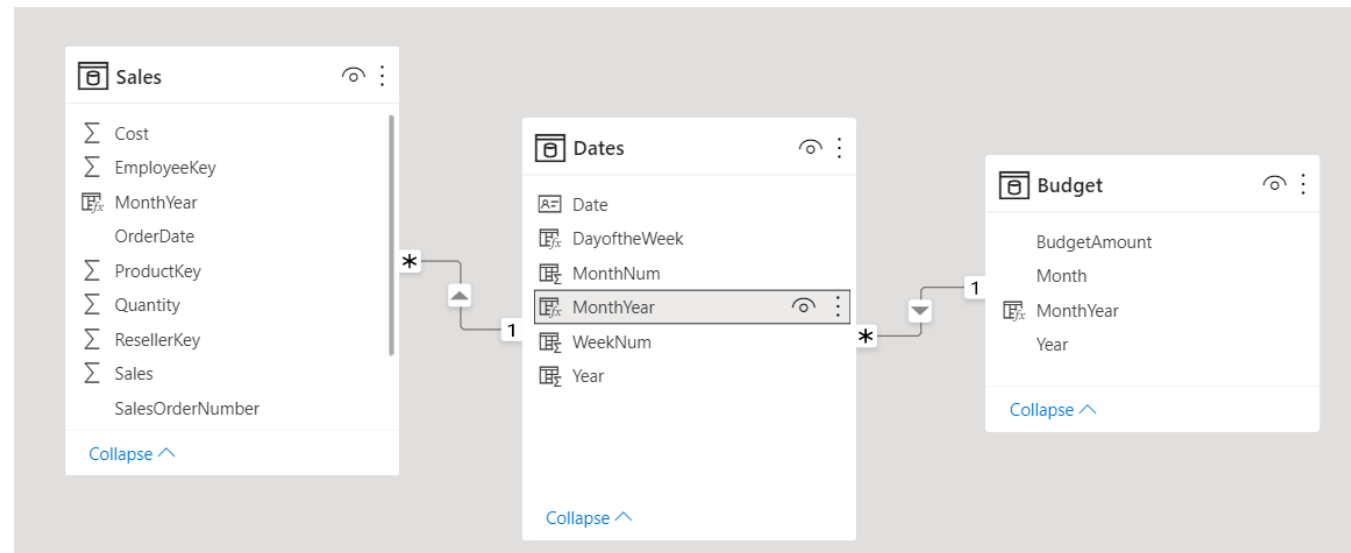
# Define data granularity



| Year   | Month | Date       |
|--------|-------|------------|
| 2013   | 1     | 01/01/2011 |
| 2013   | 2     | 01/02/2011 |
| 2013   | 3     | 01/03/2011 |
| 2013   | 4     | 01/04/2011 |
| 2013   | 5     | 01/05/2011 |
| 2013   | 6     | 01/06/2011 |
| 2013   | 7     | 01/06/2011 |
| 2013   | 8     | 01/07/2011 |
| Budget |       | Calendar   |

# Define data granularity

| Year | Month | BudgetAmount | MonthYear |
|------|-------|--------------|-----------|
| 2011 | 1     | 9493         | 2011-1    |
| 2011 | 2     | 2490         | 2011-2    |
| 2011 | 3     | 16585        | 2011-3    |
| 2011 | 4     | 5575         | 2011-4    |
| 2011 | 5     | 11611        | 2011-5    |
| 2011 | 6     | 14017        | 2011-6    |
| 2011 | 7     | 11919        | 2011-7    |
| 2011 | 8     | 3832         | 2011-8    |
| 2011 | 9     | 7094         | 2011-9    |
| 2011 | 10    | 6216         | 2011-10   |
| 2011 | 11    | 7769         | 2011-11   |
| 2011 | 12    | 12552        | 2011-12   |
| 2012 | 1     | 12781        | 2012-1    |
| 2012 | 2     | 7889         | 2012-2    |
| 2012 | 3     | 10329        | 2012-3    |
| 2012 | 4     | 5350         | 2012-4    |
| 2012 | 5     | 19216        | 2012-5    |
| 2012 | 6     | 5849         | 2012-6    |
| 2012 | 7     | 19269        | 2012-7    |
| 2012 | 8     | 3080         | 2012-8    |
| 2012 | 9     | 2381         | 2012-9    |



TotalSales = SUM(Sales[Total Sales])

BudgetAmount = SUM (Budget[BudgetAmount])

# Define data granularity

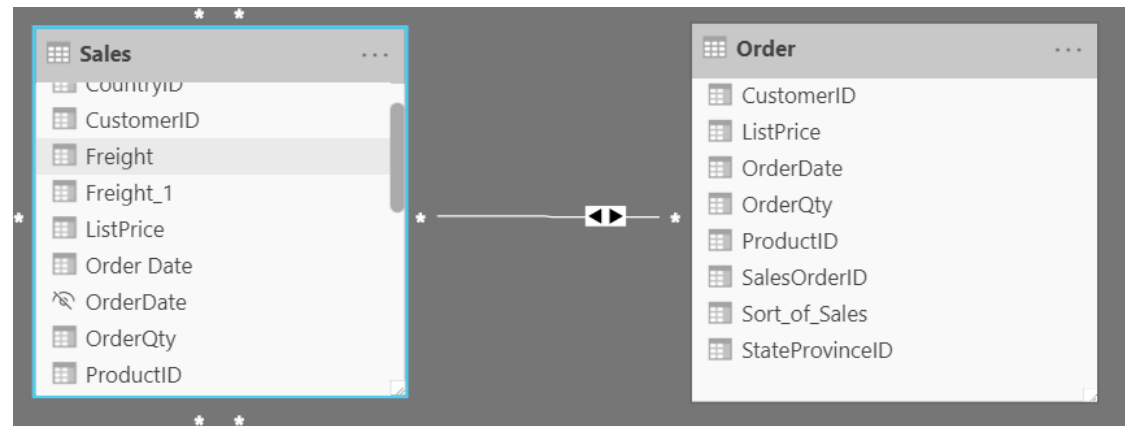
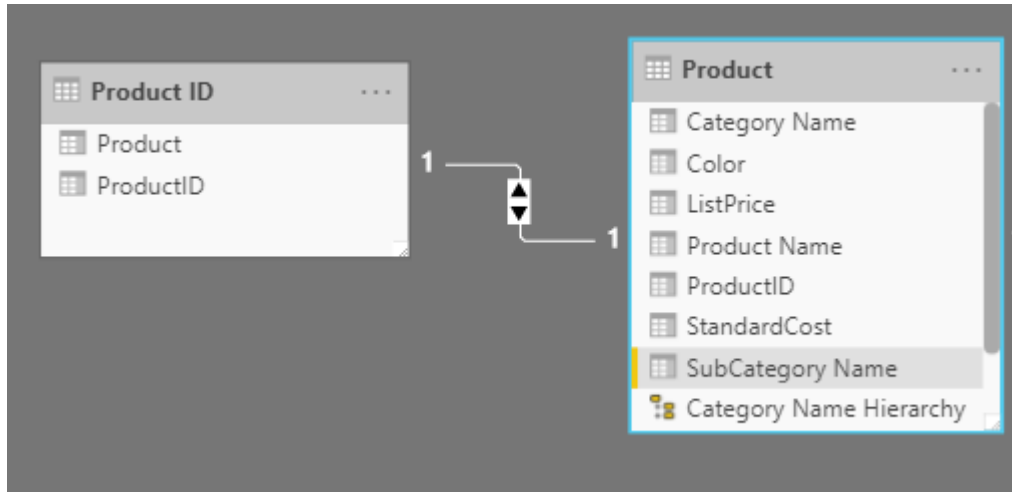
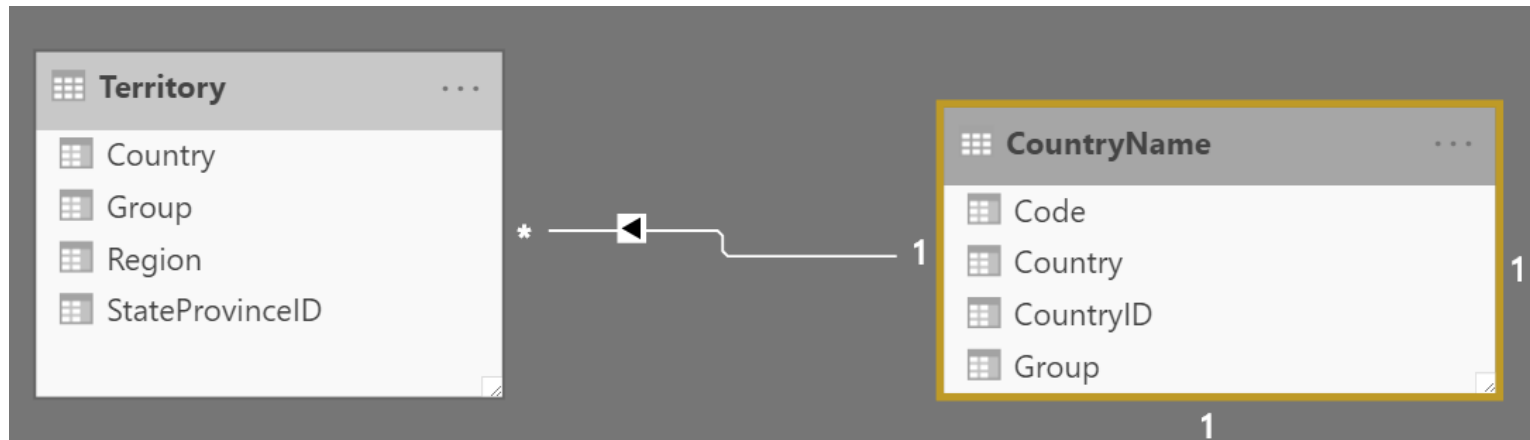
TotalSales = SUM(Sales[Total Sales])

BudgetAmount = SUM (Budget[BudgetAmount])

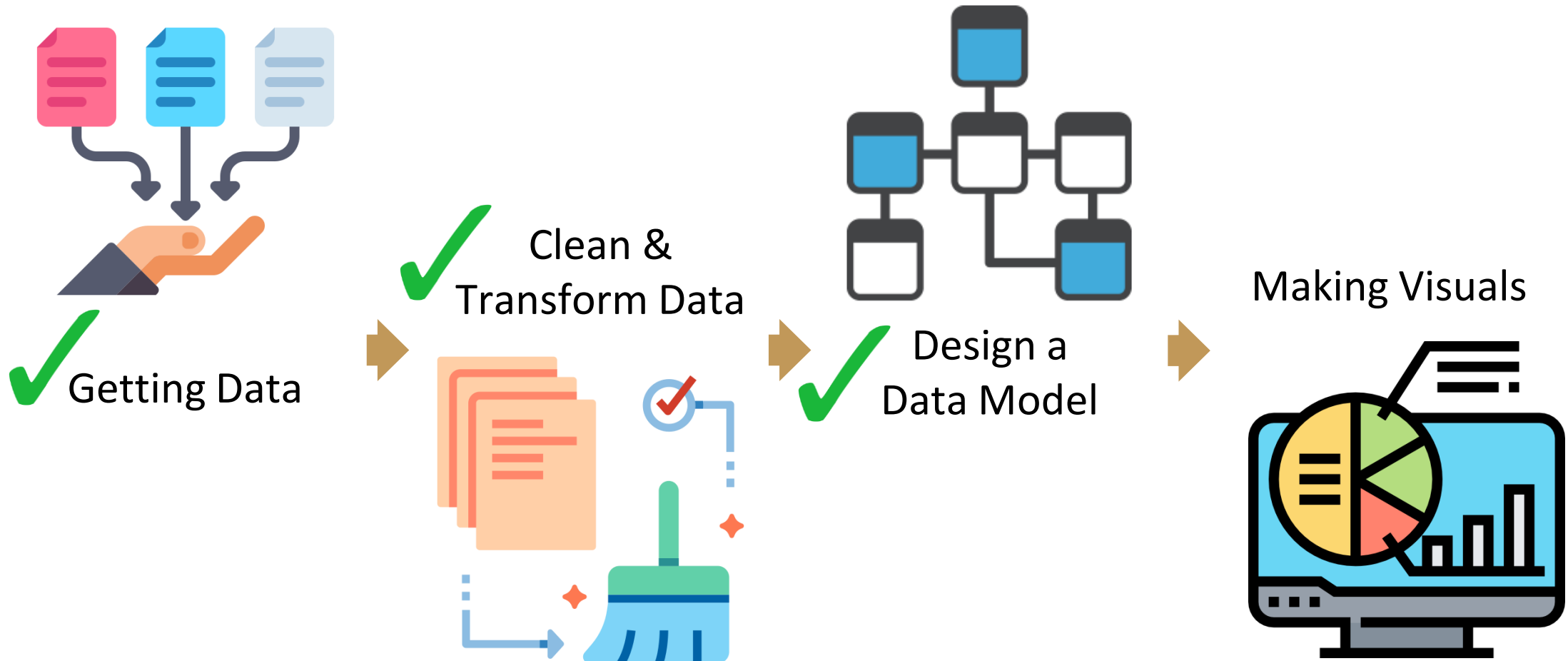
| Year  | Quarter | Month    | TotalSales | Budget Amount |
|-------|---------|----------|------------|---------------|
| 2014  | Qtr 1   | January  | 61,274     | 6,500         |
| 2014  | Qtr 1   | February | 13,396     | 6,000         |
| 2014  | Qtr 1   | March    | 110,484    | 5,000         |
| 2014  | Qtr 2   | April    | 17,980     | 7,000         |
| 2014  | Qtr 2   | May      | 77,477     | 9,000         |
| 2014  | Qtr 2   | June     | 490        | 11,000        |
| 2013  | Qtr 1   | January  | 33,438     | 5,000         |
| 2013  | Qtr 1   | February | 38,068     | 4,000         |
| Total |         |          | 1,709,647  | 120,500       |

The screenshot shows the Power BI interface with the Visualizations pane on the left and the Fields pane on the right. The Visualizations pane shows a table visualization with columns for Year, Quarter, Month, TotalSales, and Budget Amount. The Fields pane shows the 'Date' field expanded with 'Year', 'Quarter', and 'Month' selected. The 'Values' pane shows 'TotalSales' and 'Budget Amount' selected.

# Work with relationships and cardinality



# Our RoadMap

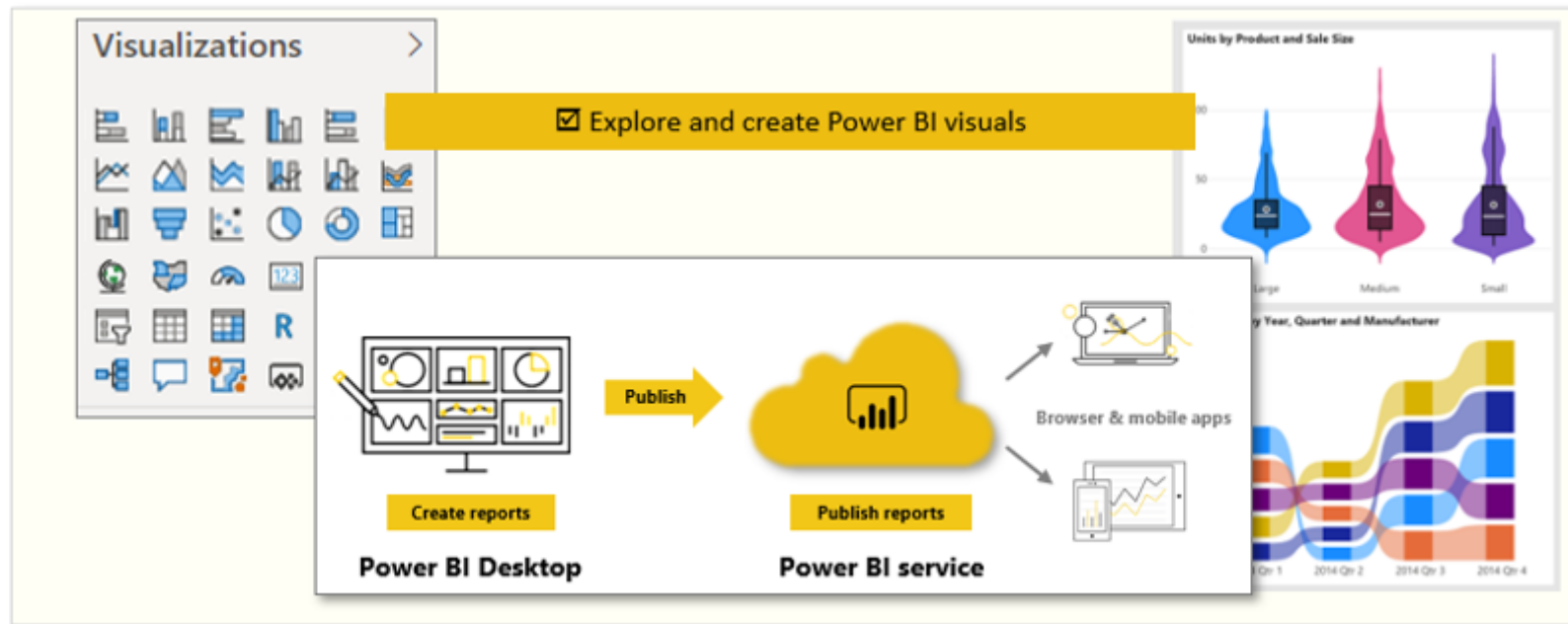


## 4<sup>th</sup> Step

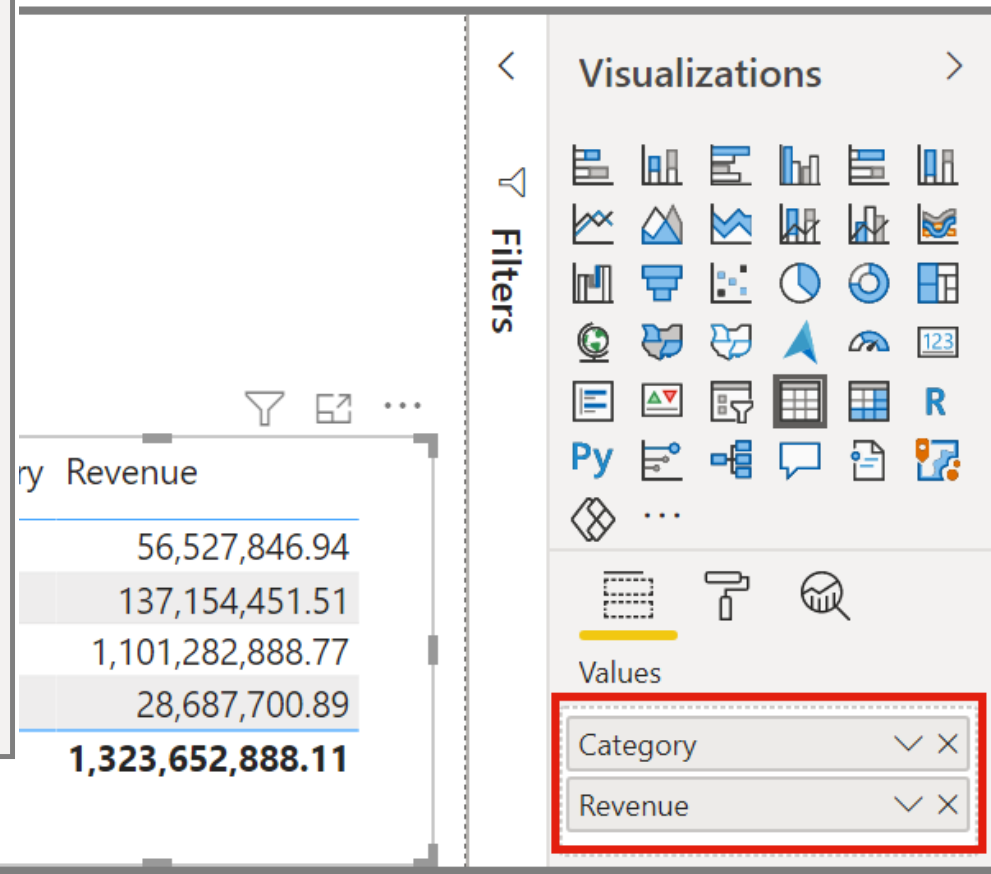
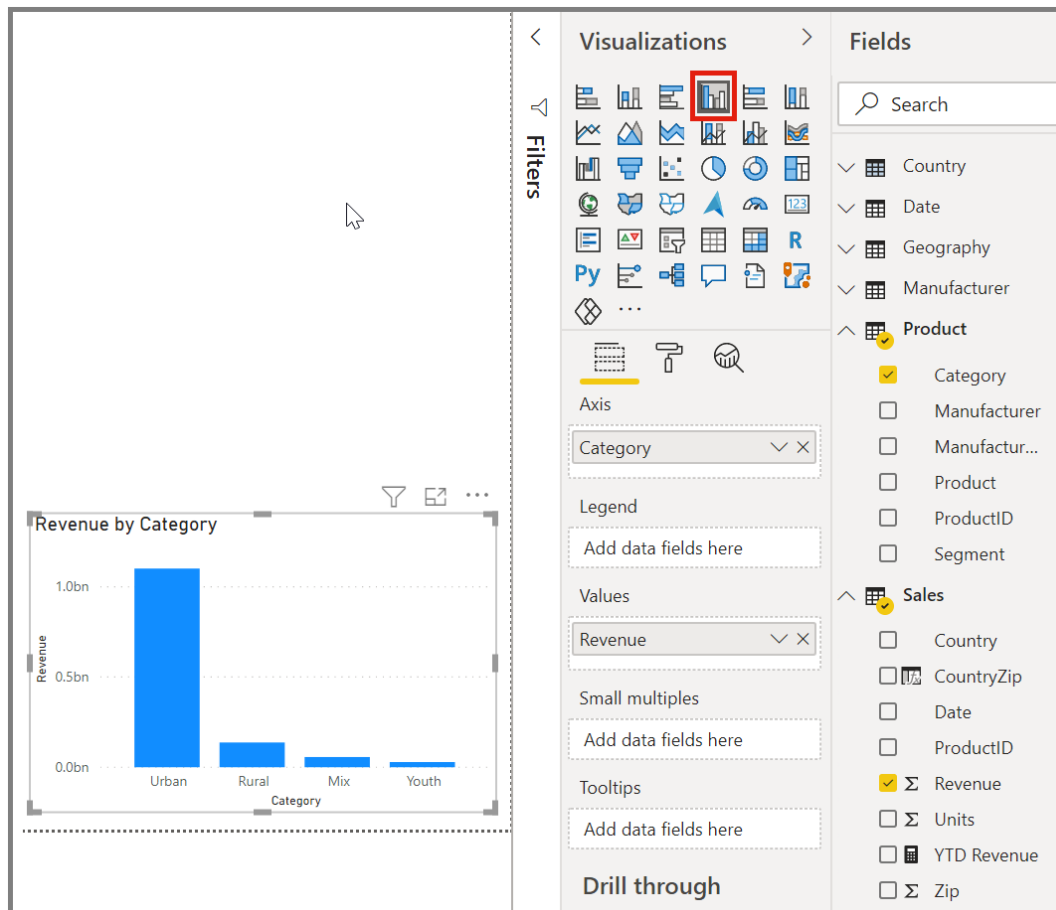


Making  
Visuals

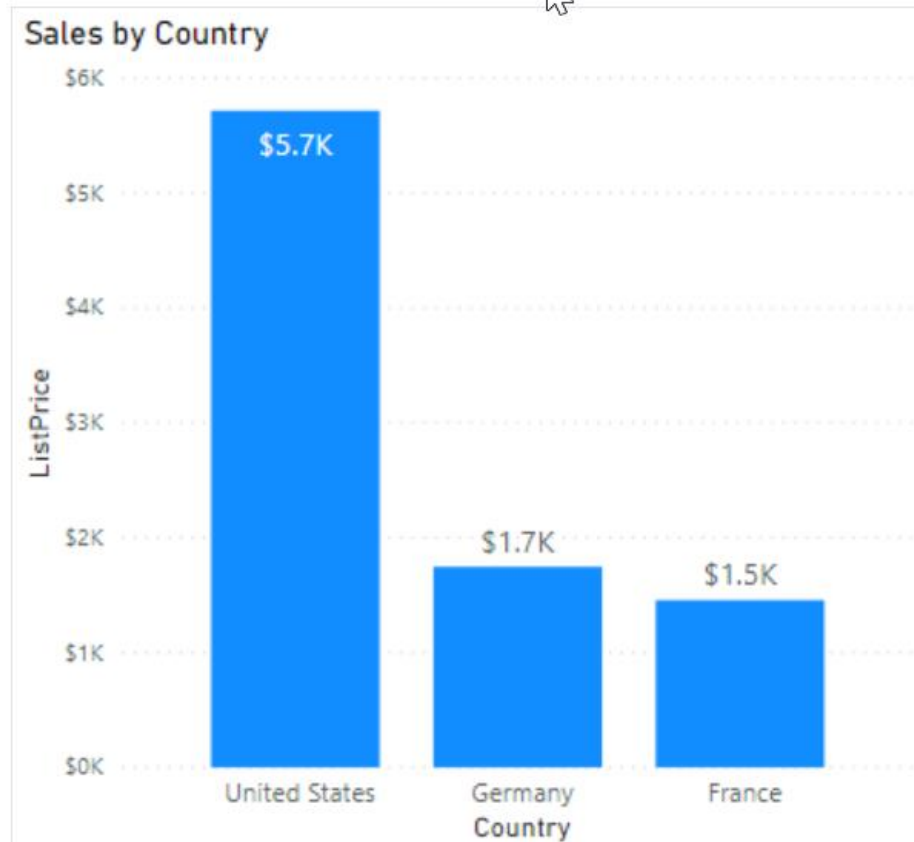
# Work with Power BI visuals



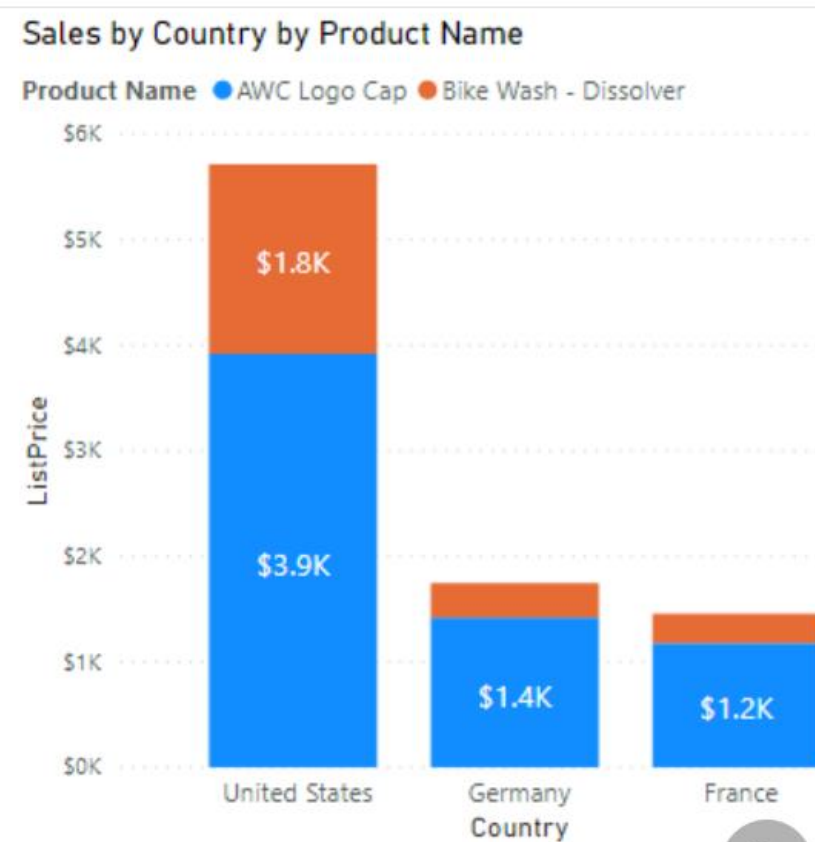
# Selecting and changing visuals



# Bar and Column charts

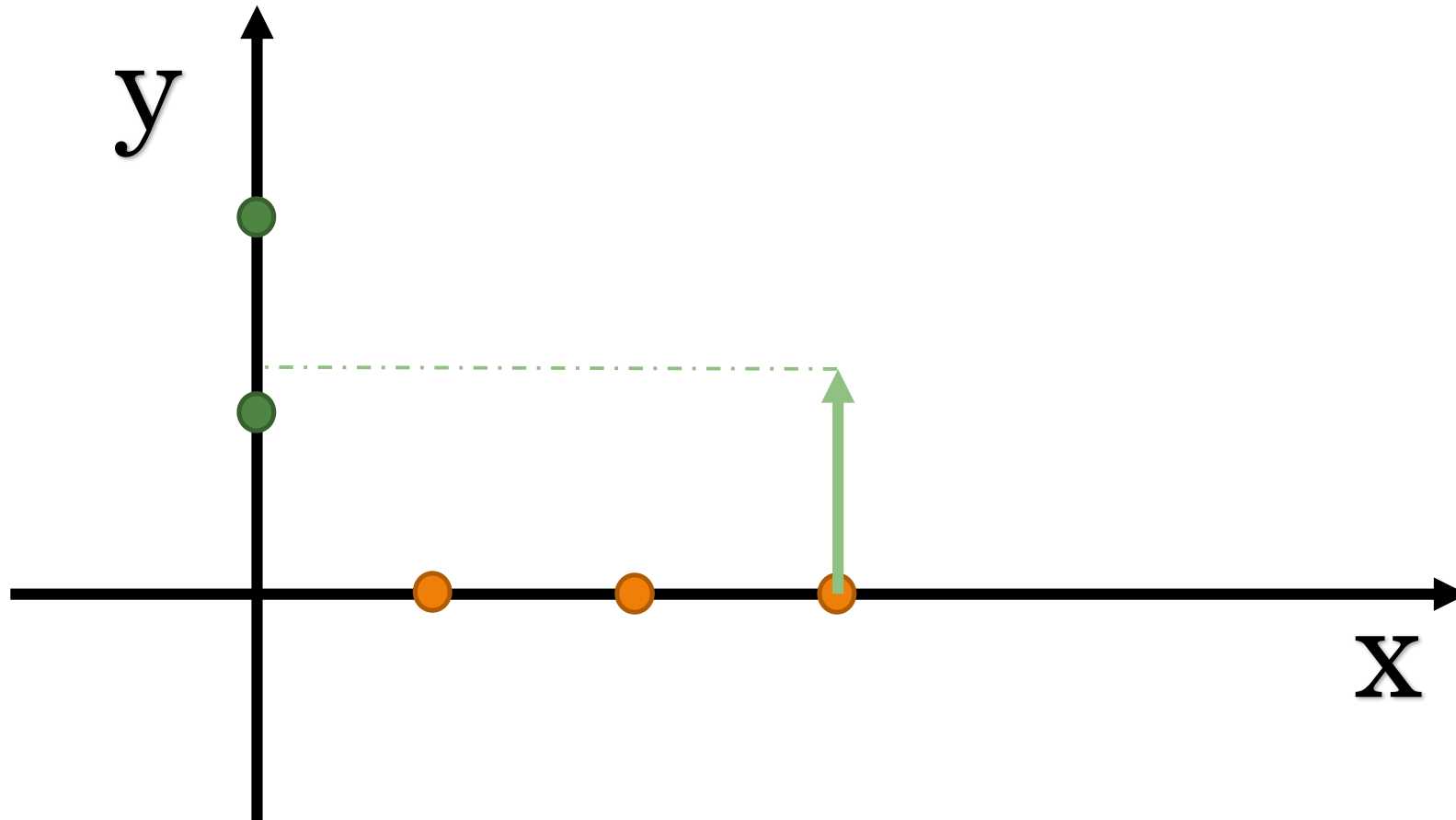


**Clustered Column Chart**



**Stacked Column Chart**

# Bar and Column charts



# Table & Matrix visuals

Visualizations

Filters

Country Sales Amount

|               |                    |
|---------------|--------------------|
| United States | \$7,390,464        |
| France        | \$773,445          |
| Germany       | \$493,628          |
| <b>Total</b>  | <b>\$8,550,077</b> |

Visualizations

Fields

Search

Territory

- ☐ City
- ☐ City (groups)
- ☒ Country
- ☐ Europe
- ☐ Group
- ☐ Norte America
- ☐ Other
- ☒ Sales Amount

Values

Country

Sales Amount

Visualizations

Fields

Category Name

- ☒ Product Color
- ☐ ListPrice
- ☐ Product Name
- ☐ Product Name (groups)
- ☐ ProductID
- ☐ StandardCost
- ☐ SubCategory Name

Territory

- ☐ City
- ☐ City (groups)
- ☒ Country
- ☐ Europe
- ☐ Group
- ☐ Norte America
- ☐ Other
- ☒ Sales Amount

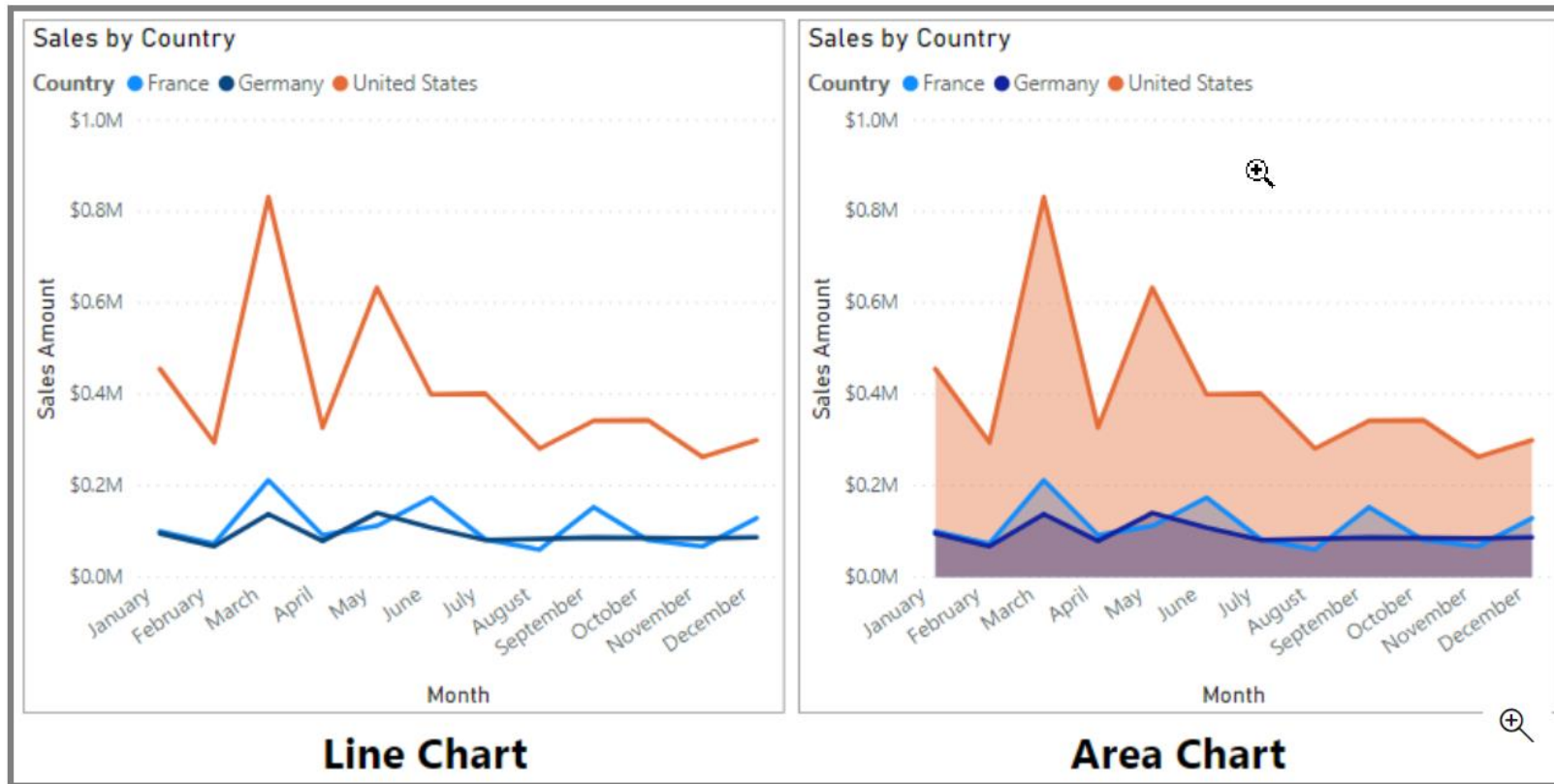
Country

Product Color

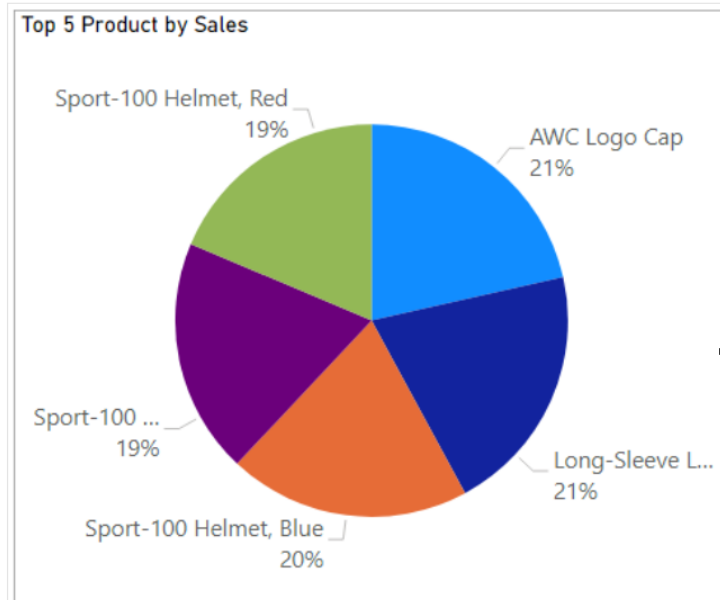
Sales Amount

| Country       | Blue               | Red                | Silver             | Total              |
|---------------|--------------------|--------------------|--------------------|--------------------|
| United States | \$4,078,994        | \$4,788,860        | \$4,052,394        | \$7,390,464        |
| France        | \$517,538          | \$429,920          | \$382,679          | \$773,445          |
| Germany       | \$301,269          | \$309,874          | \$298,159          | \$493,628          |
| <b>Total</b>  | <b>\$4,868,908</b> | <b>\$5,483,012</b> | <b>\$4,700,307</b> | <b>\$8,550,077</b> |

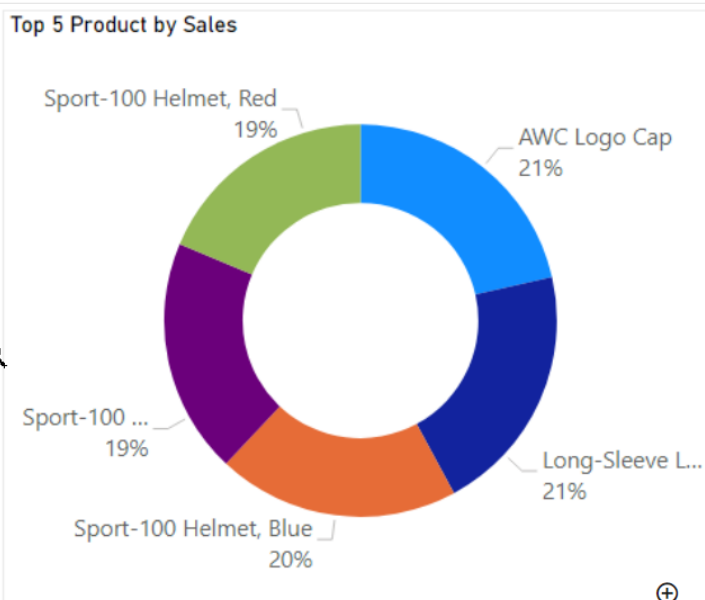
# Line and Area charts



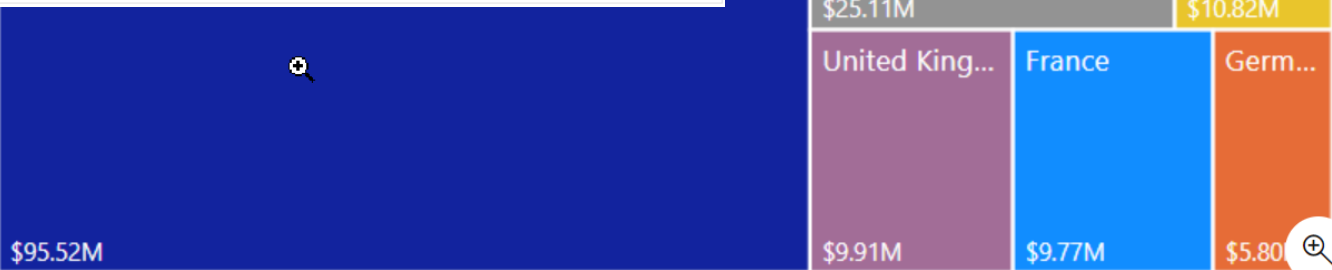
# Pie chart, donut chart, and Treemaps



Pie Chart



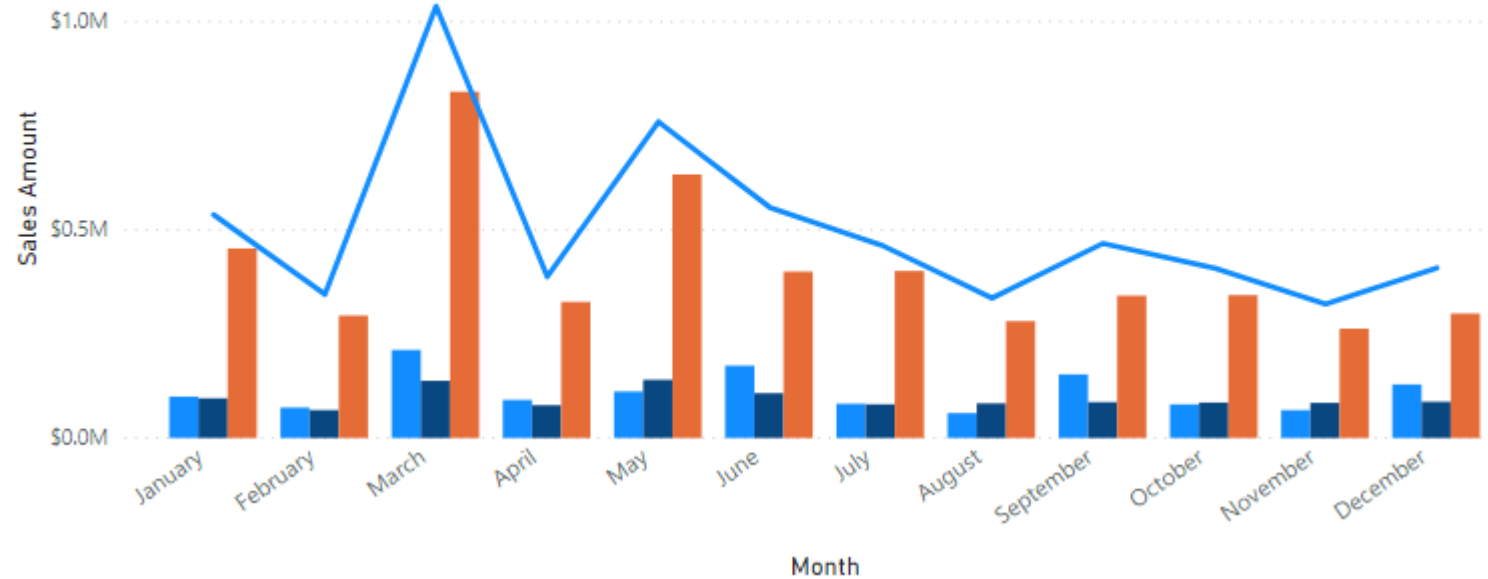
Donut Chart



# Combo Charts

Sales by Country

Country ● France ● Germany ● United States ● Sales Amount



# KPI & Card Visuals

Sales and Target by MonthKey



33M

Sales

\$156,775,000

Target

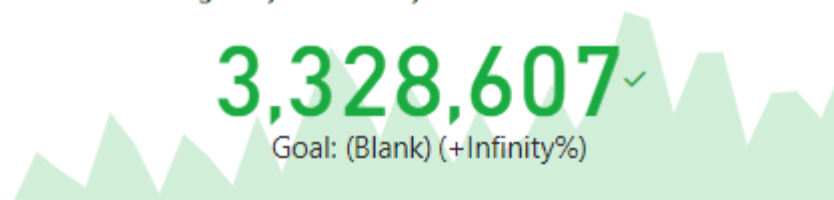
\$32,371,233.34

Cost

\$436.8

Standard Cost

Sales and Target by MonthKey



# Filters & Slicers

 **Filters**  

 Search

Filters on this visual

...

Add data fields here

Filters on this page

...

Add data fields here

Filters on all pages

...

Add data fields here

Year, Quarter, Month, Day

- ▼ ☐ 2017
- ▼ ☐ 2018
- ^ ☒ 2019
  - ▼ ☒ Qtr 1
  - ▼ ☒ Qtr 2
  - ▼ ☒ Qtr 3
  - ▼ ☒ Qtr 4
- ▼ ☐ 2020
- ▼ ☐ 2021

Day

|   |    |    |    |    |    |    |
|---|----|----|----|----|----|----|
| 1 | 6  | 11 | 16 | 21 | 26 | 31 |
| 2 | 7  | 12 | 17 | 22 | 27 |    |
| 3 | 8  | 13 | 18 | 23 | 28 |    |
| 4 | 9  | 14 | 19 | 24 | 29 |    |
| 5 | 10 | 15 | 20 | 25 | 30 |    |

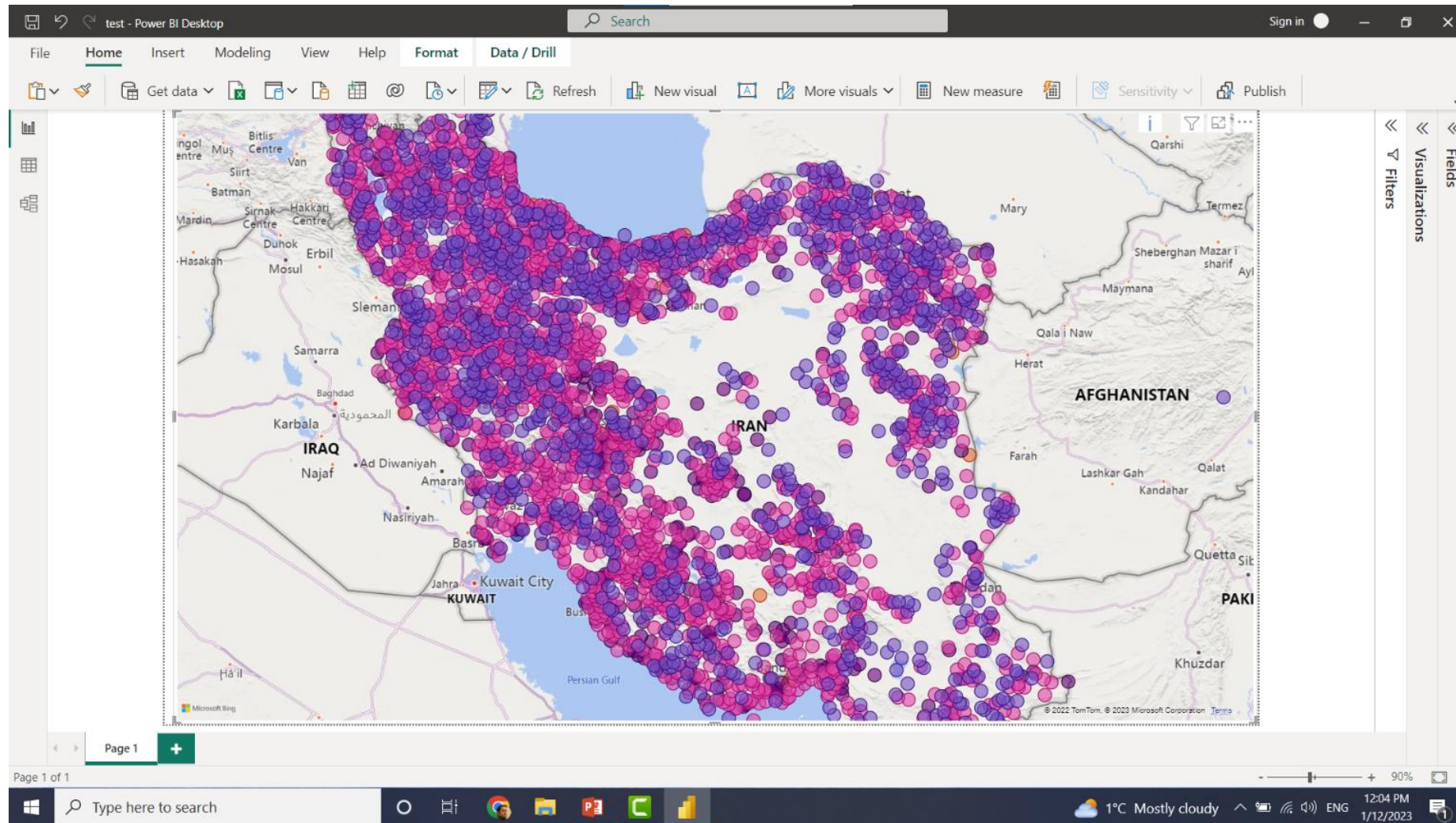
Day

1

31



# Map Visuals



# Filled Map



# Format

## Default color - Fill colors

Format s Default color - Fill colors

Gradient

What field should we base this on?

Count of province

Summarization

Count

How should we format empty values?

As zero

Minimum

Lowest value

Enter a value

☐ Add a middle color

Maximum

Highest value

Enter a value

[Learn more about conditional formatting](#)

OK

Cancel

## Visualizations

Format page



Search

> Page information

> Canvas settings

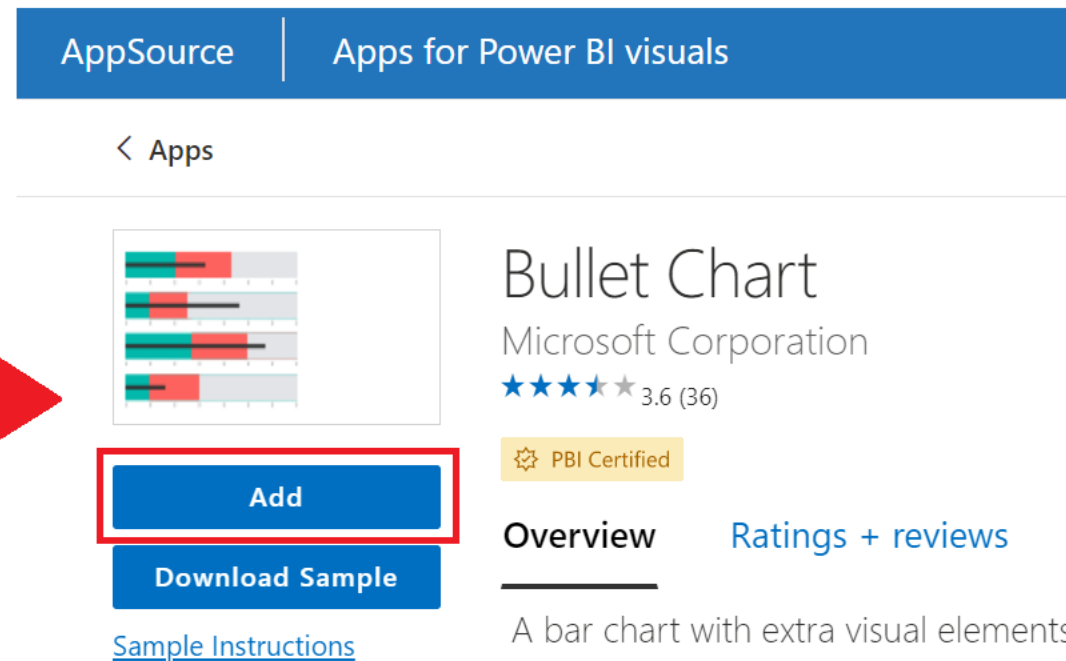
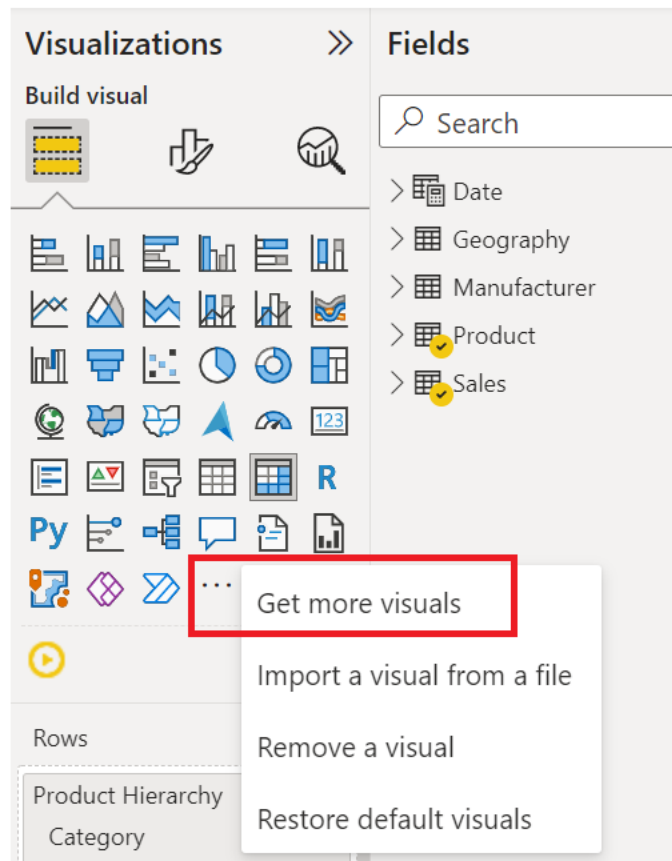
> Canvas background

> Wallpaper

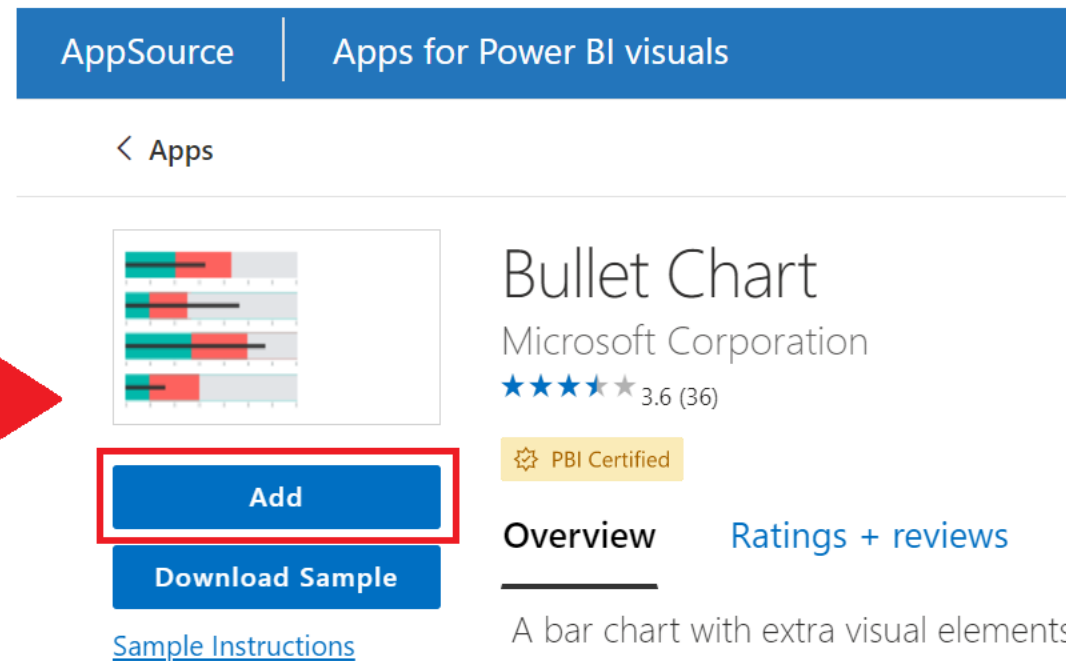
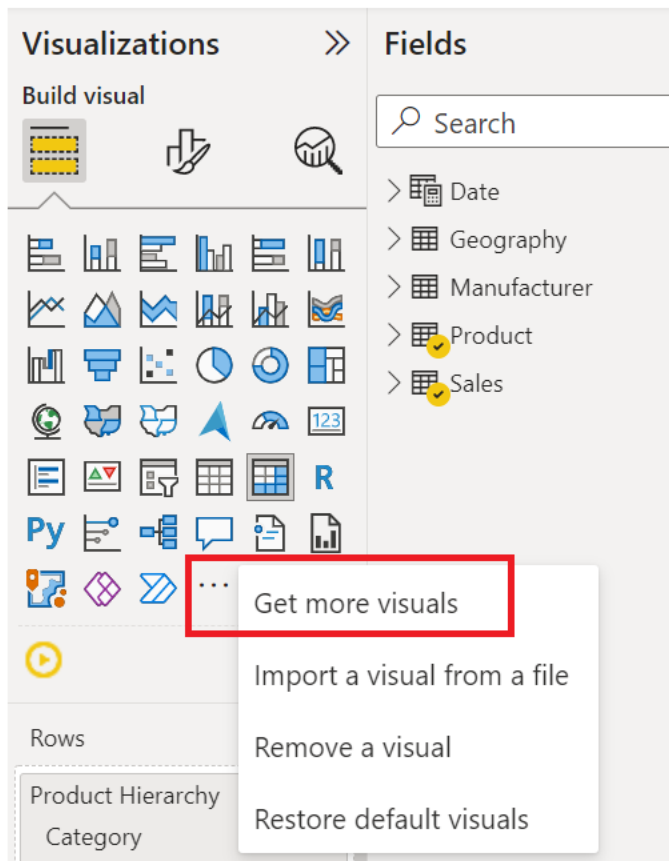
> Filter pane

> Filter cards

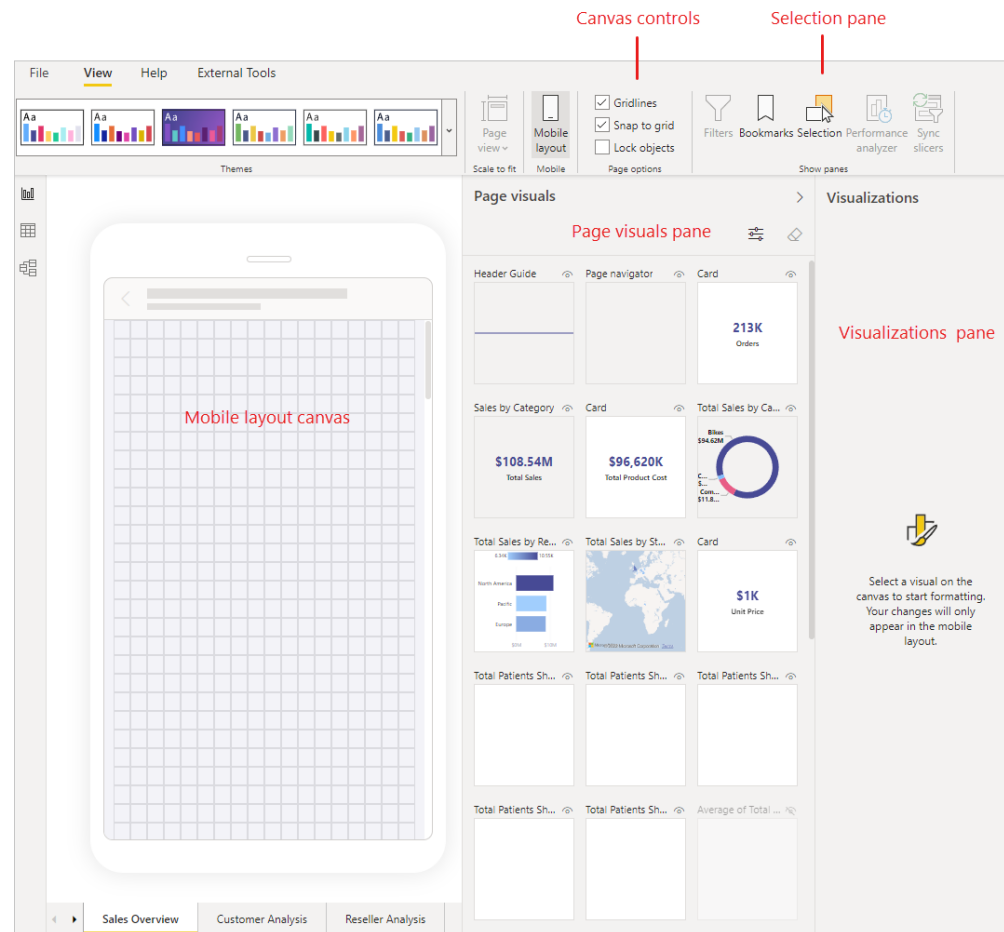
# Importing Custom Visuals



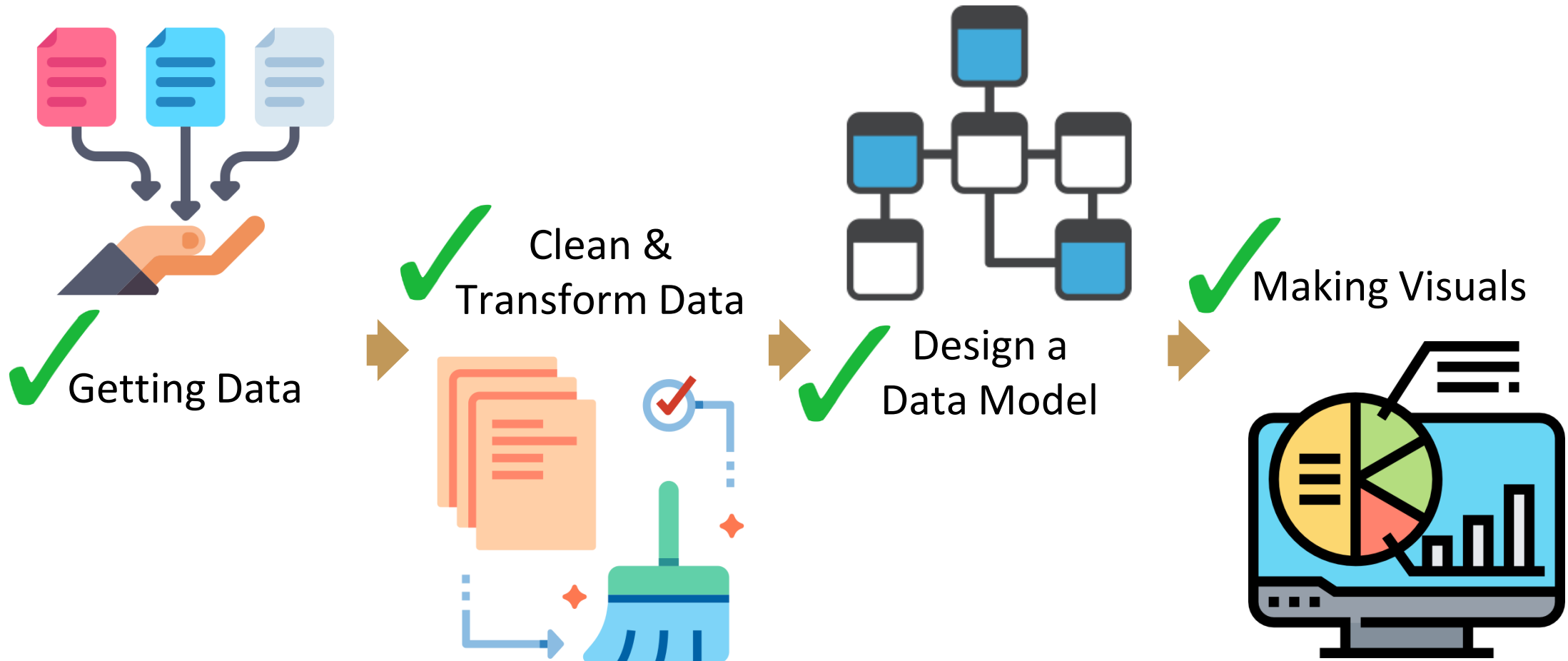
# Importing Custom Visuals



# Desktop & Mobile Layout View



# Our RoadMap



# Course Index

- What is Power BI
- Power BI installation
- My first dashboard in Power BI
- Get data in Power BI
- Clean, transform, and load data in Power BI
- Design a data model in Power BI
- Work with Power BI visuals



# Course Index

- Shamsi dates & dates dimension in Power Bi
- Introduction to creating measures using DAX in Power BI
- Create a data-driven story with Power BI reports
- Create dashboards in Power BI
- Perform analytics in Power BI
- Power BI advanced topics



# Shamsi dates & Dates dimension

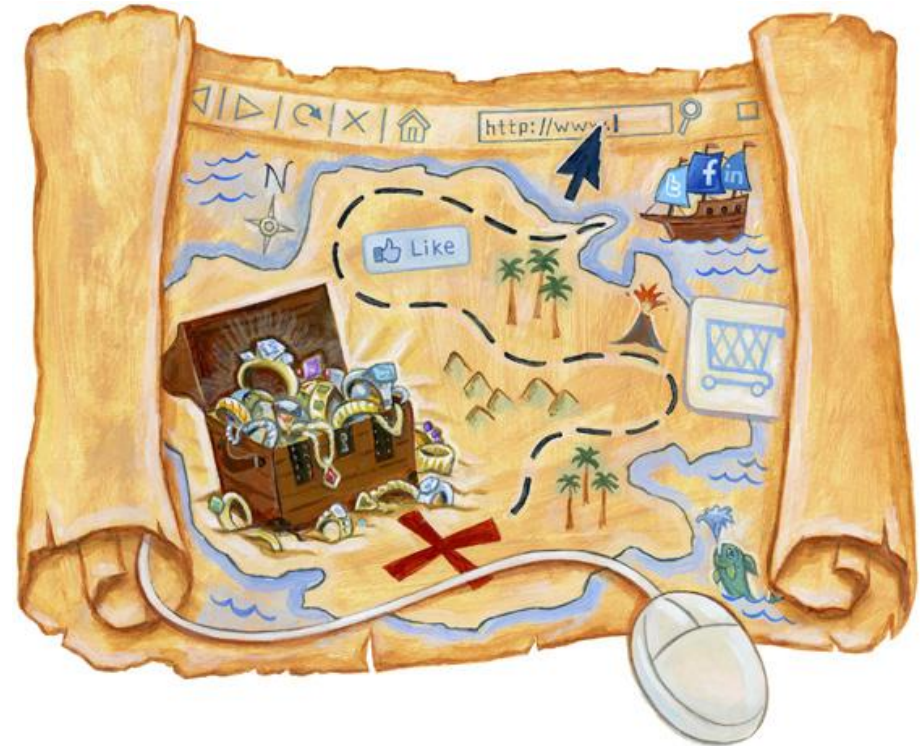
|                       |       |       |       |       |       |       |
|-----------------------|-------|-------|-------|-------|-------|-------|
| فروردین 1402          |       |       |       |       |       |       |
| March 2023 April 2023 |       |       |       |       |       |       |
| شعبان 1444 رمضان 1444 |       |       |       |       |       |       |
| ش ی د س ج پ ج         |       |       |       |       |       |       |
| 4                     | 3     | 2     | 1     |       |       |       |
| 2 24                  | 1 23  | 29 22 | 28 21 |       |       |       |
| 11                    | 10    | 9     | 8     | 7     | 6     | 5     |
| 9 31                  | 8 30  | 7 29  | 6 28  | 5 27  | 4 26  | 3 25  |
| 18                    | 17    | 16    | 15    | 14    | 13    | 12    |
| 16 7                  | 15 6  | 14 5  | 13 4  | 12 3  | 11 2  | 10 1  |
| 25                    | 24    | 23    | 22    | 21    | 20    | 19    |
| 23 14                 | 22 13 | 21 12 | 20 11 | 19 10 | 18 9  | 17 8  |
|                       | 31    | 30    | 29    | 28    | 27    | 26    |
|                       | 29 20 | 28 19 | 27 18 | 26 17 | 25 16 | 24 15 |

# Course Index

✓ Shamsi dates & dates dimension in Power Bi

➔ Introduction to creating measures using DAX in Power BI

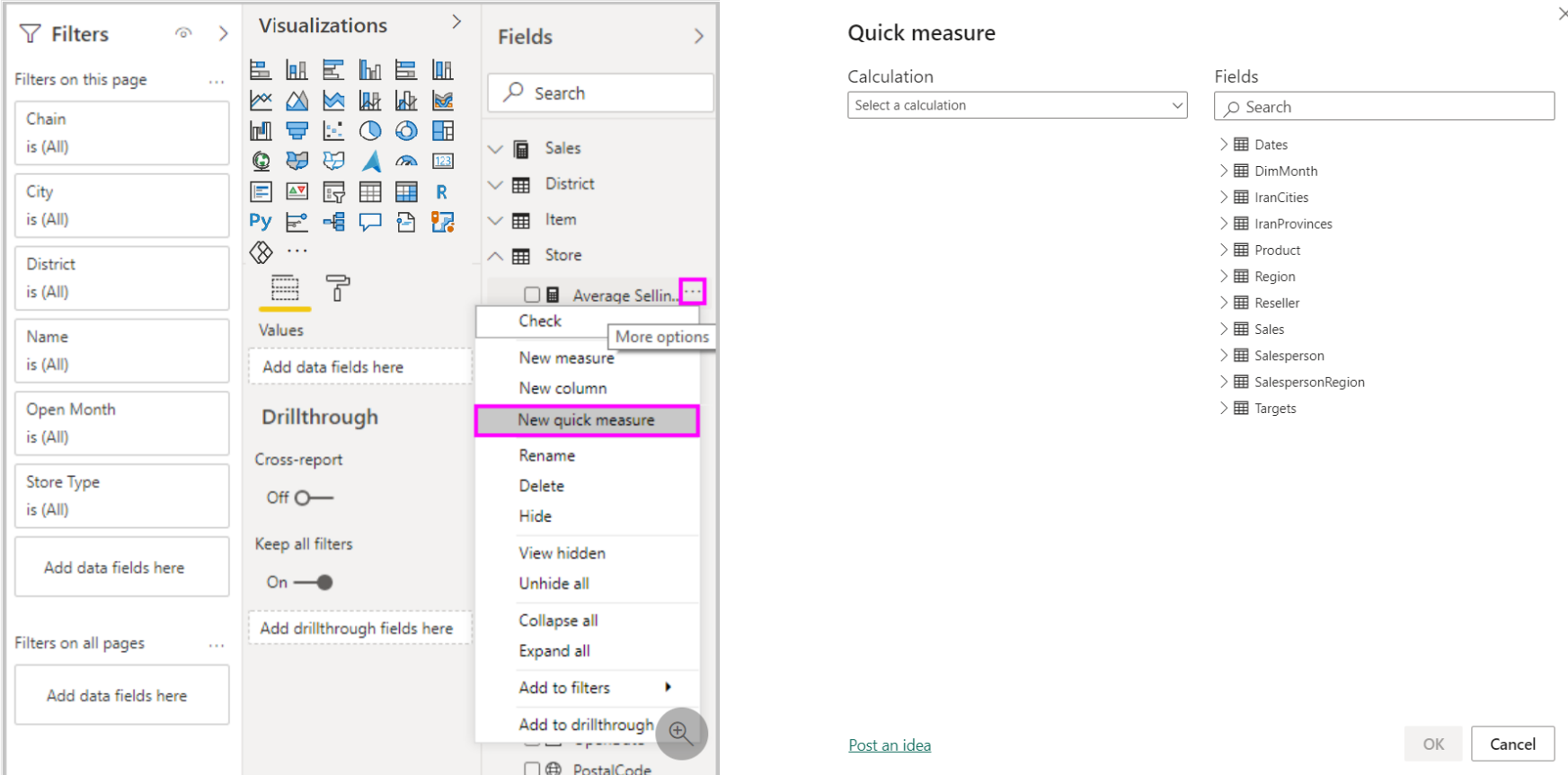
- Create a data-driven story with Power BI reports
- Create dashboards in Power BI
- Perform analytics in Power BI
- Power BI advanced topics



# DAX

- Build quick measures.
- Create calculated columns.
- Use DAX to build measures.
- Discover how context affects DAX measures.
- Use the CALCULATE function to manipulate filters.
- Implement time intelligence by using DAX.

# Create a quick measure

- 

The screenshot illustrates the process of creating a quick measure in Power BI. On the left, the 'Fields' pane shows a list of tables: Sales, District, Item, and Store. The 'Average Selling Price' measure is selected, and a context menu is open with 'New quick measure' highlighted. On the right, the 'Quick measure' dialog box is displayed, showing a search bar and a list of fields: Dates, DimMonth, IranCities, IranProvinces, Product, Region, Reseller, Sales, Salesperson, SalespersonRegion, and Targets. The 'OK' button is visible at the bottom right of the dialog.

# Create calculated columns

The screenshot displays the Power BI Desktop interface. On the left, a data table is visible with the following data:

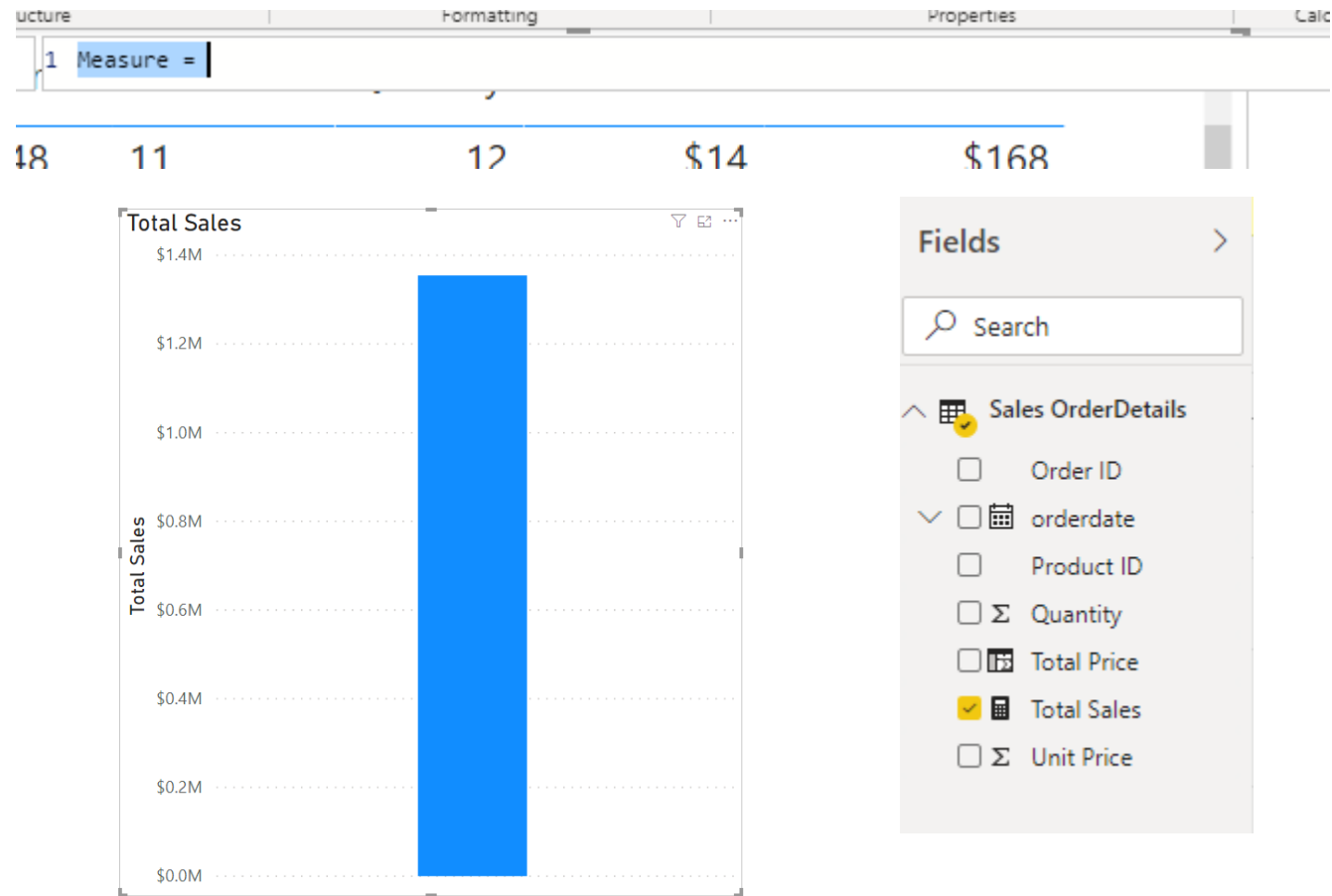
| Column 1 | Column 2 | Column 3 | Column 4 |
|----------|----------|----------|----------|
| 10248    | 11       | 12       | \$1.     |
| 10248    | 42       | 10       | \$9.     |
| 10248    | 72       | 5        | \$34.    |
| 10249    | 14       | 9        | \$18.    |
| 10249    | 51       | 40       | \$42.    |
| 10250    | 41       | 10       | \$7.     |

On the right, the 'Fields' pane is open, showing a list of options for creating a new measure. The 'New column' option is highlighted with a red box.

The 'Fields' pane options include:

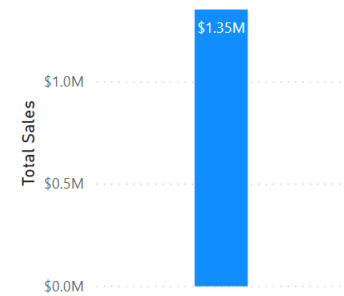
- New measure
- New column**
- New quick measure
- Refresh data
- Edit query
- Incremental refresh
- Manage aggregations
- Rename
- Delete
- Hide
- Mark as date table
- View hidden
- Unhide all
- Collapse all
- Expand all

# Create a measure

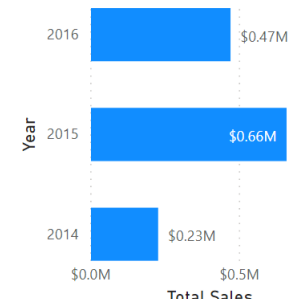


# Understand context

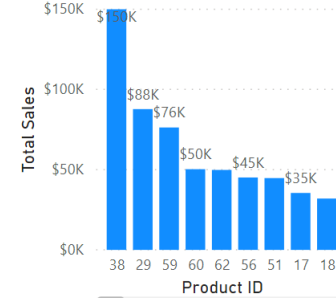
Total Sales



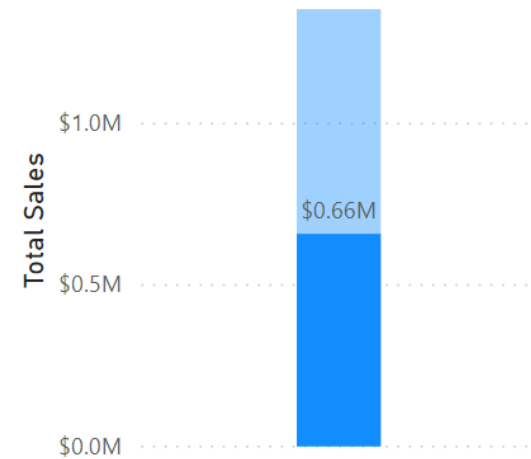
Total Sales by Year



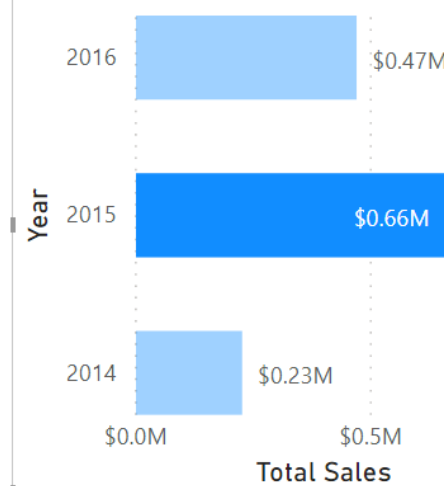
Total Sales by Product ID



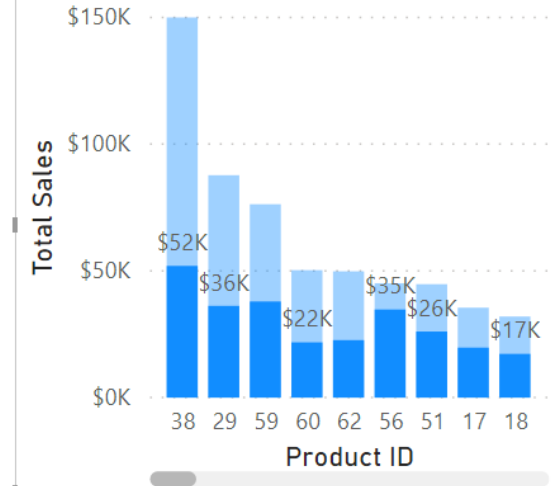
Total Sales



Total Sales by Year



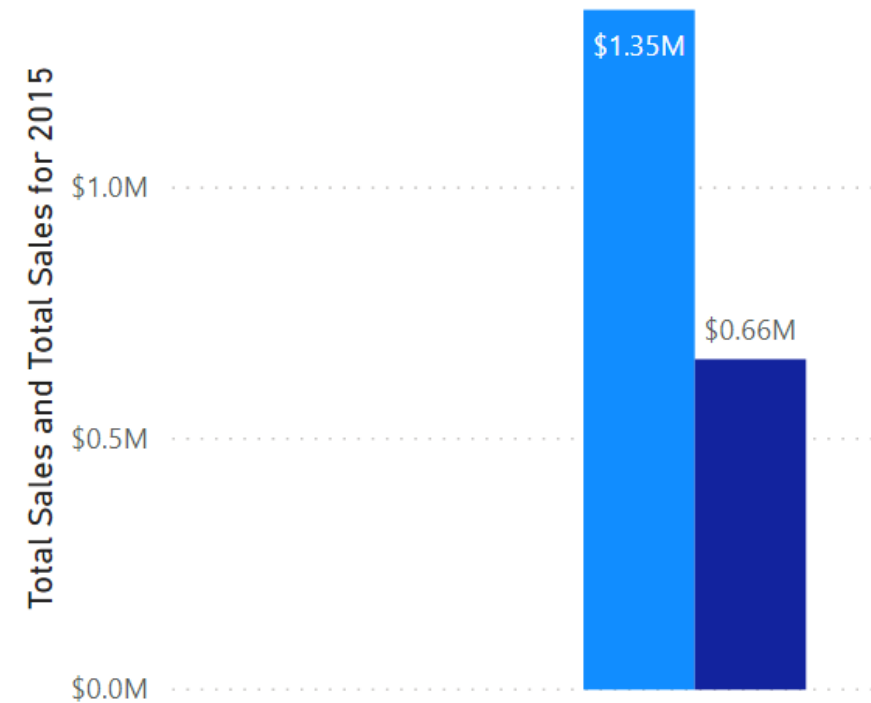
Total Sales by Product ID



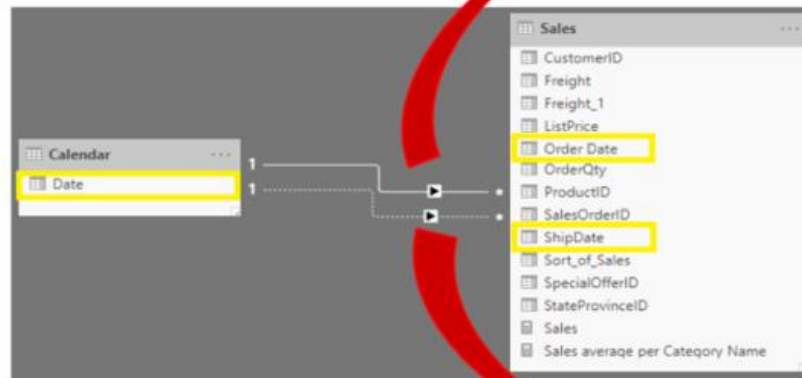
# Use the Calculate function

Total Sales and Total Sales for 2015

● Total Sales ● Total Sales for 2015



# Use relationships effectively



Edit relationship

Select tables and columns that are related.

Sales

| SalesOrderID | OrderDate    | Sort_of_Sales | Freight | Freight_1 | ProductID | OrderQty | SpecialOfferID |
|--------------|--------------|---------------|---------|-----------|-----------|----------|----------------|
| 52242        | 07 July 2013 | Internet      | 0.1248  | 0.1248    | 870       | 1        | 1              |
| 52530        | 14 July 2013 | Internet      | 0.1248  | 0.1248    | 870       | 1        | 1              |
| 52634        | 16 July 2013 | Internet      | 0.1248  | 0.1248    | 870       | 1        | 1              |

Calendar

Date

01/01/2011 00:00:00  
02/01/2011 00:00:00  
03/01/2011 00:00:00

Cardinality: Many to one ("\*:1")

Cross filter direction: Single

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Edit relationship

Select tables and columns that are related.

Sales

| Freight_1 | ProductID | OrderQty | SpecialOfferID | StateProvinceID | CustomerID | ListPrice | ShipDate        |
|-----------|-----------|----------|----------------|-----------------|------------|-----------|-----------------|
| 0.1248    | 870       | 1        | 1              | 9               | 18102      | 4.90      | 10 July 2013 07 |
| 0.1248    | 870       | 1        | 1              | 9               | 17014      | 4.90      | 17 July 2013 14 |
| 0.1248    | 870       | 1        | 1              | 9               | 17165      | 4.90      | 19 July 2013 16 |

Calendar

Date

01/01/2011 00:00:00  
02/01/2011 00:00:00  
03/01/2011 00:00:00

Cardinality: Many to one ("\*:1")

Cross filter direction: Single

☐ Make this relationship active

☐ Apply security filter in both directions

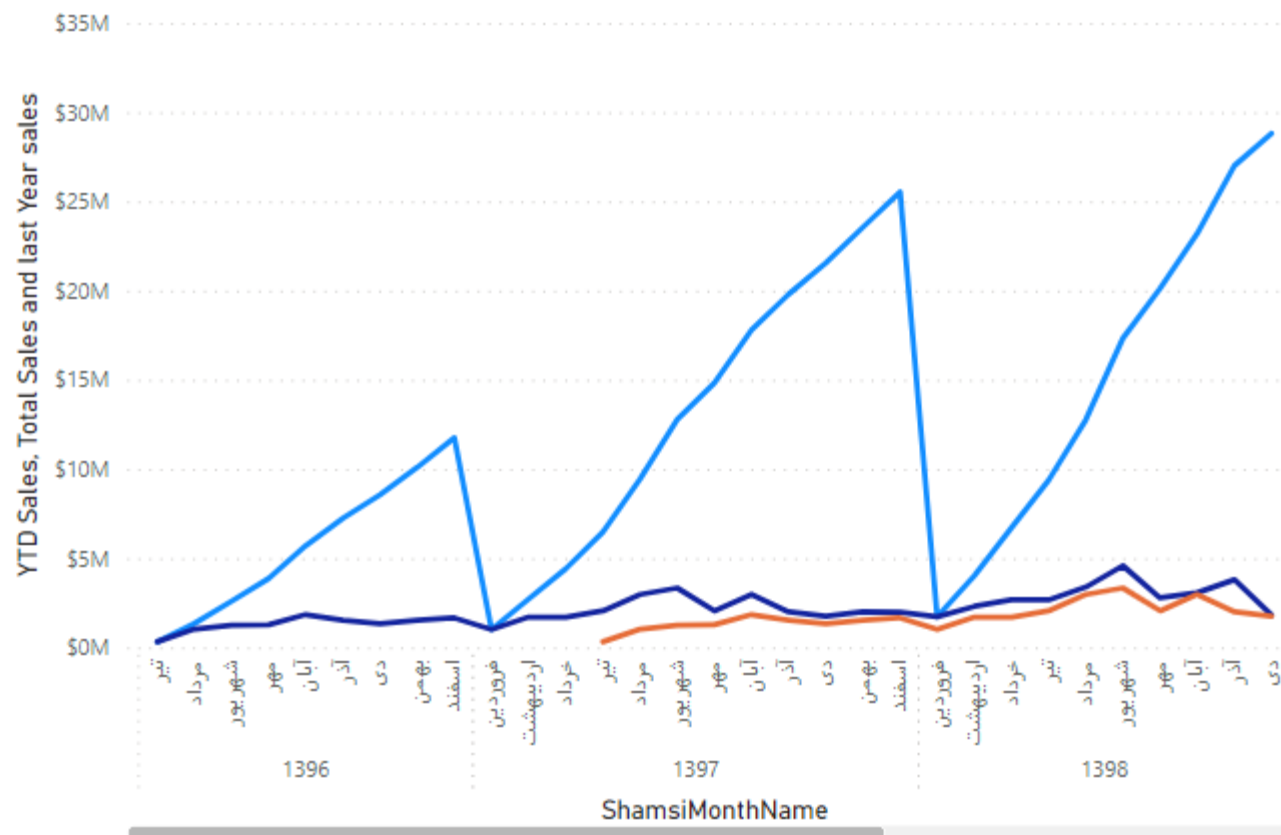
☐ Assume referential integrity

# Work with time intelligence

| ShamsiYear   | YTD Sales              | Total Sales            | last Year sales        |
|--------------|------------------------|------------------------|------------------------|
| ☐ 1396       | <b>\$11,755,905.67</b> | <b>\$11,755,905.67</b> |                        |
| تیر          | \$319,691.11           | \$319,691.11           |                        |
| مرداد        | \$1,349,988.05         | \$1,030,296.94         |                        |
| شهریور       | \$2,589,267.45         | \$1,239,279.4          |                        |
| مهر          | \$3,867,164.56         | \$1,277,897.11         |                        |
| آبان         | \$5,712,421.53         | \$1,845,256.97         |                        |
| آذر          | \$7,242,774.05         | \$1,530,352.52         |                        |
| دی           | \$8,564,725.74         | \$1,321,951.69         |                        |
| بهمن         | \$10,094,426.1         | \$1,529,700.36         |                        |
| اسفند        | \$11,755,905.67        | \$1,661,479.57         |                        |
| ☐ 1397       | <b>\$25,542,277.74</b> | <b>\$25,542,277.74</b> | <b>\$11,755,905.67</b> |
| فروردین      | \$1,016,406.81         | \$1,016,406.81         |                        |
| اردیبهشت     | \$2,723,206.72         | \$1,706,799.91         |                        |
| خرداد        | \$4,416,091.34         | \$1,692,884.62         |                        |
| تیر          | \$6,467,980.42         | \$2,051,889.08         | \$319,691.11           |
| مرداد        | \$9,438,215.61         | \$2,970,235.19         | \$1,030,296.94         |
| شهریور       | \$12,771,921.31        | \$3,333,705.7          | \$1,239,279.4          |
| مهر          | \$14,825,688.78        | \$2,053,767.47         | \$1,277,897.11         |
| آبان         | \$17,804,215.25        | \$2,978,526.47         | \$1,845,256.97         |
| آذر          | \$19,799,774.52        | \$1,995,559.27         | \$1,530,352.52         |
| دی           | \$21,551,580.69        | \$1,751,806.17         | \$1,321,951.69         |
| بهمن         | \$23,555,591.95        | \$2,004,011.26         | \$1,529,700.36         |
| اسفند        | \$25,542,277.74        | \$1,986,685.79         | \$1,661,479.57         |
| <b>Total</b> |                        | <b>\$77,548,570.2</b>  | <b>\$77,548,570.2</b>  |

YTD Sales, Total Sales and last Year sales by ShamsiYear and ShamsiMonthName

● YTD Sales ● Total Sales ● last Year sales

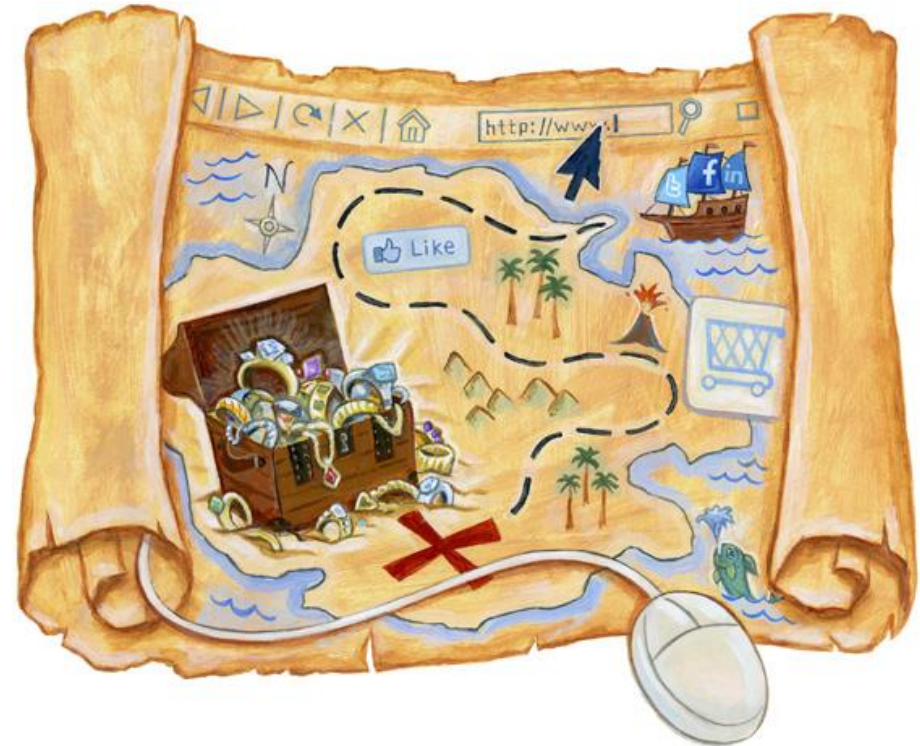


# Course Index

- ✓ Shamsi dates & dates dimension in Power Bi
- ✓ Introduction to creating measures using DAX in Power BI

➔ Create a data-driven story with Power BI reports

- Create dashboards in Power BI
- Perform analytics in Power BI
- Power BI advanced topics



# Story telling



# Power BI Best Practices



Simpson Associates

Best Practices for Power BI Visualisations | Simpson A...



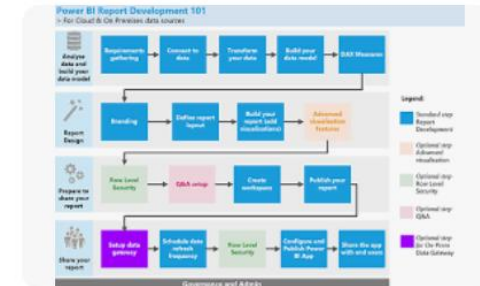
Romain Casteres

Power BI Best Practice Analyzer – Pulsweb – Romain Cast...



Blog de Bismart

The 21 Power BI Dashboard Design Best Practices

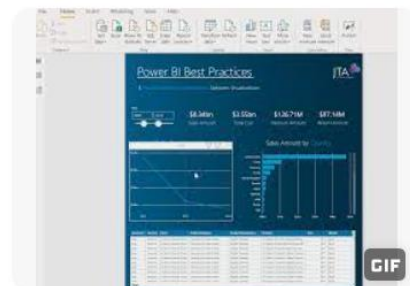


Datapears

Power BI - Report performance best practices



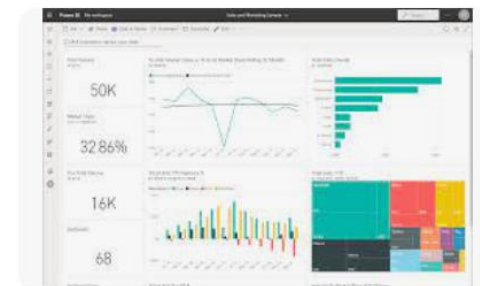
hConcepts



ITA - The Data Scientists

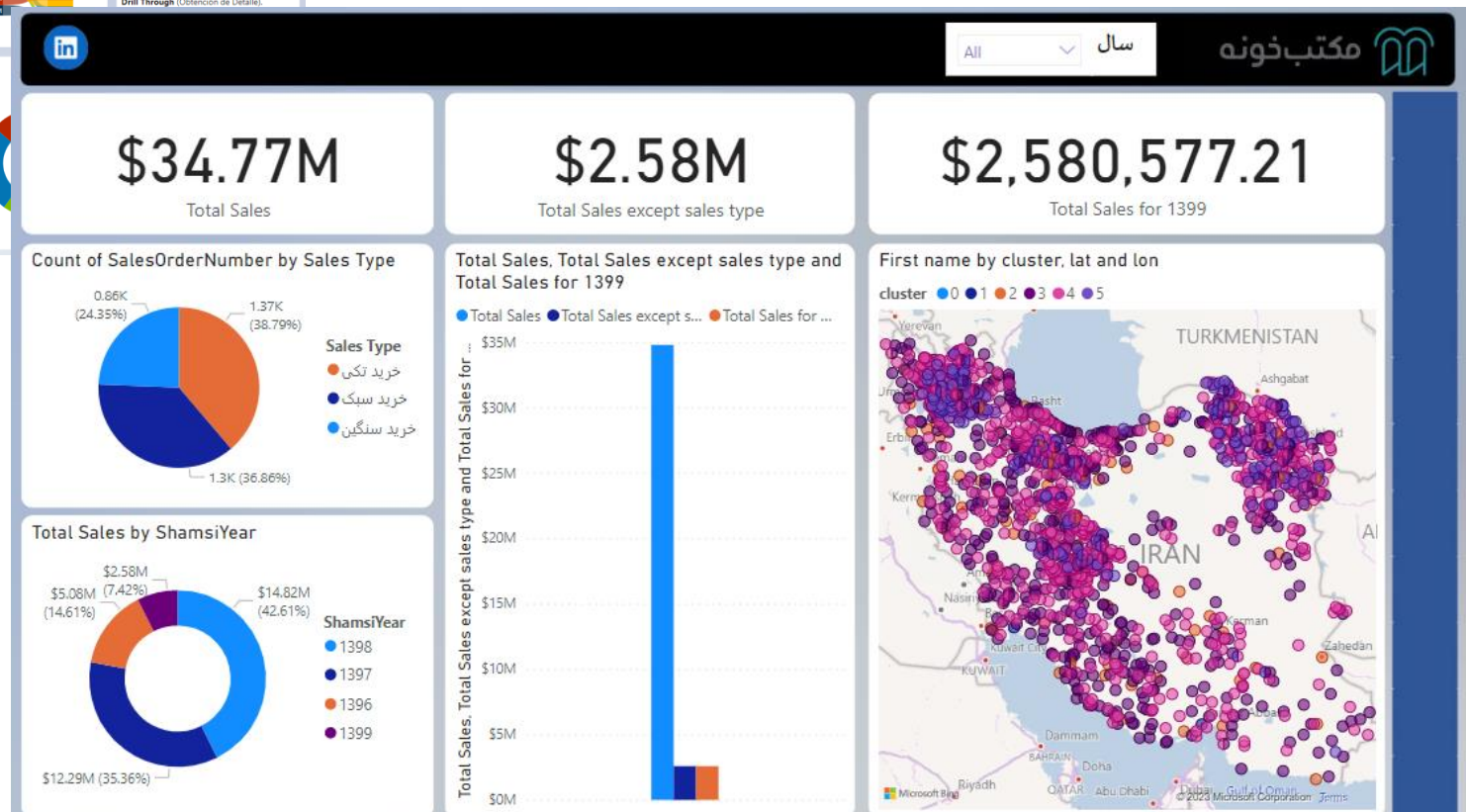
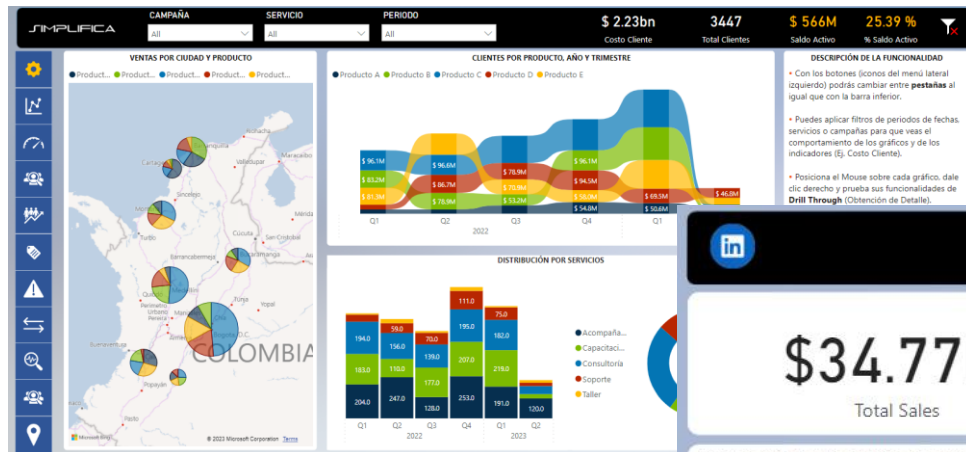


LinkedIn

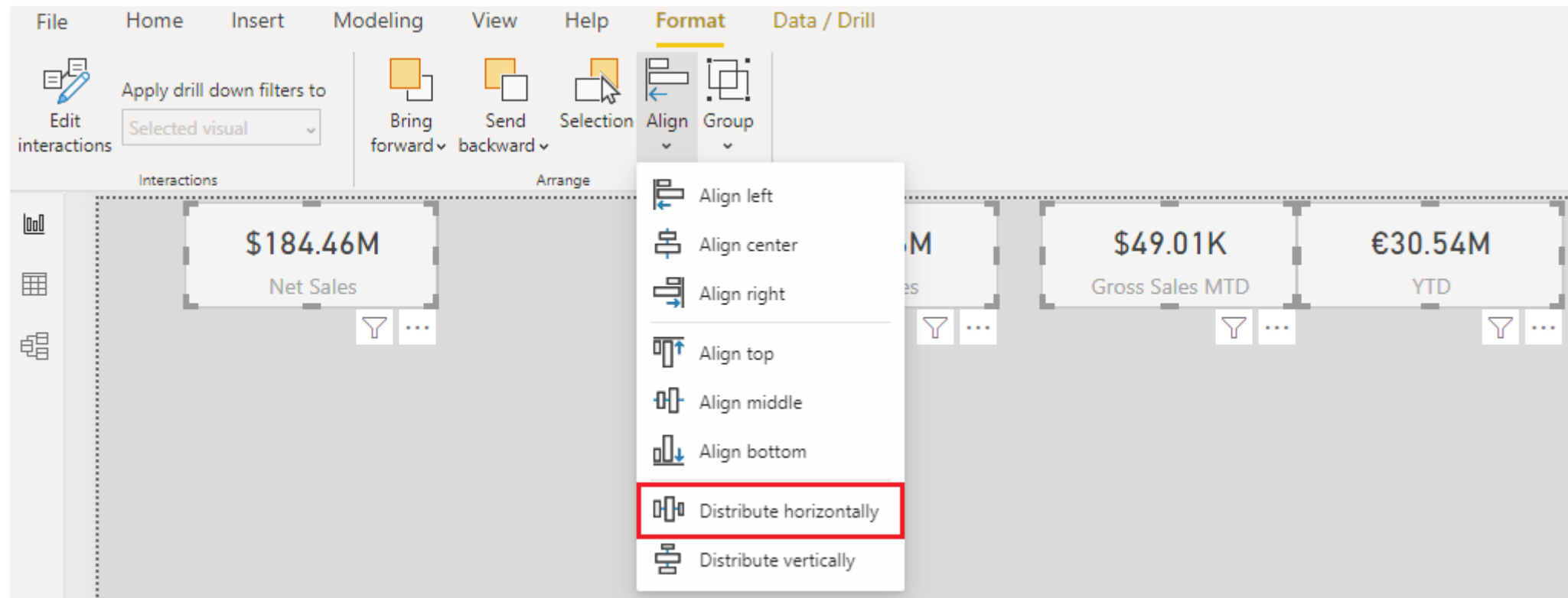


Microsoft Learn

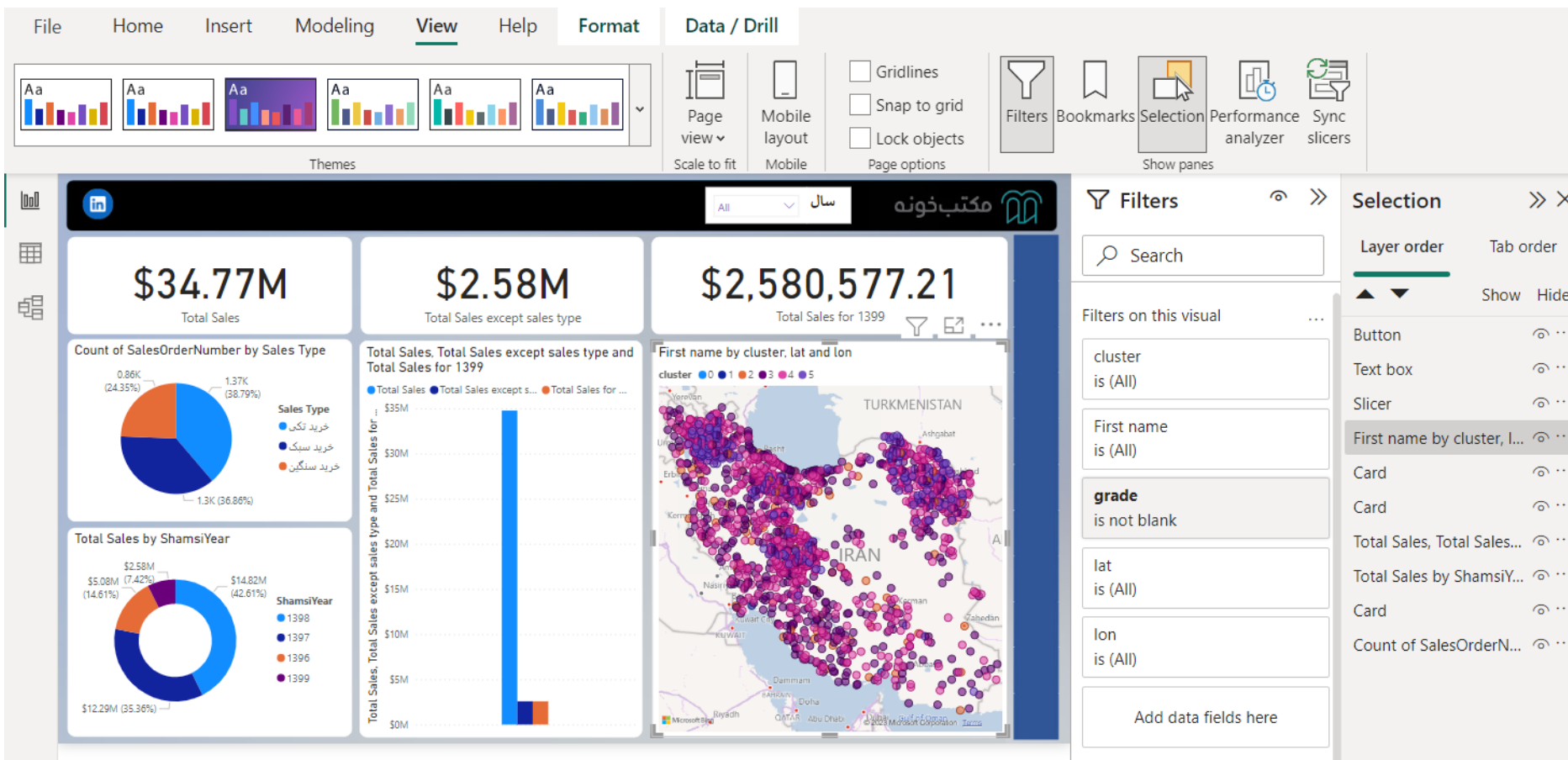
# Power BI Best Practices



# Design a report layout



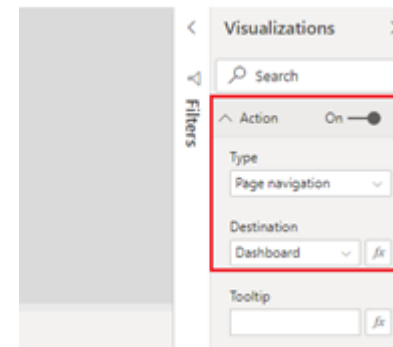
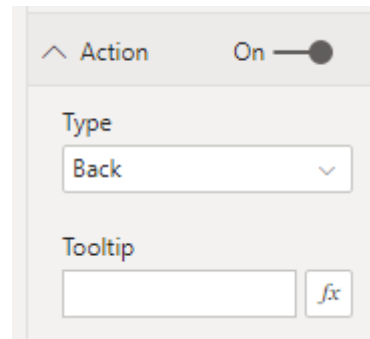
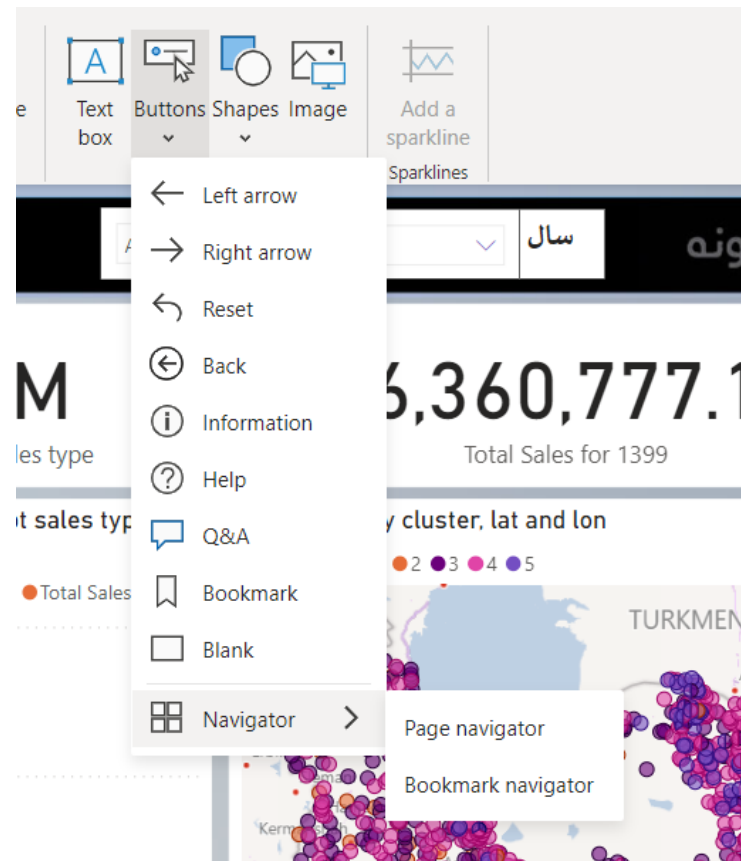
# Selections



# Page Bookmarks



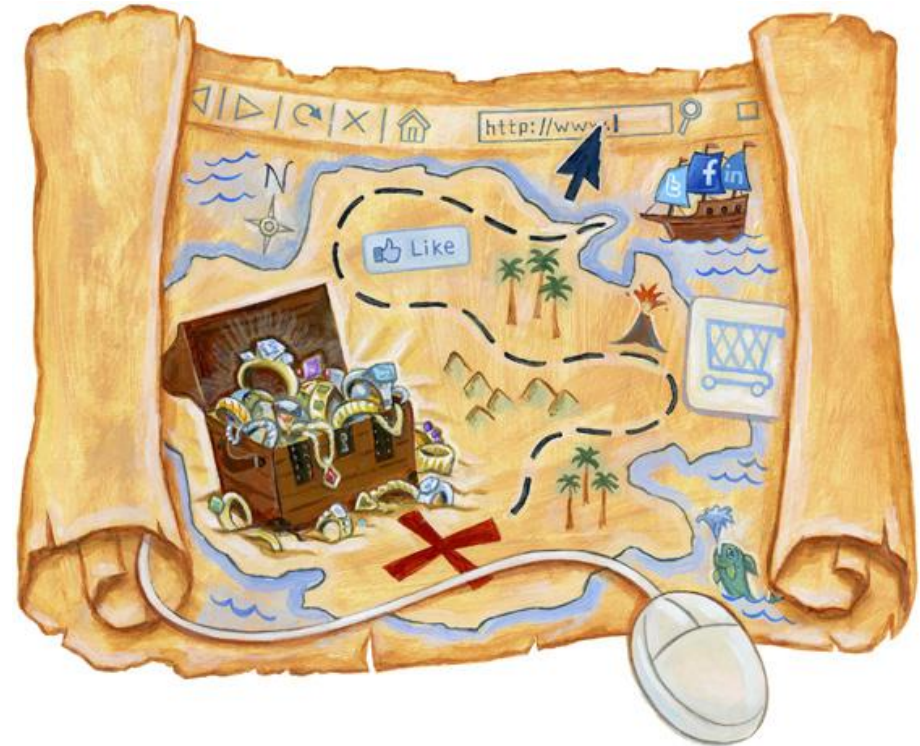
# Navigation



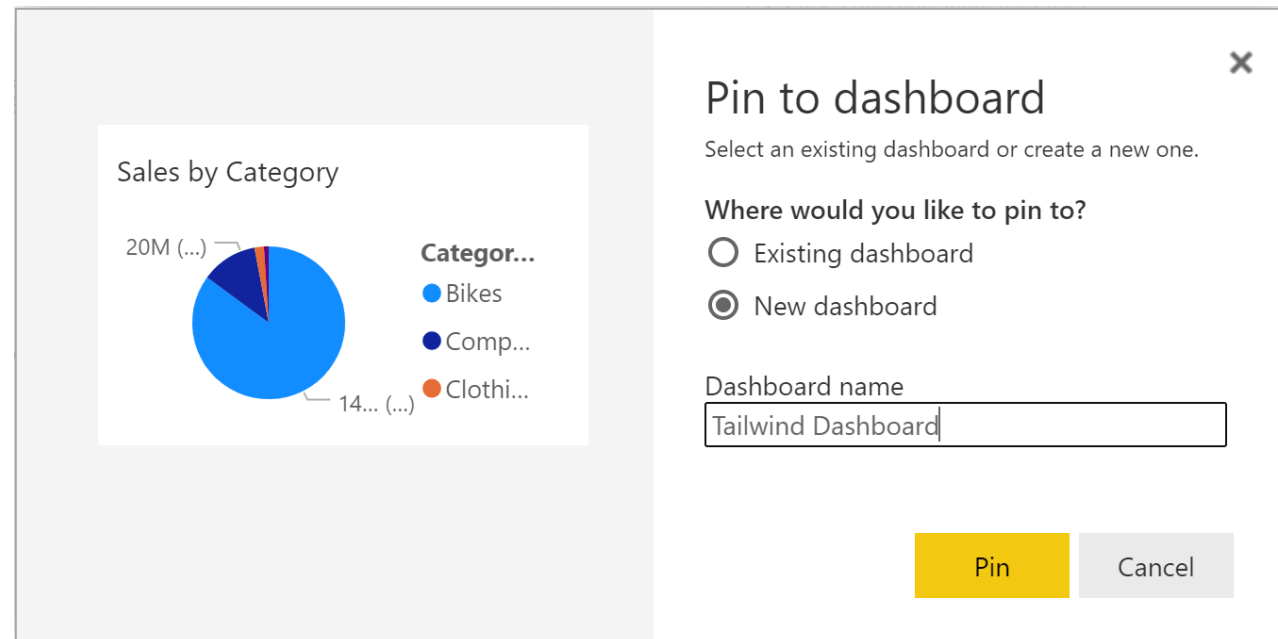
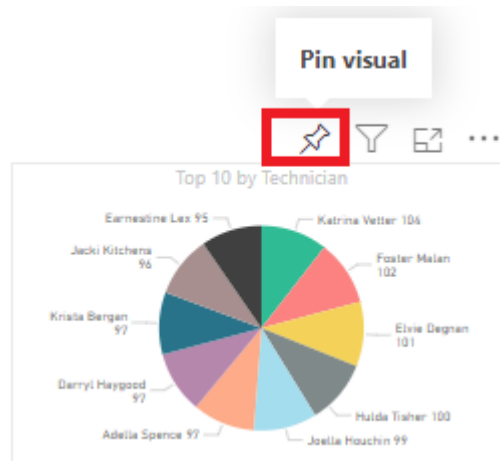
# Course Index

- ✓ Shamsi dates & dates dimension in Power Bi
- ✓ Introduction to creating measures using DAX in Power BI
- ✓ Create a data-driven story with Power BI reports

- ➔ Create dashboards in Power BI
- Perform analytics in Power BI
  - Power BI advanced topics



# Introduction to dashboards



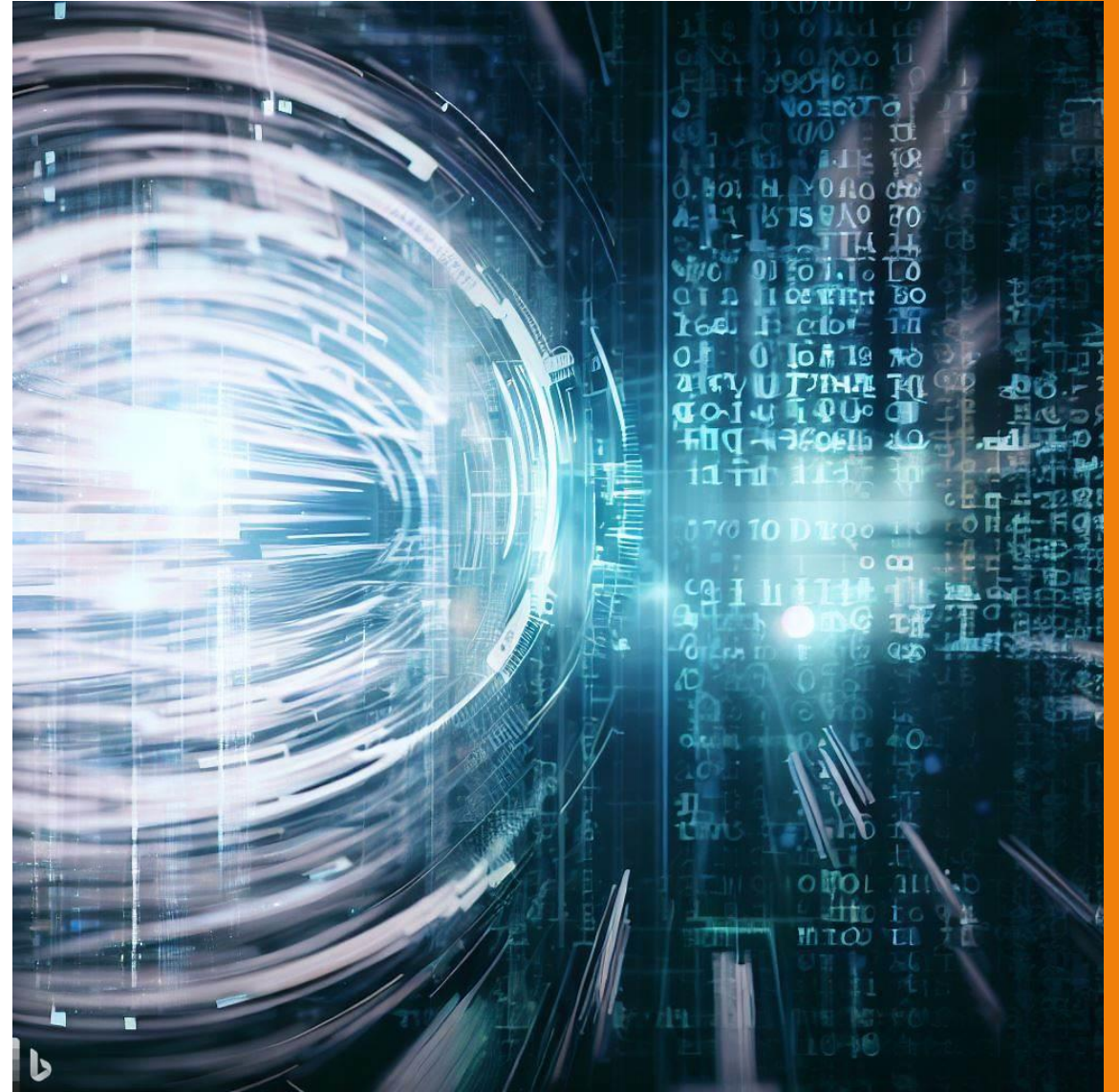
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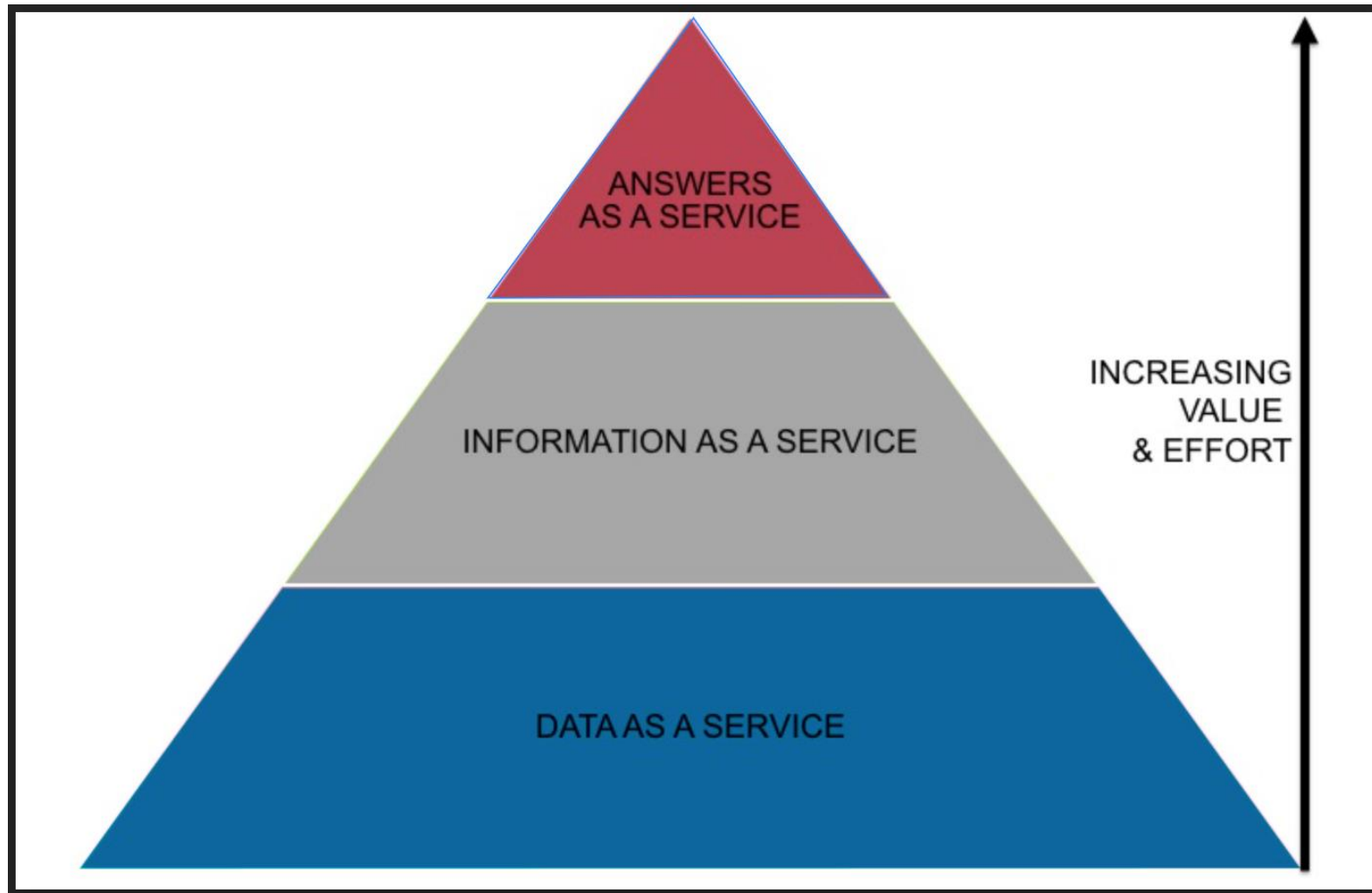


# Introduction to analytics



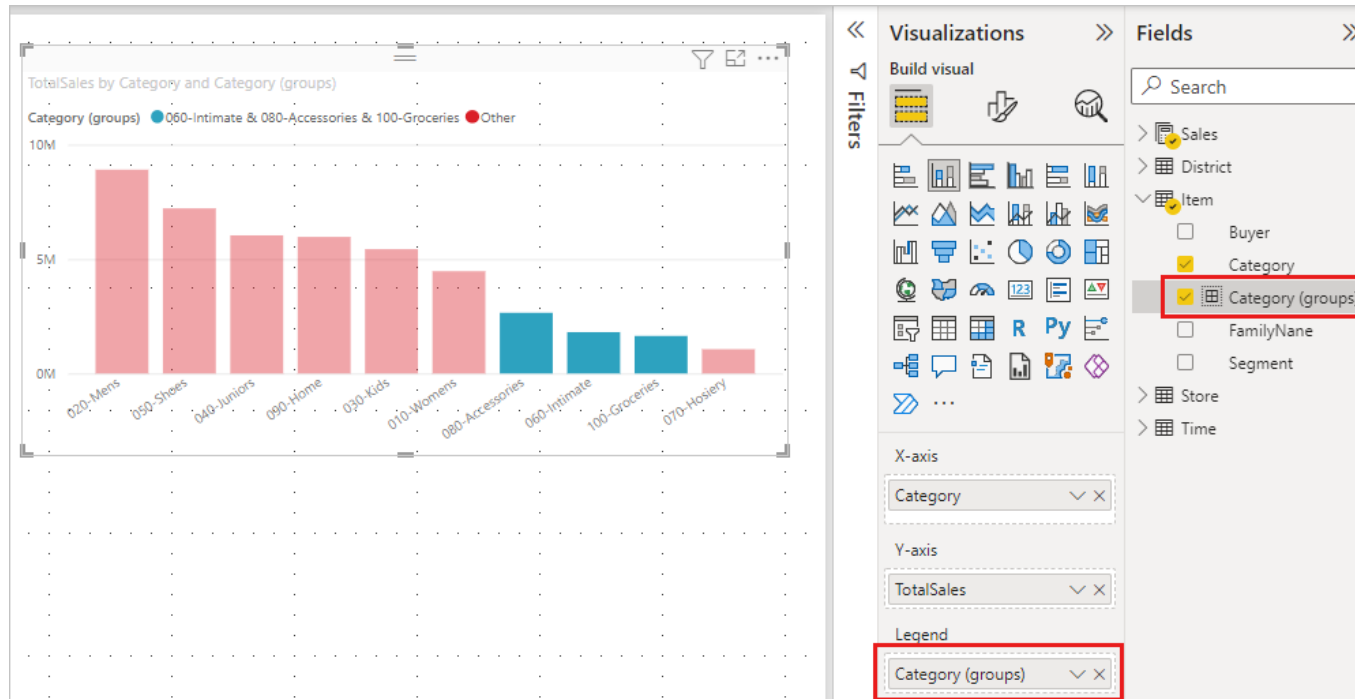


# Data As A Service (DaaS)



# Statistical functions

# Using grouping



Visualizations

Build visual

Fields

Search

Sales

District

Item

Buyer

Category

Category (groups)

FamilyName

Segment

Store

Time

X-axis: Category

Y-axis: TotalSales

Legend: Category (groups)

Small multiples

Add data fields here

Tooltips

Add data fields here

Drill through

Remove field

Rename for this visual

Move to

New quick measure

Show items with no data

New group

Edit groups

# Use a Top N filter type

Top 10 Product by Sales



**Filters**

Search

Filters on this visual

Cost of Sales is (All)

**Product Name**  
top 10 by Cost of Sales  
Filter type ⓘ  
Top N  
Show items:  
Top 10  
By value  
Cost of Sales  
Apply filter

Add data fields here

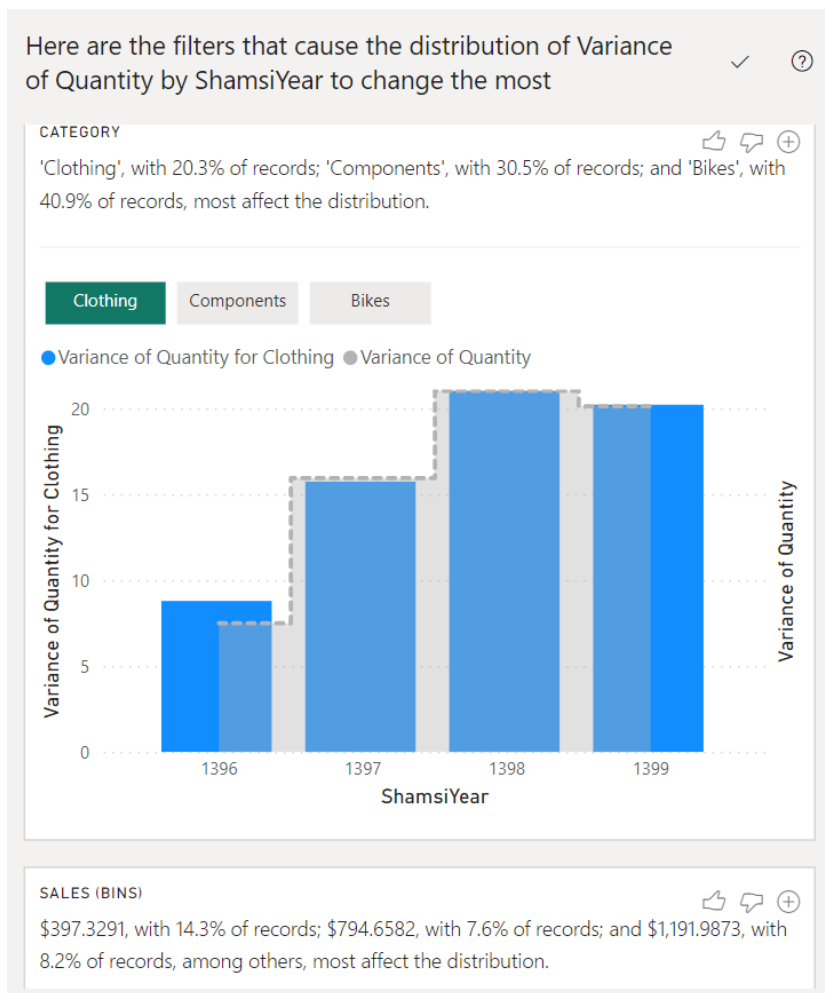
**Visualizations**

Axis  
Product Name

Legend  
Add data fields here

Values  
Cost of Sales

# Use the Analyze & Summarize features



# Course Index

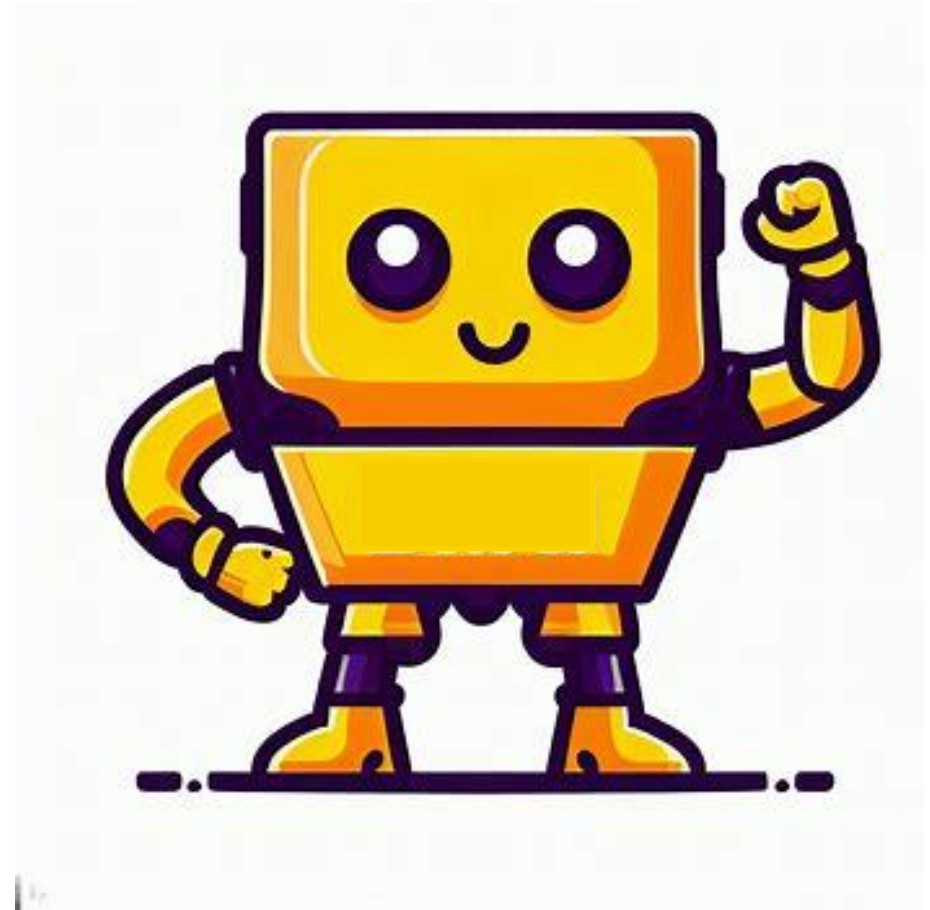
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# Advanced Topics

- Scatter Chart
- Drill Through
- Advanced tooltips
- Dax X functions
- Q&A
- AI features of Power Query

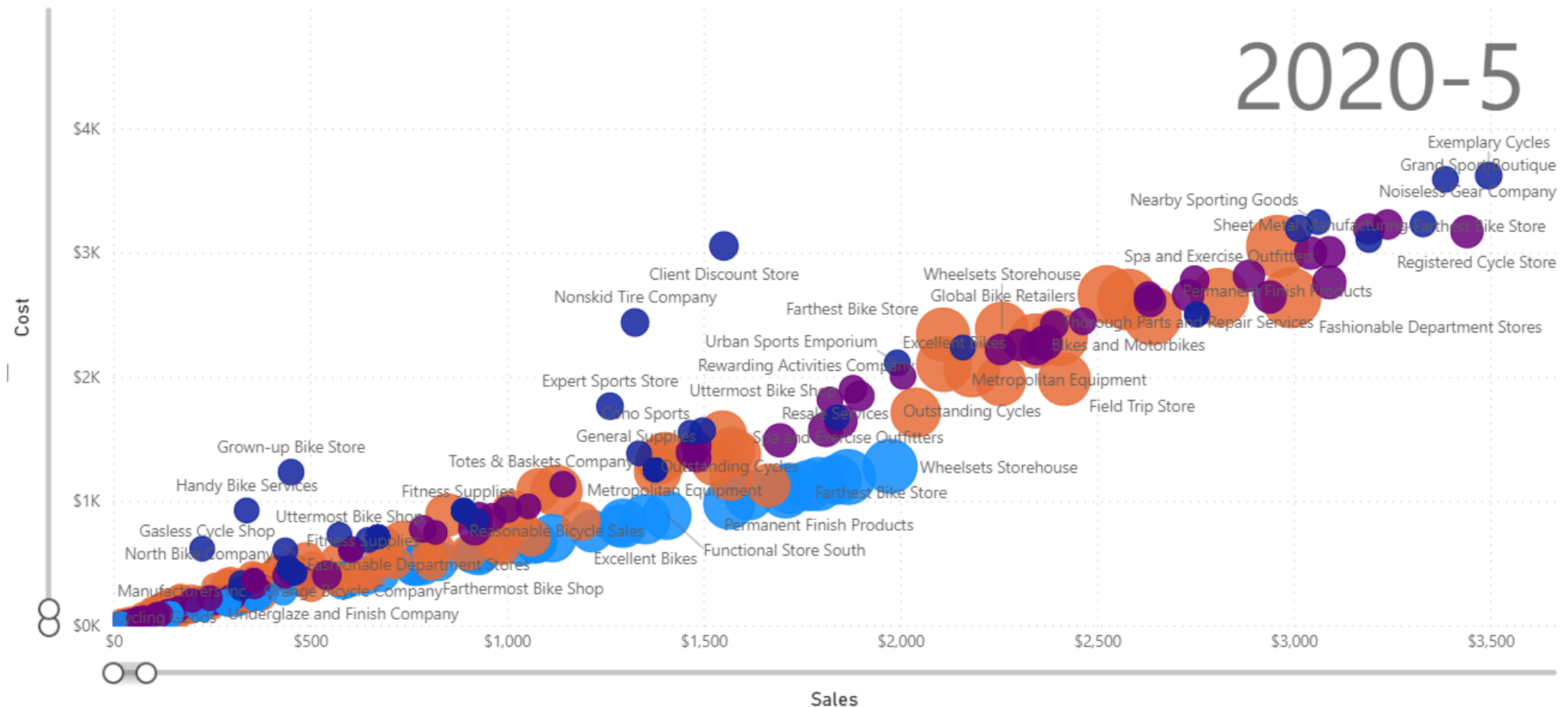


# Scatter Chart

Sales, Cost and Quantity by Reseller, Category and MonthKey



Category Accessories Bikes Clothing Components



# DAX X Functions

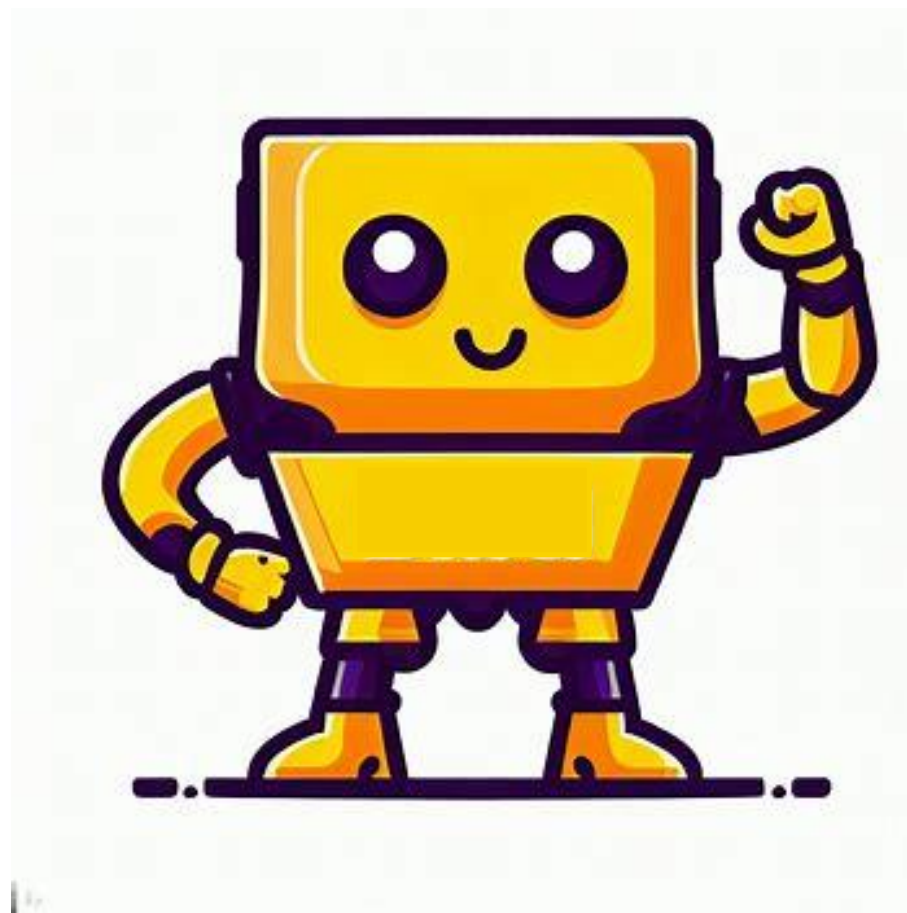
| "Non-X" Function | "X" Function |
|------------------|--------------|
| AVERAGE          | AVERAGEX     |
| COUNTA           | COUNTAX      |
| COUNT            | COUNTX       |
| MAX              | MAXX         |
| MIN              | MINX         |
| SUM              | SUMX         |
| GEOMEAN          | GEOMEANX     |
| MEDIAN           | MEDIANX      |
| STDEV.S          | STDEVX.S     |

# DAX X Functions

| ID | Price | Tax | Price + Tax |
|----|-------|-----|-------------|
| 1  | 1000  | 10  | 1010        |
| 2  | 2000  | 19  | 2019        |
| 3  | 1500  | 11  | 1511        |
| 4  | 2500  | 22  | 2522        |

- $\text{SUMX}(\text{Table}, \text{Price} + \text{Tax}) = 7062$
- $\text{AVERAGEX}(\text{Table}, \text{Price} + \text{Tax}) = 1765.5$
- $\text{MINX}(\text{Table}, \text{Price} + \text{Tax}) = 1010$
- $\text{MAXX}(\text{Table}, \text{Price} + \text{Tax}) = 2522$

# Q&A



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